Singularity

Relational Heights and Arithmetic Visibility - Research Monograph

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Reading guide

This monograph develops a precise question: how much of a relation structure can be recovered from integer-height populations, and where does its first hidden difference appear in arithmetic? Chapters 1-4 establish the objects and finite observation hierarchy. Chapters 5-8 develop primewise assembly, local analytic comparison and singular synchronization. Chapters 9-11 give exact examples, inverse questions and the bridges to earlier work. Chapters 12-14 develop computational methodology, the research program and the literature. The implementation appendix states what version 0.1.0 actually executes. The final roadmap defines thirty-six subsequent development pushes.

The central distinction is between an observation and the correspondence it forgets. A binary census can agree for every subset size and boundary size while a nested-height census differs. The arithmetic lift retains these local distinctions and turns the first discrepancy into a controlled ratio singularity. It adds analytic structure rather than new graph information. The full marked hierarchy explicitly contains the binary sector; complete graph reconstruction remains a research question.

Definitions and theorems are supplied with hypotheses and proofs. Finite graph calculations have explicit inputs and independently replayable outputs. Established antecedents are attributed in the literature chapter and bibliography. The book is a research foundation and executable companion, not a claim of a completed priority search or unrestricted reconstruction.

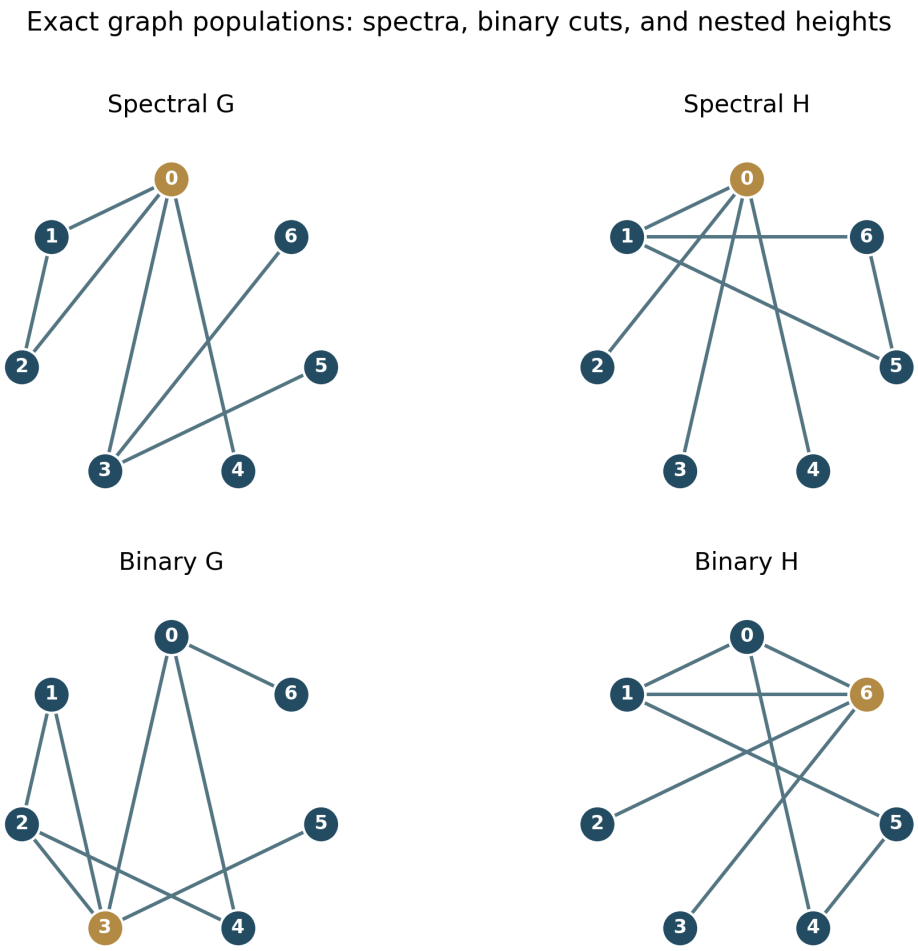


Figure 1: Exact reference graphs. Top: equal Laplacian spectra. Bottom: equal complete binary cut-volume census. Vertex labels remain original.

Chapter 1

What the population observes

1.1 A relation before a prediction

This book begins with a finite graph and a deliberately restricted experiment. Each vertex receives a nonnegative integer height. An edge pays the square of the difference between its endpoint heights. We count all height assignments having a specified total height and a specified energy. We then ask how much of the original relation between vertices survives in those counts. The question concerns the information contained in an exact mathematical observation. It does not require interpreting vertices as people, particles, assets, or physical locations.

The restriction is essential. If an experiment reports every vertex height and every labeled edge contribution, the graph can be read directly from the report. The interesting problem begins after that correspondence is removed. A histogram may tell us how many assignments have a certain energy without telling us which assignments they were. The resulting observation can preserve information that a familiar spectrum loses, while still losing information of its own. Establishing either possibility requires a proof or a concrete exact comparison.

There are consequently two distinct tasks. The forward task computes the population produced by a supplied graph. The inverse task decides what a population determines about its source. An efficient and correct forward computation does not settle the inverse question. Nor does a pair of distinguishable examples establish that every pair is distinguishable. Throughout the manuscript, reconstruction is a question to investigate, not an advertised theorem.

1.2 Objects, assignments, and the meaning of a coefficient

Let $G = (V, E)$ be a finite simple undirected graph with labeled vertex set

$$V = \{0, 1, \dots, k-1\}, \qquad k \geq 1.$$

An edge is an unordered pair of distinct vertices, and each unordered pair occurs at most once. Write d_i for the degree of vertex i, and $A_{ij} \in \{0, 1\}$ for the adjacency matrix. Connectedness will be required explicitly when needed. The basic formal counting series itself is defined for disconnected graphs as well.

A height profile is a vector $a = (a_0, \dots, a_{k-1}) \in \mathbb{Z}_{\geq 0}^k$. Its volume, energy, and span are

$$|a| = \sum_i a_i, \qquad E_G(a) = \sum_{\{i,j\} \in E} (a_i - a_j)^2, \qquad h(a) = \max_i a_i - \min_i a_i.$$

The word **volume** always means total integer height. It is not a Euclidean volume, a determinant, the number of occupied vertices, or the number of nonzero levels. A profile of volume four might put height four at one vertex, height two at two vertices, or height one at four vertices. These assignments generally have different energies.

Define the local series

$$L_G(x,q) = \sum_{a \in \mathbb{Z}^k_{\geq 0}} x^{|a|} q^{E_G(a)} = \sum_{m \geq 0} C_{G,m}(q) x^m.$$

The coefficient $[x^m q^e] L_G$ is the number of original labeled assignments of volume m and energy e. It does not count graph automorphism orbits. If a symmetry exchanges two assignments, both remain in the population. This convention makes coefficients directly comparable to enumeration and avoids inserting unannounced symmetry factors.

Theorem SNG-F01 (finite local coefficients). For every finite graph on $k \geq 1$ vertices and every $m \geq 0$, $C_{G,m}(q)$ is a polynomial in q with nonnegative integer coefficients. Its coefficient sum is

$$C_{G,m}(1) = \binom{m+k-1}{k-1}.$$

If $\Delta = \max_i d_i$, its degree is at most Δm^2.

Proof. The nonnegative integer solutions of $a_0 + \cdots + a_{k-1} = m$ are the weak compositions of m into k parts. The usual separator construction gives their number as the displayed binomial coefficient. In particular, there are finitely many terms in $C_{G,m}$. Every energy is a nonnegative integer. Moreover,

$$E_G(a) = \sum_i d_i a_i^2 - 2 \sum_{\{i,j\} \in E} a_i a_j \leq \Delta \sum_i a_i^2 \leq \Delta \Big(\sum_i a_i \Big)^2.$$

The first inequality uses nonnegativity of all heights; the second follows by expanding the square. This proves all assertions. □

Thus L_G is well defined as an element of $\mathbb{Z}[q][[x]]$. This assertion is formal: no choice of numerical x or q, and no convergence argument, is required. Later analytic interpretations must separately state the region in which a formal identity can be evaluated.

1.3 Removing a repetition that carries no relational information

Adding the same integer to every height leaves every edge difference unchanged:

$$E_G(a + t\mathbf{1}) = E_G(a), \qquad |a + t\mathbf{1}| = |a| + kt.$$

Every nonnegative profile has a unique decomposition

$$a = b + t\mathbf{1}, \qquad t = \min_i a_i, \qquad \min_i b_i = 0.$$

We therefore retain the normalized series

$$B_G(x,q) = \sum_{\substack{b \geq 0 \\ \min b = 0}} x^{|b|} q^{E_G(b)}.$$

The normalization removes the common-height tower, not an assignment-specific ambiguity. It retains one explicitly specified representative of each integer profile modulo common translation.

To distinguish assignments with one nonzero level from assignments with several levels, introduce the full marked series

$$\mathcal{B}_G(x,q,z) = \sum_{\substack{b\geq 0\\ \min b=0}} x^{|b|} q^{E_G(b)} z^{\max b}.$$

The symbol $B_G(x,q,z)$ may be used for this same three-variable object when its arguments are displayed. In these foundational chapters, $\mathcal{B}_G$ makes the extra mark visually explicit, while $B_G(x,q) = \mathcal{B}_G(x,q,1)$. On normalized profiles, maximum height equals span. Consequently z records a property that common translation does not change.

Theorem SNG-F02 (normalization and span marking). For every finite graph on $k \geq 1$ vertices,

$$B_G(x,q) = (1-x^k)L_G(x,q).$$

More generally, if

$$\widetilde{L}_G(x,q,z) = \sum_{a\geq 0} x^{|a|} q^{E_G(a)} z^{h(a)},$$

then

$$\mathcal{B}_G(x,q,z) = (1-x^k)\widetilde{L}_G(x,q,z).$$

Both normalized series have nonnegative integer coefficients despite the subtraction in these formulas.

Proof. Use the unique decomposition $a = b + t\mathbf{1}$. Energy and span are unchanged, whereas volume increases by kt. Hence $\widetilde{L}_G = \mathcal{B}_G \sum_{t\geq 0} x^{kt} = \mathcal{B}_G/(1-x^k)$. Specializing $z = 1$ gives the first identity. Each coefficient of $\mathcal{B}_G$ counts an explicitly defined finite set, proving nonnegativity without relying on cancellation. □

Keeping k fixed, B_G and L_G contain exactly the same formal information: either is recovered from the other by multiplication or division by $1-x^k$. The span mark is a different matter. Specializing $z = 1$ merges populations of different spans. We do not assume that this merging can be inverted.

Observation	Coefficient meaning	Correspondence deliberately removed
$L_G(x,q)$	All profiles by total height and energy	Identity of each profile
$B_G(x,q)$	Translation-normalized profiles by total height and energy	Identity of each normalized profile
$\mathcal{B}_G(x,q,z)$	Normalized profiles by total height, energy, and span	Identity of each profile within a marked cell
$[z]\mathcal{B}_G$	Binary profiles, excluding the two constant profiles	Identity of the vertex subset
$B_G(1,q)$, when defined	Translation classes by energy alone	Total height and span

1.4 Labeled populations can define unlabeled observations

Labels specify the population being counted, but the aggregate series does not privilege a particular labeling. Let $\sigma : V(G) \to V(H)$ be a graph isomorphism. Transport a profile by $a'_{\sigma(i)} = a_i$. This operation permutes the coordinates; it does not merge assignments.

Theorem SNG-F03 (isomorphism invariance with original multiplicity). Graph isomorphisms preserve L_G, B_G, and $\mathcal{B}_G$.

Proof. Coordinate permutation is a bijection on profiles and on normalized profiles. It preserves the sum, minimum, and maximum of their coordinates. Since it is a bijection on edges, it preserves the sum of squared differences. Each coefficient population is therefore carried bijectively to the corresponding population for H. □

The theorem proves one implication: isomorphic graphs have equal observations. It does not prove the converse. An invariant is allowed to identify nonisomorphic graphs. The inverse program investigates whether this particular invariant does so, and, if it does, which features remain recoverable despite that blindness.

For example, a single edge has energy $(a_0 - a_1)^2$. A normalized nonzero profile is either $(h, 0)$ or $(0, h)$, so

$$\mathcal{B}_{K_2}(x, q, z) = 1 + 2 \sum_{h \geq 1} x^h q^{h^2} z^h.$$

The coefficient two records the two original assignments. It is not replaced by one because the edge has an endpoint-exchanging symmetry. This elementary example is a useful check on every later convention.

For a slightly richer example, take the three-vertex path $0 - 1 - 2$. Its degrees are $(1, 2, 1)$, and direct enumeration gives

$$[x]B_G = 2q + q^2, \qquad [x^2]B_G = 2q + q^2 + 2q^4 + q^8.$$

At volume two, placing both units at an endpoint contributes q^4, placing both at the middle contributes q^8, and placing one unit at each of two vertices contributes the remaining terms. The full mark separates the two mechanisms:

$$[x^2]\mathcal{B}_G = z(2q + q^2) + z^2(2q^4 + q^8).$$

The span-one term consists of three binary subsets. The span-two term consists of three profiles concentrated at one vertex. Setting $z = 1$ adds these two populations without recording which mechanism produced each contribution.

At volume three, the unnormalized count contains the all-one profile, whose energy is zero. The normalized coefficient removes that one translation of the zero profile:

$$[x^3]B_G = 2q^2 + 4q^5 + 2q^9 + q^{18}.$$

Its coefficient sum is nine, compared with ten weak compositions of three into three parts. Thus normalization changes an original population by a precisely specified quotient rule; it does not arbitrarily discard an inconvenient solution. A useful calculation record should make this distinction visible whenever normalized and unnormalized outputs appear together.

For comparison, the triangle has

$$[x]B_{K_3} = 3q^2, \qquad [x^2]B_{K_3} = 3q^2 + 3q^8.$$

Already the first coefficient distinguishes the two graphs because their degree histograms differ. Such an example checks conventions but is not a demanding test of an inverse method. Later comparisons should deliberately match the simpler observations before attributing a discrepancy to a deeper mechanism.

1.5 What a result in this book means

Several mathematical activities appear together, and their conclusions should not be conflated. A proved identity between formal series applies coefficientwise to every allowed input. An exact finite computation establishes the cells actually enumerated. A comparison through volume M establishes equality or discrepancy through that volume. A numerical graph search may suggest a pattern without proving that no later counterexample exists. An analytic theorem requires its own convergence and continuation hypotheses even when its starting Euler factors have exact coefficients.

The book therefore separates mathematical propositions, software behavior, finite examples, and research targets. The numbered foundational statements below are proved from their displayed hypotheses. They are mathematical statements, not assertions that a proof assistant checked the implementation. Software validation supplies another kind of evidence: it tests whether a program realizes the stated mathematics on the cases examined.

Type of statement	Appropriate evidence	Scope of conclusion
Exact forward coefficient	Enumeration, recurrence, or an exact algebraic identity	The stated coefficient or family
Prefix agreement	Equal coefficient dictionaries through a declared volume	That finite prefix
Strict separation	One exact unequal coefficient	The two observations differ
Reconstruction theorem	A proof recovering isomorphism from equality	Every graph in the proved class
Reconstruction conjecture	A precise proposed implication	A question, not an established result
Analytic continuation	A convergent factorization on a stated domain	That domain and chosen branches

There is also a useful distinction between a new question for this project and a historically new mathematical construction. Integer graph lattices, Ising-type observations, graph partition functions, and inverse problems all have substantial literatures. The graph cut-lattice viewpoint belongs to established background, including [BHN1997]. Binary energy populations overlap directly with graph Ising polynomials with an external field; [AM2009] is relevant background. General graph reconstruction from families of observations must be compared carefully with the much richer families studied in [Lovasz2012]. No historical priority claim is needed to make the present calculations worthwhile.

The particular program here is to track exactly which relational information survives a succession of increasingly marked arithmetic height observations. Its central difficulty is **lost correspondence**. Counts can be exact while the identities responsible for them are absent. An inverse theorem must recover what that loss hides; an impossibility theorem must demonstrate that the lost information cannot be recovered from the declared observation. Both are substantive mathematical outcomes.

The next chapter puts the translation quotient on an integral lattice. The third chapter explains why multiple nested boundaries retain more information than a single binary cut and why repeated levels cannot be omitted. The fourth chapter turns coefficient order into a precise hierarchy of visibility, beginning with degrees and degree mixing and ending with the open reconstruction question.

Chapter 2

Graphs, integral cut lattices, and the height mark

2.1 The quadratic form and its kernel

Let G have adjacency matrix A, degree matrix $D = \mathrm{diag}(d_i)$, and combinatorial Laplacian $\Delta_G = D - A$. Expanding each square in the definition of energy gives

$$E_G(a) = a^\top \Delta_G a = \sum_i d_i a_i^2 - 2 \sum_{\{i,j\} \in E} a_i a_j.$$

The Laplacian is symmetric and positive semidefinite. This is stronger than a numerical spectral observation: the displayed edge-sum expression proves positivity for every real vector.

Theorem SNG-F04 (zero energy and component constants). For $a \in \mathbb{R}^k$, $E_G(a) = 0$ if and only if a is constant on each connected component. In particular, for connected G, $\ker \Delta_G = \mathbb{R}\mathbf{1}$.

Proof. A finite sum of nonnegative squares vanishes precisely when every summand vanishes. Thus adjacent vertices must have equal coordinates. Equality propagates along paths, proving constancy on every component. The converse follows because every edge lies within a component. Finally, $\Delta_G a = 0$ implies $a^\top \Delta_G a = 0$; component constancy also directly implies $\Delta_G a = 0$. □

For connected graphs, the zero direction is common translation. For disconnected graphs there are additional zero directions: different components can be shifted independently. This difference matters when a formal series is evaluated without its height mark. Quotienting only by global translation does not remove all zero modes of a disconnected graph.

2.2 An orientation is a bookkeeping choice

Choose an arbitrary orientation of every edge. Let ∂ be the $|E|$-by-k matrix whose row for an oriented edge $i \to j$ has -1 in column i, $+1$ in column j, and zero elsewhere. Then

$$(\partial a)_{i \to j} = a_j - a_i, \qquad \partial^\top \partial = \Delta_G.$$

The image

$$\mathcal{C}(G) = \partial \mathbb{Z}^k \subseteq \mathbb{Z}^E$$

is the integral cut, or tension, lattice in this convention. It consists of edge differences produced by integer vertex potentials. An arbitrary integer edge vector need not lie in this image: differences around every oriented cycle must sum to zero.

Reversing an edge orientation multiplies the corresponding coordinate by -1. This is an integral orthogonal transformation of $\mathbb{Z}^E$. It preserves inner products, squared lengths, and every energy count. An orientation is therefore needed to write vectors, but none of the observations depends on it.

Theorem SNG-F05 (integer quotient and cut-lattice coordinates). If G is connected, $a \mapsto \partial a$ induces an isomorphism of abelian groups

$$\mathbb{Z}^k/\mathbb{Z}\mathbf{1} \;\cong\; \mathcal{C}(G).$$

Under this isomorphism, squared lattice length is $E_G(a)$. If a root vertex r is fixed and $\Delta_G^{(r)}$ is obtained by deleting row and column r, the energy in rooted coordinates is the positive-definite integral form

$$u^\top \Delta_G^{(r)} u, \qquad u \in \mathbb{Z}^{k-1}.$$

Proof. The map is surjective onto its stated image. Its integer kernel consists of constant integer vectors by Theorem SNG-F04. The first isomorphism theorem gives the quotient description. The equality of squared length and energy follows from $\partial^\top \partial = \Delta_G$. Every integer translation class has a unique representative with coordinate $a_r = 0$, obtained by subtracting $a_r\mathbf{1}$. Substitution gives the reduced quadratic form. If that form vanishes, its full representative is constant and has root coordinate zero, so it is zero. The form is therefore positive definite. □

The root-zero representative is particularly convenient for linear algebra. It may have negative coordinates. The minimum-zero representative used by B_G is particularly convenient for a nonnegative height mark. These are two coordinate choices on the same quotient, not two different populations of translation classes.

2.3 A determinant that counts trees

The determinant of the reduced Laplacian gives a useful consistency check connecting the integer form with ordinary graph structure. We include a proof because it makes clear which combinatorial data enter the lattice and because no floating-point eigenvalue calculation is needed.

Theorem SNG-F06 (matrix-tree determinant). For connected G on $k \geq 2$ vertices and any root r,

$$\det \Delta_G^{(r)} = \tau(G),$$

where $\tau(G)$ is the number of spanning trees of G.

Proof. Delete column r of ∂, obtaining M. Then $\Delta_G^{(r)} = M^\top M$. The Cauchy–Binet formula gives

$$\det(M^\top M) = \sum_{F \subseteq E,\ |F|=k-1} \det(M_F)^2,$$

where M_F restricts M to the selected edge rows. If (V, F) is disconnected, choose a component not containing r. Its indicator vector, restricted to nonroot coordinates, is a nonzero vector killed by M_F, so the determinant is zero. A connected graph with

k vertices and $k-1$ edges is a tree. For a tree, the determinant is ± 1: repeatedly choose a nonroot leaf, expand in its column, and delete the leaf and its incident edge. Each expansion contributes a sign and an entry ± 1; the final empty determinant is one. Thus exactly the spanning-tree subsets contribute one to the sum. □

The theorem identifies a lattice determinant. It does not say that this determinant determines the lattice or the graph. Many nonisomorphic graphs have the same number of spanning trees. Even a complete list of real Laplacian eigenvalues describes the quadratic form only up to real orthogonal change of coordinates, whereas an integer height population depends on where the integer lattice sits inside that real space.

This distinction is elementary but decisive. A real orthogonal matrix preserves lengths of all real vectors. It need not map $\mathbb{Z}^{k-1}$ to itself. An integral change of lattice basis must lie in $\mathrm{GL}_{k-1}(\mathbb{Z})$, a much more restrictive group. Consequently equal real spectra alone do not supply a bijection between integer height profiles.

2.4 Unmarked theta and the extra information in a mark

For connected G, the unmarked cut-lattice theta series is

$$\Theta_G(q) = \sum_{u \in \mathbb{Z}^{k-1}} q^{u^\top \Delta_G^{(r)} u}.$$

Changing the root changes the integral coordinates but not the translation classes being counted. Likewise, an integral isometry of cut lattices preserves this series. Graph cut lattices and their relationship with graph structure have a substantial established theory; [BHN1997] provides relevant background. No converse classification from theta-series equality is assumed here.

Theorem SNG-F07 (finite energy shells and convergence). If G is connected and $k \geq 2$, every energy coefficient of Θ_G is finite, and the series converges absolutely for $|q| < 1$. Moreover,

$$\Theta_G(q) = B_G(1, q)$$

as a coefficientwise formal energy series and, for $|q| < 1$, as a convergent sum.

Proof. In root-zero coordinates, join r to each vertex i by a simple path of length at most $k-1$. The telescoping sum of edge differences and Cauchy–Schwarz give

$$u_i^2 \leq (k-1) E_G(u), \qquad \sum_{i \neq r} u_i^2 \leq (k-1)^2 E_G(u).$$

An energy shell $E_G(u) = e$ therefore lies in a finite integer box. If $0 < \rho < 1$, then

$$\rho^{E_G(u)} \leq \rho^{\|u\|_2^2/(k-1)^2},$$

and the sum of the right side is a product of convergent one-dimensional Gaussian sums. The case $q = 0$ is immediate. Finally, root-zero and minimum-zero profiles each select exactly one representative of every translation class and preserve energy. Their energy sums coincide. □

For a disconnected graph with at least two nonempty components, $B_G(1, q)$ generally diverges, even for $0 < q < 1$: set one component to height zero and another to arbitrary common height. These are distinct normalized zero-energy profiles. The true cut lattice quotients by componentwise constants, while our normalization quotients by one global constant. Those constructions must not be silently identified.

The total-height mark retains a feature that the lattice norm alone does not name. For an arbitrary integer representative a, define

$$\mu([a]) = \sum_i a_i - k \min_i a_i, \qquad \eta([a]) = \max_i a_i - \min_i a_i.$$

Both are well defined on translation classes. They equal volume and span on the minimum-zero representative. The marked theta population is therefore a lattice together with extra functions on its elements. An abstract norm-preserving lattice bijection need not preserve those functions.

2.5 The height mark is an asymmetric gauge

The function μ is not a linear functional on the quotient. A change in which vertex attains the minimum changes its formula. This nonlinearity is part of what the mark observes, rather than a defect in the normalization.

Theorem SNG-F08 (gauge identities and complement reciprocity). On $\mathbb{Z}^k/\mathbb{Z}\mathbf{1}$, the functions μ and η satisfy

$$\mu(\alpha+\beta) \le \mu(\alpha) + \mu(\beta), \qquad \mu(t\alpha) = t\mu(\alpha) \quad (t \in \mathbb{Z}_{\ge 0}),$$

$$\mu(\alpha) + \mu(-\alpha) = k\eta(\alpha).$$

For fixed span h, put $B_{G,h}(x,q) = [z^h]\mathcal{B}_G(x,q,z)$. This is a polynomial and obeys

$$B_{G,h}(x,q) = x^{kh} B_{G,h}(x^{-1},q).$$

Proof. For representatives a, b, $\min_i(a_i+b_i) \ge \min_i a_i + \min_i b_i$; subtracting k times this inequality from the coordinate sum proves subadditivity. Nonnegative integer homogeneity is immediate. Also $\min(-a) = -\max(a)$, so adding the formulas for $\mu([a])$ and $\mu([-a])$ gives $k(\max a - \min a)$. For reciprocity, every normalized span-h profile has coordinates in $\{0,\dots,h\}$, so the set is finite. The involution $a \mapsto h\mathbf{1} - a$ preserves energy, exchanges minimum and maximum, and changes volume from m to $kh - m$. Summing over the involution proves the identity. □

The reciprocity is spanwise. After setting $z = 1$, profiles with many different h are merged, and there is no single factor x^{kh} applying to their sum. This is a concrete example of a relation visible with the full mark but not immediately expressible after that mark is forgotten.

Structure retained	Natural coordinate change	Quantity necessarily preserved
Real quadratic space	Real orthogonal isometry	Real energy
Integral cut lattice	Integral lattice isometry	Integer energy shells
Height-marked lattice population	Bijection preserving energy and μ	Joint volume-energy counts
Fully marked population	Bijection preserving energy, μ, and η	Joint volume-energy-span counts

These rows are preservation requirements, not a claim that every bijection in a lower row arises from graph relabeling. The inverse problem asks precisely how much of the source graph is forced by increasingly restrictive observations.

2.6 Two endpoint calculations

At $q = 1$, energy is forgotten completely. Every coordinate contributes its own geometric series, so

$$L_G(x,1) = \frac{1}{(1-x)^k}, \qquad B_G(x,1) = \frac{1-x^k}{(1-x)^k}.$$

No edge appears in these expressions. At $q = 0$, only zero-energy assignments remain. If the connected components have sizes $k_1, \dots, k_c$, Theorem SNG-F04 gives

$$L_G(x,0) = \prod_{j=1}^{c} \frac{1}{1-x^{k_j}}, \qquad B_G(x,0) = (1-x^k)\prod_{j=1}^{c} \frac{1}{1-x^{k_j}}.$$

For connected graphs this reduces to $L_G(x,0) = 1/(1-x^k)$ and $B_G(x,0) = 1$. Thus both endpoints erase most graph structure: one ignores every edge energy, and the other retains only exact agreement within components. An inverse experiment must retain intermediate energy information rather than relying on these endpoint values alone.

2.7 Disjoint unions and recoverable component sizes

The unnormalized series factors naturally over independent components. Normalization by one global minimum does not factor in the same way, because choosing the minimum of the combined profile couples the choice of representatives in its components. This is a useful instance in which an exact quotient must be carried through an otherwise simple product rule.

Theorem SNG-F19 (component factorization and its endpoint inverse). For graphs G, H on $k, \ell \geq 1$ vertices,

$$L_{G \sqcup H}(x,q) = L_G(x,q) L_H(x,q),$$

$$B_{G \sqcup H}(x,q) = \frac{1-x^{k+\ell}}{(1-x^k)(1-x^\ell)} B_G(x,q) B_H(x,q).$$

Moreover, the zero-energy series $L_G(x,0)$ determines the multiset of connected-component sizes.

Proof. A profile on a disjoint union is uniquely a pair of profiles on its components. Both volume and energy add, proving the first identity. Substituting $L_G = B_G/(1-x^k)$, and the corresponding formulas for H and the union, proves the second identity in formal series. For the final assertion, let c_j be the number of components of size j. Then

$$L_G(x,0) = \prod_{j \geq 1} (1-x^j)^{-c_j}.$$

Formal logarithms are legitimate because the constant term is one. Expanding $-\log(1-x^j) = \sum_{r \geq 1} x^{jr}/r$, put $b_n = n[x^n] \log L_G(x,0)$. Then

$$b_n = \sum_{j \mid n} j c_j.$$

Recover $c_1 = b_1$ and recursively

$$c_n = \frac{1}{n}\left(b_n - \sum_{\substack{j \mid n \\ j<n}} j c_j\right).$$

This uniquely determines every component-size multiplicity. Only finitely many are nonzero. □

The inverse statement is deliberately modest. Zero energy knows component sizes but not the edges within a component. A path and a complete graph on the same connected vertex set have identical zero-energy series. The formal logarithm extracts exactly the available component information without pretending that it contains finer internal structure.

There is a related warning about the full span mark. The span of a concatenated pair of arbitrary profiles depends on their relative vertical placement, not just on the two individual spans. Consequently one should not multiply $\mathcal{B}_G$ and $\mathcal{B}_H$ and declare the result to be $\mathcal{B}_{G \sqcup H}$. A correct marked factorization would need enough additional data to retain minima, maxima, and their relative shifts. The scalar unnormalized product avoids that missing correspondence because it sums actual coordinates before applying a quotient.

The cut-lattice description clarifies why integer heights are a useful meeting point. They retain the arithmetic placement of a quadratic form, while their minimum-zero representatives supply a nonlinear order mark. The next chapter expresses that order mark through nested threshold sets. This reveals precisely what lies beyond a single binary cut.

Chapter 3

Nested heights and the interaction of boundaries

3.1 A height profile is a stack of threshold sets

A binary profile chooses a subset of vertices: height one on the subset and zero outside. A general integer profile chooses several nested subsets. This description is exact, but only if consecutive threshold sets are allowed to be equal. A jump of three at one vertex crosses three integer levels even if the set of vertices above those levels never changes.

For a normalized profile $a \geq 0$ with $\min a = 0$ and $\max a = h$, define

$$S_t(a) = \{i \in V : a_i \geq t\}, \qquad 1 \leq t \leq h.$$

Then

$$\emptyset \neq S_h \subseteq S_{h-1} \subseteq \cdots \subseteq S_1 \subsetneq V, \qquad a = \sum_{t=1}^{h} \mathbf{1}_{S_t}.$$

The inclusions are weak. Replacing them by strict inclusions changes the population by discarding legitimate profiles.

Theorem SNG-F09 (threshold-chain bijection). Normalized profiles of span $h \geq 1$ are in bijection with weakly nested chains of h nonempty proper vertex subsets. The inverse map is $a_i = \#\{t : i \in S_t\}$, and

$$|a| = \sum_{t=1}^{h} |S_t|.$$

Proof. Threshold sets of a profile have the stated nesting. Because some coordinate is zero, S_1 is proper; because some coordinate is h, S_h is nonempty. Conversely, nesting implies that, for each vertex, membership occurs at an initial segment of the thresholds. Thus the formula for a_i gives exactly the original membership sets when thresholds are taken again. A vertex outside S_1 has height zero and a vertex in S_h has height h, proving normalization and span. The volume identity follows by counting all incidences (i, t) with $i \in S_t$ in two orders. □

The chain is not an auxiliary encoding with multiple extensions. It is uniquely determined by the original profile, and it reconstructs that profile uniquely. Consequently a chain-based counting algorithm preserves original assignment multiplicity without an additional correction factor.

3.2 Squared differences do not add level by level

Write $c_G(S) = |\partial S|$ for the number of edges with one endpoint in S and one outside. For disjoint vertex sets U, W, write $e_G(U, W)$ for the number of edges joining them. The ordinary integer coarea identity says

$$\sum_{\{i,j\}\in E} |a_i - a_j| = \sum_{t=1}^{h} c_G(S_t).$$

Our energy uses squares, so this identity alone is insufficient. An edge crossing two levels contributes four, not two. Cross terms record which nested boundaries are crossed by the same edge.

Theorem SNG-F10 (nested-boundary energy formula). For the threshold chain of a normalized profile,

$$E_G(a) = \sum_{t=1}^{h} c_G(S_t) + 2 \sum_{1\le r<t\le h} e_G(S_t, V \setminus S_r).$$

Equivalently,

$$E_G(a) = \sum_{r,t=1}^{h} e_G(S_{\max(r,t)}, V \setminus S_{\min(r,t)}).$$

Proof. Use the oriented edge-difference matrix ∂ from Chapter 2 and substitute $a = \sum_t \mathbf{1}_{S_t}$ into $E_G(a) = \|\partial a\|_2^2$. Diagonal inner products equal $c_G(S_t)$. If $S_t \subseteq S_r$, an edge contributes to both gradients exactly when it joins S_t to $V \setminus S_r$. On such an edge the two gradient signs agree, irrespective of the chosen orientation; otherwise at least one gradient is zero. Their inner product is therefore $e_G(S_t, V \setminus S_r)$. Expanding the squared norm proves the first formula, and pairing the two orders of each off-diagonal term proves the second. □

An edgewise proof gives the same mechanism. If endpoint heights are $u < v$, the edge crosses exactly $v - u$ thresholds. The ordered pairs of crossed thresholds number $(v - u)^2$. The double sum counts those pairs edge by edge. This argument also explains why repeated threshold sets must remain: an edge can cross several numerically different thresholds whose vertex subsets happen to coincide.

The formula concerns pairwise edge energies throughout. The cross terms are interactions between observations of nested boundaries; they do not turn the graph Hamiltonian into a new three-body or higher-body edge law. Several boundaries can jointly reveal information that isolated boundary histograms lose, even though the underlying energy remains a sum over edges.

3.3 Two examples where repetition matters

Consider the path $0 - 1 - 2$. For $a = (0, 2, 2)$, the threshold chain is

$$S_1 = S_2 = \{1, 2\}.$$

Each set has cut size one. The sum of the two cut sizes is two, but the actual energy is $(0 - 2)^2 + (2 - 2)^2 = 4$. The cross term contributes the missing two, because one edge joins S_2 to $V \setminus S_1$. A strict-chain enumeration would omit this profile entirely.

For $a = (0, 1, 3)$, the same path gives

$$S_1 = \{1, 2\}, \qquad S_2 = S_3 = \{2\}.$$

The diagonal cut sum is three. The boundaries of S_1 and S_2 lie on different edges, so their overlap term is zero; the same holds for S_1, S_3. The repeated pair S_2, S_3 contributes two. The total is five, agreeing with $1^2 + 2^2$.

Path profile	Volume	Span	Diagonal cut sum	Cross-term contribution	Energy
$(0,1,1)$	2	1	1	0	1
$(0,2,2)$	4	2	2	2	4
$(0,1,3)$	4	3	3	2	5
$(0,1,2)$	3	2	2	0	2

The last row is also useful. Distinct boundaries can have no overlap, making the squared energy equal the coarea sum. There is no universal rule replacing the cross terms by a fixed function of the individual cut sizes. Their value depends on how the boundaries sit together in the same graph.

A four-cycle makes this failure especially explicit while holding both individual sizes and individual cut counts fixed. Let the cycle be $0-1-2-3-0$, and let $S = \{0, 1, 2\}$. Consider either $T = \{0\}$ or $T' = \{1\}$ as the inner threshold set. Both chains have outer size three, inner size one, and cut counts two and two. However,

$$e_G(T, V \setminus S) = 1, \qquad e_G(T', V \setminus S) = 0.$$

The two profiles are $(2, 1, 1, 0)$ and $(1, 2, 1, 0)$; both have volume four and span two. Their energies are respectively six and four. Thus even the ordered list of threshold sizes and cut counts does not determine the height energy. One must also retain which boundary edges are shared.

This example does not prove that two graphs with equal complete binary cut polynomials must have different height series. Both profiles occur in the same graph, and a global inverse implication could in principle use information unavailable in this two-boundary summary. What the example proves is the narrower and useful fact that the displayed chain energy cannot be computed from its separate boundary marginals alone.

3.4 An exact chain expansion of the marked series

Combining the bijection and energy formula gives

$$\mathcal{B}_G(x, q, z) = 1 + \sum_{h \geq 1} z^h \sum_{\emptyset \neq S_h \subseteq \cdots \subseteq S_1 \subsetneq V} x^{\sum_t |S_t|} q^{\sum_t c_G(S_t) + 2\sum_{r<t} e_G(S_t, V \setminus S_r)}.$$

This is an identity of formal series. At a specified total volume m, only chains of length at most m can contribute, so the coefficient remains a finite sum. The expression is also a useful specification for an independent implementation: direct weak-composition enumeration and threshold-chain enumeration must return the same marked cells.

The expansion shows exactly what a binary observation omits. The binary sector reports one subset at a time, through its size and cut. A chain term refers to several

nested subsets in the same graph and additionally uses their boundary-overlap counts. Separate histograms of subset sizes and cuts do not retain the correspondence needed to form these overlaps.

This statement is directional. The full marked series certainly contains the binary histogram because its span-one coefficient extracts it. It does not follow that the two-variable specialization $B_G(x,q) = \mathcal{B}_G(x,q,1)$ permits that same extraction. After span is forgotten, a cell of given volume and energy can receive both binary and nonbinary contributions. No inversion of that mixture is proved here.

3.5 The binary sector and its reciprocity

Define the binary cut polynomial

$$I_G(x,q) = \sum_{S \subseteq V} x^{|S|} q^{c_G(S)}.$$

Its height interpretation overlaps the external-field Ising polynomial studied in [AM2009], subject to the stated normalization of the energy and spin variables. The present definition is explicit, so no external convention is needed to compute it.

Theorem SNG-F11 (binary sector extraction). For $k \geq 2$,

$$[z]\mathcal{B}_G(x,q,z) = I_G(x,q) - 1 - x^k.$$

Also $I_G(x,q) = x^k I_G(x^{-1},q)$.

Proof. A minimum-zero span-one profile takes values zero and one and contains both values. It is uniquely the indicator of a nonempty proper subset. Its volume is subset size and its energy is cut size. The omitted empty and full subsets give 1 and x^k. Complementation preserves cut size and sends subset size to $k - |S|$, proving reciprocity. □

The symmetry pairs the two sides of a cut. We count both sides as different profiles. A side of size r and its complement of size $k-r$ appear at their respective powers of x; when the sizes are equal they still contribute two unless other multiplicities happen to coincide.

3.6 Minimum cuts are the first positive energy shell

For a connected graph on $k \geq 2$ vertices, define edge connectivity by

$$\lambda(G) = \min_{\emptyset \neq S \subsetneq V} c_G(S) > 0.$$

This definition suffices here; no equivalence with a disjoint-path formulation is needed.

Theorem SNG-F12 (minimum-energy shell). Every nonzero normalized profile of span h obeys

$$E_G(a) \geq \lambda(G)h.$$

The smallest positive energy is $\lambda(G)$, and its full marked population is

$$[q^{\lambda(G)}]\mathcal{B}_G(x,q,z) = z \sum_{\substack{\emptyset \neq S \subsetneq V \\ c_G(S) = \lambda(G)}} x^{|S|}.$$

Proof. For every nonnegative integer d, $d^2 \geq d$. Hence the energy is at least the absolute-difference sum, which by coarea is $\sum_t c_G(S_t)$. Every threshold set is nonempty and proper, so each cut is at least $\lambda(G)$. This gives the bound. An indicator of a minimum cut has energy $\lambda(G)$, so the bound is attained. If a nonzero profile has that energy, the bound forces $h = 1$, and then its subset must be a minimum cut. □

For a path, $\lambda = 1$, so the first positive shell consists of the two orientations of every bridge cut. For a triangle, $\lambda = 2$, and its six nonempty proper subsets give

$$[q^2]\mathcal{B}_{K_3}(x, q, z) = 3z(x + x^2).$$

These formulas identify the very first nonconstant energy sector. They do not recover arbitrary boundary overlaps at higher energy.

3.7 Support restrictions useful for exact computation

Several finite bounds follow directly from the profile interpretation. They identify impossible marked cells before any enumeration and provide consistency checks afterward.

Theorem SNG-F13 (volume, span, and extreme energy). Let $k \geq 2$, and let a be a nonzero normalized profile of volume m and span h. Then

$$h \leq m \leq (k-1)h.$$

If G is connected,

$$E_G(a) \geq \max\left\{\lambda(G)h, \left\lceil \frac{h^2}{\operatorname{diam}(G)} \right\rceil\right\}.$$

For any graph with maximum degree $\Delta > 0$, the largest energy among normalized volume-m profiles is Δm^2, attained exactly by $a = me_i$ at maximum-degree vertices.

Proof. Some coordinate equals h, so the sum is at least h. At least one coordinate is zero and every other coordinate is at most h, giving the upper bound. The minimum-cut bound was proved above. Choose a shortest path between a minimum-height and a maximum-height vertex. If its length is ℓ, telescoping and Cauchy–Schwarz give $h^2 \leq \ell \sum_{e \text{ on path}} (\partial a)_e^2 \leq \operatorname{diam}(G) E_G(a)$. Energy is integral, giving the ceiling. Finally, the bound $E_G(a) \leq \Delta \sum_i a_i^2 \leq \Delta m^2$ was proved in Chapter 1. The profile me_i at a maximum-degree vertex attains it. If equality holds and $\Delta > 0$, then $\sum_i a_i^2 = m^2$. Expanding m^2 shows that all products $a_i a_j$, $i \neq j$, vanish; exactly one coordinate is nonzero. Its degree must be Δ. □

Thus only

$$\left\lceil \frac{m}{k-1} \right\rceil \leq h \leq m$$

can occur in a nonzero volume-m normalized population. For fixed h, complementation from Chapter 2 pairs volumes m and $kh - m$. A computation violating either restriction has lost a normalization condition or a multiplicity.

Nested thresholds provide the human explanation for the hierarchy studied next. One occupied unit reveals a degree. Two units reveal degree mixing. Several units can occupy different heights and thereby test several linked boundaries. The information comes from their shared placement in one relation, not from replacing the original edge law with an unrelated interaction.

3.8 Primitive height scale and exact dilation

Repeated sets also encode a common integer scale. If every nonzero coordinate of a normalized profile is divisible by d, division by d gives another normalized profile. Volumes and spans divide by d, while energy divides by d^2. This is an exact quadratic scaling law, not an approximation valid only at large height.

Call a nonzero normalized profile primitive when the greatest common divisor of its coordinates is one; zeros do not change that greatest common divisor. Let

$$\mathcal{P}_G(x,q,z) = \sum_{\substack{a\geq 0,\ \min a=0\\ a\neq 0,\ \gcd(a_0,\dots,a_{k-1})=1}} x^{|a|}q^{E_G(a)}z^{\max a}.$$

The zero profile is excluded because its common divisor is not a positive scale.

Theorem SNG-F20 (primitive-height decomposition). For every graph on $k \geq 2$ vertices,

$$\mathcal{B}_G(x,q,z) - 1 = \sum_{d\geq 1} \mathcal{P}_G(x^d, q^{d^2}, z^d).$$

If μ is the arithmetic Möbius function, characterized by $\sum_{d|n}\mu(d) = 1$ for $n = 1$ and zero otherwise, then

$$\mathcal{P}_G(x,q,z) = \sum_{d\geq 1}\mu(d)\big(\mathcal{B}_G(x^d, q^{d^2}, z^d) - 1\big).$$

Both identities are coefficientwise finite in every positive volume.

Proof. Every nonzero integer profile has a unique positive greatest common divisor d, and a unique primitive quotient $b = a/d$. Normalization is preserved. Direct substitution gives $|a| = d|b|$, $E_G(a) = d^2E_G(b)$, and $\max a = d\max b$, proving the first identity. For inversion, write $T_dF = F(x^d, q^{d^2}, z^d)$; then $T_dT_e = T_{de}$. Substitute the first identity in the proposed inverse. The coefficient multiplying $T_n\mathcal{P}_G$ is $\sum_{d|n}\mu(d)$, leaving only $n = 1$. At volume $m > 0$, only $d \mid m$ can contribute, so the rearrangement is finite coefficientwise. □

For the path example $(0,2,2)$, the primitive quotient is $(0,1,1)$: volume two becomes four, energy one becomes four, and span one becomes two. This decomposition distinguishes a genuinely new shape of height profile from an integral dilation of an earlier shape. It provides another exact observation hierarchy that can be implemented independently of threshold-chain enumeration.

Primitive height here refers only to common divisibility of integer coordinates. It does not mean a primitive graph, a primitive conductor phase, or an element primitive in a different lattice. Those notions may interact in later calculations, but their definitions must not be merged because they share a word. The decomposition itself uses no connectedness assumption, so it also applies when nonconstant componentwise profiles have zero energy.

Chapter 4

Visibility depth and the loss of correspondence

4.1 Increasing volume as an observation hierarchy

The coefficient $C_{G,m}(q) = [x^m]L_G(x, q)$ reports every energy attainable with exactly m units of nonnegative integer height, together with original assignment multiplicities. Increasing m enlarges the set of placements that can be tested. It does not simply increase numerical precision in an unchanged observation. New occupancy patterns become available: repeated occupation of one vertex, occupation of several vertices, and different heights on linked supports.

For equal-order graphs, normalization does not change the first place at which their scalar-height observations differ. If k is fixed, multiplication by $1 - x^k$ has constant term one. A first nonzero coefficient cannot be removed by that multiplication because no earlier nonzero coefficient exists to cancel it.

The hierarchy must be specified before it is interpreted. A volume prefix of the two-variable series is one hierarchy; a span prefix of the fully marked series is another. The binary sector is one span layer, not one volume layer: a binary subset of size ten has volume ten and span one. Conversely, a one-vertex profile of height ten has the same volume and span ten. Keeping those distinctions prevents an observation about a binary cut polynomial from being silently transferred to an unmarked height specialization.

4.2 The first coefficients can be read by hand

Write e_i for the unit profile at vertex i. At volume one the only profiles are e_i, and

$$E_G(e_i) = d_i.$$

At volume two the profiles are $2e_i$ and $e_i + e_j$, $i < j$. The latter subtracts two from the sum of endpoint degrees precisely when the endpoints are adjacent. This is the first layer that couples local degrees to a relation between vertices.

Theorem SNG-F14 (coefficients through volume three). For every simple graph,

$$C_{G,0}(q) = 1, \qquad C_{G,1}(q) = \sum_i q^{d_i},$$

$$C_{G,2}(q) = \sum_i q^{4d_i} + \sum_{i<j} q^{d_i+d_j-2A_{ij}},$$

and

$$C_{G,3}(q) = \sum_i q^{9d_i} + \sum_{i\neq j} q^{4d_i+d_j-4A_{ij}} + \sum_{i<j<\ell} q^{d_i+d_j+d_\ell-2e_G(\{i,j,\ell\})}.$$

Here $e_G(S)$ denotes the number of edges induced by S. Empty index sums are zero. For normalized coefficients,

$$[x^m]B_G = C_{G,m} - C_{G,m-k},$$

where $C_{G,t} = 0$ for $t < 0$.

Proof. Partition the weak compositions of one, two, and three by their positive parts. The possibilities are respectively 1; 2 or $1+1$; and 3, $2+1$, or $1+1+1$. Substitute each profile in

$$E_G(a) = \sum_i d_i a_i^2 - 2\sum_{\{i,j\}\in E} a_i a_j.$$

The pattern $2e_i + e_j$ is indexed by ordered distinct vertices because the heights distinguish the two positions. The three-unit pattern is indexed by unordered triples. These are disjoint exhaustive cases, giving the formulas without missing multiplicities. The normalized identity is coefficient extraction from Theorem SNG-F02. □

For example, $C_{G,1}$ determines the degree histogram exactly: its coefficient of q^d is the number of vertices of degree d. It does not identify which vertex has each degree, because coordinate labels have been summed out. That missing assignment of degrees to positions is already a small instance of lost correspondence.

Normalization matters at low volume in small graphs. For $k = 2$, $[x^2]B_G = C_{G,2} - 1$, removing the constant profile $(1, 1)$. For $k = 3$, the corresponding first subtraction occurs at volume three. The uniform formula is safer than assuming that normalized and unnormalized coefficients agree at every displayed low order.

4.3 What the second coefficient adds

Define the edge degree-sum polynomial

$$J_G(q) = \sum_{\{i,j\}\in E} q^{d_i+d_j}.$$

This is one measure of how degrees mix across edges. It is coarser than the full joint degree matrix because different degree pairs can have the same sum, but it is already more relational than the degree histogram alone.

Theorem SNG-F15 (second-layer degree mixing). For every graph,

$$C_{G,2}(q) = \sum_i q^{4d_i} + \sum_{i<j} q^{d_i+d_j} + (q^{-2}-1)J_G(q).$$

The expression is a polynomial: every edge has $d_i + d_j \geq 2$. For graphs G, H with the same degree histogram,

$$C_{G,2} = C_{H,2} \iff J_G = J_H.$$

Proof. Separate the pair sum in Theorem SNG-F14 into edges and nonedges. Start by assigning $q^{d_i+d_j}$ to every pair; on an edge replace it by $q^{d_i+d_j-2}$. The change is the displayed correction. If degree histograms agree, both of the first two sums agree, so the difference of second coefficients is $(q^{-2}-1)(J_G-J_H)$. A nonzero Laurent polynomial cannot annihilate another nonzero Laurent polynomial over the integers. Thus the difference vanishes exactly when $J_G - J_H$ does. □

This proposition prevents overinterpreting an attractive example. Suppose an exact comparison gives

$$C_{G,2} - C_{H,2} = 2q^2(1-q)^3(1+q).$$

If the graphs have the same degree histogram, then

$$J_G - J_H = 2q^4(1-q)^2.$$

The observation has detected a difference in degree mixing. That is a valid strict separation from any matched invariants, but it is not evidence that a genuinely high-order height interaction was needed. A stronger next example should first match J_G, then look for a later discrepancy.

Volume	Allowed positive height patterns	Direct information visible in the formula
0	Empty support	One zero profile
1	1	Degree histogram
2	2, 1+1	Degree data and edge degree-sum mixing
3	3, 2+1, 1+1+1	Ordered degree-adjacency weights and degree-decorated triple edge counts
m	Integer partitions of m placed on vertices	Aggregated degree-decorated supports of size at most m

The final row deliberately says “aggregated.” It is not a claim that every decorated induced-subgraph count can be independently recovered from one coefficient polynomial. Different patterns can contribute the same power of q, and their multiplicities are then added.

4.4 A precise finite-depth comparison

For graphs G, H of the same order, define their scalar-height visibility depth by

$$r(G,H) = \min\{m \geq 0 : C_{G,m}(q) \neq C_{H,m}(q)\},$$

with $r = \infty$ when no discrepancy exists. The same definition using $[x^m]B_G$ gives the same value. A depth records how much total height is required to separate the two exact observations. It is not a claim about computation time, and it is not automatically a graph-theoretic reconstruction radius.

Theorem SNG-F16 (prefix equivalence and an ultrapseudometric). For each $M \geq 0$, agreement of $C_{G,m}$ for all $m \leq M$ is an equivalence relation on graphs of fixed order. Visibility depths satisfy

$$r(G,H) \geq \min\{r(G,K), r(K,H)\}.$$

Consequently $d(G,H) = 2^{-r(G,H)}$, with $2^{-\infty} = 0$, is an ultrapseudometric. It is a metric on the quotient obtained by identifying graphs with identical full scalar-height series.

Proof. Equality of finite coefficient lists is reflexive, symmetric, and transitive. If both pairs (G,K) and (K,H) agree below an index r_0, then G,H agree below that index as well. This gives the displayed inequality, including infinite depths. Exponentiating reverses the inequality and gives $d(G,H) \le \max\{d(G,K), d(K,H)\}$. Zero distance means equality of the full series, which is exactly the relation used in the stated quotient. □

Calling this a pseudometric is substantive caution. If nonisomorphic graphs share the entire series, their distance is zero. Proving that this cannot happen in a graph class would be a reconstruction theorem for that class. The present definitions do not assume it.

4.5 What an occupancy coefficient can depend on

A profile of volume m has at most m occupied vertices. Its energy depends on the induced adjacency among those vertices and on their ambient degrees. Edges from an occupied vertex to an unoccupied vertex contribute through its degree; no further structure among unoccupied vertices enters that profile's energy.

To state this precisely, regard the induced graph $G[S]$ as decorated by the ambient degree d_i at every vertex of S. Two decorated induced graphs are isomorphic when a graph isomorphism also matches those degree labels. Let the decorated t-deck record how many subsets of size t have each such type.

Theorem SNG-F17 (decorated-deck sufficiency for a prefix). If two graphs of the same order have equal decorated t-decks for every $t \le M$, then their scalar-height coefficients agree for every $m \le M$. Their fully marked coefficients agree at those volumes as well.

Proof. Fix a nonempty support S of size $t \le m$. Profiles with that exact support assign positive integers b_i to its vertices, summing to m. Their energies are

$$\sum_{i \in S} d_i b_i^2 - 2 \sum_{\{i,j\} \in E(G[S])} b_i b_j.$$

A degree-preserving induced-graph isomorphism gives a coordinate-permutation bijection on these positive assignments, preserving volume, energy, and span. For normalized profiles, exact supports are proper subsets; the full-support case is excluded on both graphs because their common order is the same. Summing over matching decorated support types proves equality at each $m \le M$. The zero profile contributes equally. □

This is a one-way implication. The coefficient hierarchy can be coarser than the decorated decks because it aggregates different supports and positive assignments into the same energy cells. The proposition identifies a sufficient source of indistinguishability and explains the local nature of a bounded-volume experiment without asserting an unproved inversion.

It suggests several distinct research questions. Can nonisomorphic graph families have arbitrarily large finite visibility depth? Which natural graph classes admit a uniform depth bound? Can decorated-deck constructions produce exact equal-series examples, or do higher heights eventually distinguish them? These are precise targets.

Finite searches can provide examples and exclusions within their declared range, but cannot settle the unrestricted questions merely by growing a table.

4.6 Why labeled probes make reconstruction easy

Consider instead the fully vertex-marked series

$$\mathcal{L}_G(z_0, \dots, z_{k-1}; q) = \sum_{a \geq 0} \Big(\prod_i z_i^{a_i} \Big) q^{E_G(a)}.$$

The variables now identify where each unit was placed. This is a much stronger observation than scalar total height.

Theorem SNG-F18 (labeled reconstruction at degree two). The coefficients of $\mathcal{L}_G$ through total z-degree two determine the labeled adjacency matrix of G.

Proof. The coefficient of z_i is the single monomial q^{d_i}, recovering d_i. For distinct i, j, the coefficient of $z_i z_j$ is $q^{d_i + d_j - 2A_{ij}}$. Its exponent, together with the two recovered degrees, gives

$$A_{ij} = \frac{d_i + d_j - \deg_q([z_i z_j]\mathcal{L}_G)}{2}.$$

Every coefficient used is a single monomial with coefficient one, so its exponent is unambiguous. □

Thus a theorem saying that arbitrary vertex-marked probes determine the graph would not resolve the difficult scalar-height inverse problem. The correspondence that makes reconstruction easy has been retained explicitly. The interesting restriction is the specialization $z_i = x$ for all i, which merges those individually identifiable monomials into common volume cells.

This is the same reason that general graph-observation frameworks must be compared with their actual permitted test families. Results for all graph homomorphism counts, as discussed in [Lovasz2012], concern an enormously rich set of probes. One particular partition function or one particular family of height weights does not acquire the same reconstruction power by resemblance.

4.7 Marks, specializations, and the conclusions they permit

For any observation, forgetting a mark is a deterministic map on its data. Equality before forgetting implies equality afterward. A discrepancy afterward also proves a discrepancy beforehand. The reverse implications need not hold. In particular, full span-marked equality implies binary-profile equality by Theorem SNG-F11; scalar-height equality alone is not presently proved to imply it.

Numerical specialization of q can lose information as well. The endpoints are extreme examples: at $q = 1$ every graph of the same order has the same scalar-height series, while at $q = 0$ every connected graph has $B_G = 1$. More generally, a nonzero polynomial discrepancy can vanish at a selected algebraic value of q. Exact polynomial comparison is stronger than comparison at one such value. If q_0 is transcendental, no nonzero integer polynomial vanishes there, so an exact evaluation would preserve coefficient distinctions; that logical observation does not provide an effective exact arithmetic method for an arbitrary transcendental input.

Established implication	Reason	Converse available here?
Graph isomorphism $\Rightarrow$ full marked equality	Coordinate-permutation bijection	Not proved
Full marked equality $\Rightarrow$ scalar-height equality	Set $z{=}1$	Not proved
Full marked equality $\Rightarrow$ binary-profile equality	Extract $[z]$	Not proved
Connected scalar-height equality $\Rightarrow$ unmarked theta equality	Sum finite energy shells	Not proved
Same decorated decks through size $M \Rightarrow$ height-prefix equality through volume M	Support decomposition	Not proved
Exact coefficient discrepancy $\Rightarrow$ different graph observations	Definition of equality	Yes, at some coefficient if the formal series differ

Finally, a later primewise Euler lift of L_G should be understood as an analytic reorganization. Its coefficient at a prime power recovers the corresponding local coefficient, and the local coefficients specify every Euler factor. That lift can expose useful singularities and arithmetic asymptotics, but it does not restore correspondence already discarded by the local histogram. The forward computation, the observation hierarchy, and the inverse reconstruction problem remain distinct throughout.

The resulting program is deliberately open in the right place. We can prove what the first layers observe, identify exact sources of finite-depth agreement, and formulate complete-series reconstruction as a concrete question. We can also search for persistent blindness and treat an exact counterexample as a mathematical result. What we cannot do is promote a successful prefix calculation into a conclusion about every unseen layer. The hierarchy provides the language for making the next calculation precise.

Chapter 5

Prime arithmetic: assembling relational height populations

The height population becomes an arithmetic object when one integer height is assigned to every prime valuation. This step has a precise meaning. A vertex is no longer assigned one nonnegative integer; it is assigned a positive integer, whose exponent at each prime supplies a local height. The graph compares those exponents independently at each prime. Unique factorization then assembles the comparisons into an Euler product. No relation of the form $n_u + n_v = n_w$ is imposed, and numerical closeness of two ordinary integers is not the quantity being measured. The construction records multiplicative disagreement, prime by prime.

Throughout this chapter, $G = (V, E)$ is a finite loopless simple graph with $k \geq 2$ vertices. Connectedness is stated where needed. For a nonnegative height vector a, write

$$|a| = \sum_{v \in V} a_v, \qquad E_G(a) = \sum_{\{u,v\} \in E} (a_u - a_v)^2, \qquad L_G(x, q) = \sum_{a \geq 0} x^{|a|} q^{E_G(a)}.$$

The coefficient of every fixed power of x is a polynomial in q with nonnegative integer coefficients. Thus this definition is meaningful formally before convergence is discussed. For numerical analysis we take $0 \leq q \leq 1$, use $0^0 = 1$, and initially require $|x| < 1$. At fixed volume n, there are exactly $\binom{n+k-1}{k-1}$ labeled profiles. Consequently

$$|L_G(x, q)| \leq (1 - |x|)^{-k} \qquad (|q| \leq 1).$$

This elementary majorant will support every exchange of sums made below. The number of vertices, rather than the number of edges, controls this convenient uniform bound. Graph structure remains in the distribution of energy among profiles of the same volume.

5.1 The arithmetic weight and its exact assembly

For a positive integer tuple $n = (n_v)_{v \in V}$, define its relational valuation energy by

$$\mathcal{E}_G(n) = \sum_{p \ mprime} E_G\big((v_p(n_v))_{v \in V}\big).$$

Only finitely many terms are nonzero. The definition is invariant under simultaneously multiplying all coordinates by the same positive integer: this adds the same prime valuation to every local height and does not change any difference. Such multiplication does change the product $\prod_v n_v$, so energy and size remain separate statistics. Define

$$Z_G(s,q) = \sum_{n_v \geq 1} \frac{q^{\mathcal{E}_G(n)}}{(\prod_v n_v)^s}.$$

Theorem SNG-A01 — valuation assembly. For $0 \leq q \leq 1$ and $\Re s > 1$, the preceding Dirichlet series is absolutely convergent and

$$Z_G(s,q) = \prod_p L_G(p^{-s}, q).$$

Proof. Absolute values of the summands are bounded by $\prod_v n_v^{-\sigma}$, where $\sigma = \Re s$. Their sum is $\zeta(\sigma)^k < \infty$. Restrict first to tuples supported on a finite set of primes. Unique factorization identifies such a tuple with one height vector for each of those primes. Its denominator factors as $\prod_p p^{s|a(p)|}$, and its weight factors as $\prod_p q^{E_G(a(p))}$. Multiplication of the finitely many absolutely convergent local series gives the finite Euler product. Increasing the finite prime set exhausts all tuples. Dominated convergence, or monotone convergence for real positive s, proves the identity. The local estimate $L_G(p^{-s}, q) = 1 + O_G(p^{-\sigma})$ also establishes absolute convergence directly. □

The theorem does not claim that arithmetic assembly creates extra information about the graph. The complete local series determines the Euler product. Its value is that it transfers a local relational statistic into an analytic language in which the first surviving distinction has a sharply identifiable singularity. Subsequent chapters prove that transfer rather than treating the Euler product as decoration.

Consider a single edge with heights a, b. Its energy is $(a-b)^2$. At a prime dividing only one coordinate to exponent one, the local penalty is q. A common factor at that prime contributes no penalty. A quotient with valuation difference two contributes q^4, not q^2. Thus the energy is more sensitive to repeated unequal valuation than a simple count of unmatched prime factors. The quadratic choice is part of the definition, not a claim that it is uniquely natural. Other edge costs can be studied, but their theorems require separate hypotheses.

5.2 Removing the common factor exactly

Translation invariance of energy has a local and a global consequence. Locally it removes the common minimum height. Globally it removes the greatest common divisor of all coordinates. These are the same operation viewed through prime valuations.

Theorem SNG-A02 — common-shift factorization. Put

$$B_G(x,q) = \sum_{\substack{a \geq 0 \\ \min_v a_v = 0}} x^{|a|} q^{E_G(a)}.$$

Then, as a formal power series and for $|x| < 1$,

$$L_G(x,q) = \frac{B_G(x,q)}{1-x^k}.$$

Equivalently, on the quotient lattice $\mathbb{Z}^k/\mathbb{Z}\mathbf{1}$, define

$$\nu([a]) = \sum_v a_v - k \min_v a_v.$$

This function is well defined, and

$$B_G(x,q) = \sum_{[a]\in\mathbb{Z}^k/\mathbb{Z}\mathbf{1}} x^{\nu([a])} q^{E_G(a)}.$$

Proof. Every nonnegative profile has a unique decomposition $a = b + m\mathbf{1}$, with $m = \min a$ and $\min b = 0$. Its energy equals $E_G(b)$, and its volume is $|b| + km$. Summing x^{km} over $m \geq 0$ gives the geometric factor. Every integer profile has a unique translate with minimum zero, which proves the quotient-lattice formula. □

The quotient formula explains what the volume mark retains. The quadratic form E_G measures distances transverse to the constant direction. The function ν additionally remembers the distinguished positive cone supplied by vertex coordinates. An abstract isometry of quadratic lattices need not preserve that function. Removing common shifts does not discard the order information; it records it on a canonical representative. This distinction is central when two unmarked energy distributions agree but their volume-marked distributions differ.

Corollary SNG-A03 — primitive tuple factorization. For $\Re s > 1$,

$$\frac{Z_G(s,q)}{\zeta(ks)} = \sum_{\substack{n_v\geq 1 \\ \gcd(n_v : v\in V)=1}} \frac{q^{\mathcal{E}_G(n)}}{(\prod_v n_v)^s} = \prod_p B_G(p^{-s},q).$$

Proof. Write every integer tuple uniquely as $n_v = d\, m_v$, where the coordinates of m have common gcd one. Its energy is unchanged, and its denominator gains d^{ks}. Alternatively, common gcd one says that at every prime the minimum coordinate valuation is zero, exactly the defining local condition for B_G. Absolute convergence justifies both factorizations. □

Here “primitive” means a common gcd of one. It does not mean pairwise coprime coordinates. A tuple can share a factor between two vertices while having no factor common to every vertex. Confusing these conditions would change the local population and invalidate the shift factor.

5.3 Synchronization and independence as exact endpoints

At $q = 1$, all energy penalties disappear. At $q = 0$, only zero-energy profiles survive. For a connected graph these are precisely the constant profiles. Thus the same family interpolates between independent coordinates and complete synchronization, in an exact combinatorial sense.

Theorem SNG-A04 — endpoint identities. If G is connected, then

q	$L_G(x,q)$	$B_G(x,q)$	$Z_G(s,q)$
1	$(1-x)^{-k}$	$(1-x^k)(1-x)^{-k}$	$\zeta(s)^k$
0	$(1-x^k)^{-1}$	1	$\zeta(ks)$.

The Euler products converge respectively for $\Re s > 1$ and $\Re s > 1/k$. If the graph has connected component sizes $k_1, \dots, k_c$, the synchronized endpoint is instead $\prod_j \zeta(k_j s)$.

Proof. Independence gives a geometric series for each vertex. At zero energy every edge has equal endpoint heights, and therefore heights are constant on each connected component. The arithmetic tuple is correspondingly constant on each component. The indicated geometric and zeta factors follow. □

These endpoint identities require no limiting argument. A limit involving both q and s is a different problem because the singularity at $s = 1$ can amplify a small nonzero energy weight. Chapter 8 proves a uniform estimate that makes this distinction quantitative.

5.4 Minimum cuts are the first arithmetic correction

Assume now that the graph is connected, and let $\lambda(G)$ be its edge connectivity. A minimum cut is a nonempty proper vertex set whose edge boundary has size λ. We count both sides: S and $V \setminus S$ are separate height profiles and generally have different volumes.

Theorem SNG-A05 — first positive energy. Among nonconstant integer profiles, the least energy is λ. Among minimum-zero profiles, the profiles of energy λ are exactly $\mathbf{1}_S$ for minimum cuts S. Consequently

$$[q^\lambda] B_G(x,q) = \sum_{\substack{\emptyset \neq S \subsetneq V \\ |\partial S| = \lambda}} x^{|S|}, \qquad [q^e] B_G = 0 \quad (0 < e < \lambda).$$

Proof. For an integer difference d, $d^2 \geq |d|$. If a has minimum zero and span h, then the discrete coarea identity gives

$$\sum_{\{u,v\}} |a_u - a_v| = \sum_{t=0}^{h-1} |\partial\{v : a_v > t\}|.$$

Every level set in that sum is nonempty and proper, so $E_G(a) \geq \lambda h$. A nonzero minimum-zero profile has $h \geq 1$. A binary minimum-cut profile attains energy λ. If energy equals λ, the bound forces $h = 1$, and the profile is the indicator of a minimum cut. □

Write $P(z) = \sum_p p^{-z}$ for the prime zeta function on its initial half-plane. The first positive energy coefficient now has an exact analytic description.

Theorem SNG-A06 — minimum-cut prime-zeta coefficient. Initially for $\Re s > 1$,

$$[q^\lambda] \log \frac{Z_G(s,q)}{\zeta(ks)} = \sum_{\substack{\emptyset \neq S \subsetneq V \\ |\partial S| = \lambda}} P(|S| s).$$

Proof. The constant coefficient in q of B_G is one, and there are no positive coefficients below degree λ. Hence its first positive coefficient is unchanged by the formal logarithm. Apply Theorem SNG-A05 to $\log B_G(p^{-s}, q)$, and sum over primes. Absolute convergence on $\Re s > 1$, uniformly for q in a sufficiently small complex disk, justifies coefficient extraction. This uniformity also follows from the local majorant and the fact that each normalized factor equals one at $q = 0$. □

A cut with a large side contributes at a different scaled argument than a cut with a small side. The coefficient therefore contains the sizes of oriented minimum-cut sides, not merely their number. It also identifies the scope of this first correction: it sees minimum cuts only. Higher energy coefficients mix larger cuts with profiles having several height levels. Extracting those sectors requires retained marks or additional identities; the first coefficient alone does not reconstruct the graph.

5.5 Two small models that expose the retained information

For the graph consisting of a single edge, a minimum-zero profile is either $(0, 0)$, $(h, 0)$, or $(0, h)$, with $h \geq 1$. Thus there is a closed local formula

$$B_{K_2}(x, q) = 1 + 2\sum_{h\geq 1} q^{h^2} x^h, \qquad L_{K_2}(x, q) = \frac{1 + 2\sum_{h\geq 1} q^{h^2} x^h}{1 - x^2}. \tag{5.1}$$

The exponent h^2 records a genuinely unequal-height sector. Replacing it by h would give a different model even though the first positive-energy coefficient would agree. The first minimum-cut correction is $2P(s)$: either coordinate can be the raised endpoint. At volume two the three original profiles are $(2, 0), (1, 1), (0, 2)$, with energy histogram $1 + 2q^4$. Normalization removes the common-shift profile $(1, 1)$, leaving $2q^4$. This small calculation separates labeled population counting, common-factor removal, and energy weighting without relying on a large graph example.

For the path on three vertices, choose its middle vertex as a root and write the edge differences as $u = a_1 - a_0$ and $v = a_2 - a_0$. There is one integer profile modulo common translation for every pair $(u, v) \in \mathbb{Z}^2$. Its minimum-zero volume is

$$\nu = u + v - 3\min(0, u, v), \qquad E = u^2 + v^2.$$

Consequently

$$B_{P_3}(x, q) = \sum_{u,v\in\mathbb{Z}} q^{u^2+v^2} x^{u+v-3\min(0,u,v)}. \tag{5.2}$$

The energy has independent coordinates, but the volume does not. The minimum couples them through which vertex is lowest. This is a direct elementary instance of metric information being simpler than order information. It also illustrates why a factorization of an unmarked theta series does not automatically factor its volume-marked refinement.

Theorem SNG-A19 — all trees share the unmarked energy theta. Let T be any tree with k vertices, and let

$$\vartheta(q) = \sum_{h \in \mathbb{Z}} q^{h^2}.$$

For $0 \le q < 1$,

$$B_T(1, q) = \vartheta(q)^{k-1}. \tag{5.3}$$

Proof. Choose a root and orient every edge away from it. Sending a vertex profile modulo common integer translation to its $k-1$ oriented edge differences is a bijection with $\mathbb{Z}^{k-1}$. Injectivity follows by propagating equal differences along the unique root-to-vertex paths. Surjectivity follows by starting the root at zero and integrating the prescribed integer differences along those paths. Energy becomes the sum of squares of the independent edge differences. At $x = 1$, the minimum-zero representative introduces no extra weight, so the sum factors into $k-1$ copies of ϑ. Absolute convergence follows from the convergent one-dimensional Gaussian series. □

This is a useful family of exact collisions, not merely a single searched pair. Every tree of a fixed order has the same unmarked energy distribution. The marked series can already distinguish a four-vertex path from a four-vertex star at volume one: the first has degree polynomial $2q+2q^2$, the second $3q+q^3$. Their unmarked quadratic lattices agree, yet the coordinate cone places the lattice vectors into different volume layers. More demanding matched examples must remove this elementary degree explanation, but the family makes the logical source of the restored information unmistakable.

Connectedness matters to an unmarked specialization. If the graph has several components, removing one global common shift leaves independent relative component shifts of zero energy. The sum $B_G(1, q)$ then diverges, even when $q < 1$. The local series at $|x| < 1$ remains perfectly well defined. On disconnected graphs it is preferable to normalize one common shift per component before setting the volume variable to one. Ordinary component factorization gives

$$L_G(x, q) = \prod_{j=1}^{c} L_{G_j}(x, q),$$

while the single-shift normalized series is

$$B_G(x, q) = \frac{1 - x^k}{\prod_j (1 - x^{k_j})} \prod_j B_{G_j}(x, q).$$

The remaining denominator is the exact record of those unconstrained relative shifts. This is an example where checking the meaning of a normalization prevents a misleading finite theta interpretation.

Chapter 6

Euler analysis: formal coordinates and controlled analytic factors

An Euler product has two distinct kinds of structure in this project. Its coefficients are exact finite polynomials, so logarithms and Euler transforms can be manipulated formally. Its values are infinite products, so analytic conclusions additionally require convergence and choices of branches. The distinction is useful rather than restrictive. Formal calculations reveal which factor ought to be removed; a remainder estimate proves where that removal has analytic meaning. The results below keep those two steps explicit.

Fix a graph with k vertices, write

$$L(x,q) = 1 + \sum_{n\geq 1} \ell_n(q) x^n, \qquad \ell_n(q) = \sum_{|a|=n} q^{E(a)},$$

and initially regard q as an indeterminate. Every ℓ_n belongs to $\mathbb{Z}_{\geq 0}[q]$. Formal logarithms belong to $\mathbb{Q}[q][[x]]$, because division by positive integers enters the logarithm expansion. No numerical approximation to q, the coefficients, or the graph enters this part of the theory.

6.1 Logarithmic coordinates from finite data

Put $\log L = \sum_{n\geq 1} c_n(q) x^n$. The identity $L' = L(\log L)'$ gives a triangular recurrence that computes each logarithmic coefficient using only the local coefficients through the same volume:

$$c_n = \ell_n - \frac{1}{n} \sum_{j=1}^{n-1} j\, c_j \ell_{n-j}. \tag{6.1}$$

In particular,

$$c_1 = \ell_1, \quad c_2 = \ell_2 - \ell_1^2/2, \quad c_3 = \ell_3 - \ell_1\ell_2 + \ell_1^3/3.$$

The recurrence is a coefficient identity, not a fitted model. If two graphs agree through volume $r-1$, they also agree in logarithmic coordinates through that volume, and their difference at volume r is unchanged. This triangular property is the reason first-discrepancy singularities can be read directly from the original local coefficients.

Theorem SNG-A07 — rational polynomial Euler transform. There is a unique sequence $b_m(q) \in \mathbb{Q}[q]$ such that, formally in x,

$$L(x,q) = \prod_{m\geq 1} (1-x^m)^{-b_m(q)}. \tag{6.2}$$

It is given by

$$b_m(q) = \sum_{d|m} \frac{\mu(d)}{d}\, c_{m/d}(q). \tag{6.3}$$

Only factors with $m \leq N$ affect coefficients through volume N.

Proof. Taking formal logarithms of the proposed product gives

$$c_n = \frac{1}{n}\sum_{m|n} m b_m.$$

Ordinary Möbius inversion gives (6.3). Conversely those coefficients satisfy the displayed logarithmic identity, and formal exponential recovers L. Uniqueness follows from the same triangular relation. Rational polynomial exponents are meaningful through the formal exponential definition of $(1-x^m)^{-b_m}$. □

The first exponents are

$$b_1 = \ell_1, \qquad b_2 = \ell_2 - \frac{\ell_1^2+\ell_1}{2},$$

$$b_3 = \ell_3 - \ell_1\ell_2 + \frac{\ell_1^3-\ell_1}{3}.$$

These formulas are economical coordinates for local information. They do not make every exponent nonnegative or integral as a polynomial in q. Such claims would be false for general integer-coefficient series. The existence theorem is over $\mathbb{Q}[q]$, where the necessary divisions are legitimate.

Common-shift removal has an especially simple Euler-coordinate description. If $B = (1-x^k)L$, then

$$b_m(L) = b_m(B) + \mathbf{1}_{m=k}.$$

At the connected synchronized endpoint the only nonzero exponent is $b_k = 1$. At the independent endpoint the only nonzero exponent is $b_1 = k$. These are useful exact consistency checks, but they do not determine intermediate exponents from the endpoints.

6.2 A different transform keeps integral bidegrees

There is a second, genuinely different product decomposition. Instead of putting a polynomial exponent on $1 - x^m$, retain energy inside the individual factors:

$$L(x,q) = \prod_{m\geq 1}\prod_{e\geq 0}(1-x^m q^e)^{-N_{m,e}}. \tag{6.4}$$

At each fixed m, only finitely many $N_{m,e}$ are nonzero. The exponents can be negative. This is a formal decomposition and does not identify an actual population of positive "atoms" without a separate positivity theorem.

Theorem SNG-A08 — integral bidegree transform. For every series $L \in 1 + x\mathbb{Z}[q][[x]]$, there are unique integers $N_{m,e}$ satisfying (6.4). Their generating series is the plethystic logarithm

$$\sum_{m,e} N_{m,e} x^m q^e = \sum_{d\geq 1} \frac{\mu(d)}{d} \log L(x^d, q^d). \tag{6.5}$$

Proof. Suppose all factors of volume below m have been removed. The remaining series has the form $1 + x^m P_m(q) + O(x^{m+1})$, with integral polynomial P_m. For every coefficient $N_{m,e} = [q^e]P_m$, multiply by $(1 - x^m q^e)^{N_{m,e}}$. This removes volume m and preserves integrality. Induction gives existence and uniqueness. Taking logarithms of (6.4), and applying Möbius inversion to the simultaneous substitutions $(x, q) \mapsto (x^d, q^d)$, gives (6.5). Thus the apparently rational expression in (6.5) has integral coefficients. □

The substitution $q \mapsto q^d$ is essential. Omitting it computes the transform in Theorem SNG-A07 rather than the bidegree transform. The two descriptions organize the same formal series differently. For example, a factor $(1-xq)^{-1}$ has one bidegree atom, yet its ordinary exponents $b_m(q)$ are generally nontrivial polynomials. Coefficient-level equality permits conversion; the interpretations should remain distinct.

The corresponding weighted arithmetic factor is

$$\mathcal{Z}_{m,e}(s,q) = \prod_p (1 - q^e p^{-ms})^{-1},$$

with logarithm, on its absolutely convergent domain,

$$\log \mathcal{Z}_{m,e}(s,q) = \sum_{j\geq 1} \frac{q^{ej}}{j} P(mjs).$$

It is not equal to $\zeta(ms)^{q^e}$. Replacing one expression by the other discards all the terms involving q^{ej} for $j > 1$. The ordinary rational-exponent transform and the integral bidegree transform therefore lead to different useful analytic factorizations.

6.3 Leading divergence and what it forgets

The first local coefficient has a simple graph formula:

$$\alpha_G(q) = \ell_1(q) = \sum_{v \in V} q^{d_v}.$$

Only one vertex has height one in a volume-one profile. Its energy is the number of incident edges. Thus the dominant singularity records the degree multiset and no finer graph information.

Theorem SNG-A09 — leading real singularity. For fixed $0 \le q \le 1$, define

$$H_G(s,q) = \prod_p (1-p^{-s})^{\alpha_G(q)} L_G(p^{-s},q). \tag{6.6}$$

The product is holomorphic for $\Re s > 1/2$, using the analytic logarithm of $1-p^{-s}$ defined by its convergent power series on $\Re s > 0$. It is positive at the real point $s = 1$ and holomorphic and nonzero in a complex neighborhood of that point. Initially for $\Re s > 1$,

$$Z_G(s,q) = \zeta(s)^{\alpha_G(q)} H_G(s,q).$$

Along the positive real axis,

$$Z_G(1+\varepsilon,q) \sim H_G(1,q)\varepsilon^{-\alpha_G(q)} \qquad (\varepsilon \downarrow 0). \tag{6.7}$$

Proof. Locally $L = 1 + \alpha x + O(x^2)$, uniformly for $q \in [0,1]$. Multiplication by $(1-x)^\alpha$ cancels the linear coefficient, leaving $1 + O(x^2)$. The prime sum of the remainder converges normally on compact subsets of $\Re s > 1/2$. Each individual factor is holomorphic there; L is analytic on $|x| < 1$, and the chosen power is holomorphic on $\Re s > 0$. For real positive s, all factors are positive. The infinite tail is nonzero by absolute convergence of its logarithm; finitely many remaining factors are nonzero in a neighborhood of $s = 1$. The factorization follows on the original domain from the usual zeta Euler product. Finally $\zeta(1+\varepsilon) \sim 1/\varepsilon$. □

The word "holomorphic" in the theorem does not assert that H_G is zero-free throughout its half-plane. A finite local factor can have complex zeros. Nonvanishing is asserted near the positive real point where it is proved. Similarly, fractional powers of zeta are unambiguous on the original half-plane but need branch choices if continued around zeros or poles elsewhere.

6.4 Finite-order zeta factorization

Theorem SNG-A10 — controlled finite Euler extraction. Fix an integer $M \ge 1$ and a numerical $q \in [0,1]$. Put

$$H_{G,M}(s,q) = \prod_p L_G(p^{-s},q) \prod_{m=1}^{M} (1-p^{-ms})^{b_m(q)}. \tag{6.8}$$

This product is holomorphic on $\Re s > 1/(M+1)$. It is positive for real $s > 1/(M+1)$, and nonzero in a neighborhood of every point on that positive real interval. On the initial half-plane,

$$Z_G(s,q) = \left(\prod_{m=1}^{M} \zeta(ms)^{b_m(q)} \right) H_{G,M}(s,q). \tag{6.9}$$

Proof. Theorem SNG-A07 implies

$$L_G(x,q) \prod_{m=1}^{M} (1-x^m)^{b_m(q)} = 1 + O(x^{M+1})$$

near zero. For primes sufficiently large, this estimate is uniform on any compact subset of the indicated half-plane, and its prime sum converges. The finitely many small-prime factors are holomorphic on $\Re s > 0$; the powers use the logarithms specified above. Positivity on the real axis follows directly from (6.8), and absolute logarithmic convergence proves local nonvanishing there. Equation (6.9) follows by finite multiplication of zeta Euler products on $\Re s > 1$. □

This theorem removes any need to assume convergence of the infinite product over all m in an analytic setting. Formal identity (6.2) does not by itself license an unrestricted exchange of its two infinite products, over primes and over volumes. Small-prime local zeros can limit the radius of the logarithmic power series, and the exponents can grow. Keeping M finite is both sufficient for a prescribed visibility calculation and analytically safer.

Equation (6.9) supplies local continuations after branches have been chosen in regions avoiding relevant zeta zeros and poles. It is not a blanket meromorphic-continuation theorem. Nonintegral exponents generally produce branched functions. Even on the positive real axis below one, separate zeta factors can be negative, so assigning real fractional powers independently does not reproduce an automatically positive partition function. The positive quantity directly defined there is the remainder product (6.8); its relation to a continued Z_G must respect the chosen branches.

6.5 Why ratios are cleaner than individual continuations

If two graphs have identical low-volume coefficients, their low-order zeta exponents cancel before any continuation is attempted. The quotient Euler product can consequently converge farther left as a positive real product. Its first remaining zeta factor has argument approaching one from above. No lower-argument zeta powers are needed to state the resulting real asymptotic. This is the analytic advantage of comparing matched structures rather than interpreting every formal singularity of an individual object as an independently observable feature.

The hierarchy is exact but limited. It proves where a known local distinction appears arithmetically. It does not prove that a distinction always exists between nonisomorphic graphs, nor that all formal exponent locations are actual singularities after every cancellation. If the local series agree, every arithmetic lift in this chapter agrees. If the first discrepancy vanishes at a special numerical q, the visibility order at that parameter must be recomputed. These qualifications identify the genuine inverse problem rather than weakening a theorem that never claimed more.

6.6 A complete low-order transform example

The single-edge model makes the difference between polynomial exponents and positive constituent populations concrete. Its first coefficients are

$$\ell_1 = 2q, \qquad \ell_2 = 1 + 2q^4, \qquad \ell_3 = 2q + 2q^9.$$

Substitution into the rational Euler transform gives

$$\begin{aligned} b_1(q) &= 2q, \\ b_2(q) &= 1 + 2q^4 - 2q^2 - q, \\ b_3(q) &= 2q^9 - 4q^5 + (8q^3 - 2q)/3. \end{aligned} \tag{6.10}$$

At $q = 1/2$, these are $1, 1/8, -31/256$. Thus a valid finite extraction reads

$$Z_{K_2}(s, 1/2) = \zeta(s)\zeta(2s)^{1/8}\zeta(3s)^{-31/256} H_{K_2,3}(s, 1/2) \qquad (\Re s > 1). \tag{6.11}$$

The negative third exponent is not a negative number of arithmetic objects. It is a subtraction introduced by logarithmic and Möbius coordinates. Likewise, the exponent one eighth does not describe one eighth of an independent integer coordinate. Those are exact algebraic coordinates on a positive local series. They are useful precisely because they can cancel lower-order contributions, not because each factor has a separate population interpretation.

At the endpoints the same formulas reduce correctly: at $q = 0$, the second exponent is one and the other displayed exponents are zero; at $q = 1$, the first exponent is two and the next two are zero. Agreement at both endpoints therefore places only weak restrictions on the intermediate polynomial behavior. The coefficient recurrence supplies the actual intermediate values. A numerical plot interpolating only the endpoints would have no authority to invent them.

The bidegree decomposition has a different bookkeeping purpose. Its first volume layer contains the atom count $N_{1,1} = 2$. Removing $(1 - xq)^{-2}$ from the single-edge series leaves, at volume two, the polynomial $1 - 3q^2 + 2q^4$. Hence the second bidegree layer already contains a negative exponent, $N_{2,2} = -3$. Even integral bidegree exponents are therefore not automatically positive atoms. The notation "plethystic" supplies a formal inversion rule; it does not establish a free-generation theorem for an underlying combinatorial class.

6.7 Uniformity and the boundary of the weight parameter

For each fixed extraction order M, the remainder product in Theorem SNG-A10 is jointly continuous in real $q \in [0, 1]$ and real $s > 1/(M + 1)$. On compact subsets of that parameter region its large-prime logarithmic tail converges uniformly. To see this, the finitely many polynomials $b_m(q)$ are bounded on the compact weight interval, and the coefficient majorant bounds the Taylor remainders uniformly for x in a small fixed disk. The canceled coefficients through order M are identically zero as polynomials, so the remaining estimate is uniformly $O(x^{M+1})$. A convergent prime-power sum dominates the tail. Finitely many positive real local factors then preserve continuity and local nonvanishing.

This continuity statement is often enough to pass a weight limit through a regularized product. It does not mean that the unregularized Euler product converges uniformly as $s \downarrow 1$. The leading zeta factor carries the obstruction to that stronger statement. Chapter 8 isolates it in a quantitative estimate rather than suppressing it through a formal exchange of limits.

The chosen weight range also has a mathematical boundary. Suppose the graph has an edge, choose a vertex of positive degree d_v, and fix $0 < x < 1$. The terms coming from profiles ne_v are $q^{d_v n^2} x^n$. For every real $q > 1$, these terms fail even to tend to zero as $n \to \infty$. Thus the unbounded local height series diverges. The restriction $q \leq 1$ cannot simply be removed by calling the series a partition function. A finite height cutoff would define a different object and would require its own limiting argument.

For $|q| < 1$, the local series is analytic in the complex weight parameter. At a boundary value such as $q = 1$, derivatives from the allowed real side can still exist to every finite order, because each derivative introduces only polynomial energy factors and the geometric volume decay dominates them. Such smooth one-sided behavior should not be mistaken for an unrestricted analytic extension beyond the unit disk. These domain distinctions matter especially when symbolic exponents are evaluated numerically: exact polynomial coefficients can be evaluated anywhere, while the infinite series from which they were extracted has the domain proved for it.

Chapter 7

Visibility singularities: the first distinction survives arithmetic assembly

A comparison is strongest when simpler explanations have been removed explicitly. Two graphs can have the same degree multiset and therefore the same leading divergence. They can agree on additional finite invariants and still differ in a later local height coefficient. The arithmetic question is then not whether an Euler product can be attached to each graph: that is already automatic. The useful question is where the first unmatched local observation becomes analytically visible.

For $0 \le q \le 1$, put $\ell_{G,n}(q) = [x^n]L_G(x,q)$. In this chapter, q is fixed before an analytic comparison is made. Polynomial equality in q is stronger than numerical equality at a selected parameter. A polynomial discrepancy can vanish at an endpoint or an interior root, in which case the first visible volume at that numerical parameter can change.

7.1 A precise first-discrepancy theorem

Theorem SNG-A11 — first-discrepancy ratio law. Let G, H have the same finite number k of vertices. Fix $q \in [0,1]$. Suppose

$$\ell_{G,n}(q) = \ell_{H,n}(q) \quad (0 \le n < r), \qquad \delta = \ell_{G,r}(q) - \ell_{H,r}(q) \neq 0. \tag{7.1}$$

For real $s > 1/r$, the ratio product

$$R_{G,H}(s,q) = \prod_p \frac{L_G(p^{-s},q)}{L_H(p^{-s},q)} \tag{7.2}$$

converges absolutely to a positive number. It agrees with Z_G/Z_H on the original domain $s > 1$. Define

$$K_{G,H}(s,q) = \prod_p (1-p^{-rs})^{\delta} \frac{L_G(p^{-s},q)}{L_H(p^{-s},q)}. \tag{7.3}$$

The latter product converges absolutely and is positive for real $s > 1/(r+1)$. It is holomorphic and nonzero in a complex neighborhood of $s_0 = 1/r$. For real $s > s_0$,

$$R_{G,H}(s,q) = \zeta(rs)^{\delta} K_{G,H}(s,q), \qquad R_{G,H}(s,q) \sim C(q)(rs-1)^{-\delta}, \tag{7.4}$$

where $C(q) = K_{G,H}(1/r, q) > 0$.

Proof. Condition (7.1) gives

$$\frac{L_G(x,q)}{L_H(x,q)} = 1 + \delta x^r + O(x^{r+1})$$

near zero, because the denominator has constant term one. The prime sum of the deviation converges for $rs > 1$. No local denominator vanishes on the positive real axis: both local series have nonnegative coefficients and constant term one. Thus the product is positive, and its logarithm converges absolutely. Multiplication by $(1-x^r)^{\delta}$ cancels the first discrepancy, leaving $1 + O(x^{r+1})$. This proves convergence of (7.3) on the stated real interval.

To obtain a complex neighborhood, separate finitely many small primes from the normally convergent tail. Every small-prime denominator is positive at s_0, hence nonzero in a sufficiently small disk about that point. The tail is analytic and nonzero there by its convergent logarithm. The factors $(1 - p^{-rs})^{\delta}$ use the analytic logarithm defined on $\Re s > 0$. This gives the asserted local holomorphic nonzero function. On $s > 1/r$, the ordinary zeta Euler product yields the factorization. Finally $\zeta(1+t) \sim 1/t$ with $t = rs - 1$. □

A positive discrepancy makes the ratio diverge at its visibility point; a negative discrepancy makes it tend to zero. Interchanging the graphs reverses that interpretation. If δ is nonintegral, the complex singularity is a fractional-power branch singularity. It should not be called a pole. The theorem supplies a local branch and a positive-real asymptotic, not a globally single-valued meromorphic function.

There is also no requirement that the original partition functions converge separately at s_0. Their common low-order divergences have been canceled in the ratio product before the limit is taken. This cancellation is the main analytic mechanism. It makes a comparison meaningful farther left without pretending that either original positive Dirichlet series has acquired an absolutely convergent definition there.

7.2 A volume-two calculation and its actual meaning

At volume two there are only two kinds of profile: $2e_v$, and $e_u + e_v$ with $u < v$. Their energies give the identity

$$\ell_{G,2}(q) = \sum_v q^{4d_v} + \sum_{u<v} q^{d_u + d_v - 2A_{uv}}. \tag{7.5}$$

For graphs with the same degree multiset, define the edge degree-sum polynomial

$$M_G(q) = \sum_{\{u,v\} \in E} q^{d_u + d_v}.$$

Then

$$\ell_{G,2} - \ell_{H,2} = (q^{-2} - 1)(M_G - M_H). \tag{7.6}$$

Although written with q^{-2}, the right side is a polynomial: every edge has degree sum at least two. Formula (7.6) follows by starting with the sum over all unordered vertex pairs and correcting the exponent by minus two precisely on edges.

Calculation SNG-A12 — first reference pair. On vertices $0, \dots, 6$, take

$$E(G) = \{01, 02, 03, 04, 12, 35, 36\},$$
$$E(H) = \{01, 02, 03, 04, 15, 16, 56\}.$$

Both have degree multiset $\{4, 3, 2, 2, 1, 1, 1\}$. Exact profile enumeration gives equality at volume one and

$$\delta_2(q) = \ell_{G,2} - \ell_{H,2} = 2q^2 - 4q^3 + 4q^5 - 2q^6 = 2q^2(1-q)^3(1+q). \tag{7.7}$$

It is positive on $0 < q < 1$, so Theorem SNG-A11 applies with $r = 2$. At $q = 1/2$,

$$R_{G,H}(s, 1/2) \sim C_2(2s-1)^{-3/32}.$$

The arithmetic conclusion is exact. Its local source is also elementary: $M_G - M_H = 2q^4(1-q)^2$. Thus this example illustrates arithmetic visibility of degree-assortativity information. It does not demonstrate that volume two captures a fundamentally higher-order graph relation. Matching more demanding invariants is a separate task, and the interpretation of a collision should state precisely which invariants were matched.

7.3 A volume-three witness beyond the binary census

At volume three, the profile types are $3e_v$, $2e_u + e_v$ with $u \neq v$, and indicators of three-element subsets. Hence

$$\ell_{G,3}(q) = \sum_v q^{9d_v} + \sum_{u \neq v} q^{4d_u + d_v - 4A_{uv}} + \sum_{|S|=3} q^{|\partial S|}. \tag{7.8}$$

This identity keeps the explanation transparent. The last term is binary. The middle term tests unequal heights at two vertices. It can distinguish a joint degree-mixing pattern that the degree-sum polynomial merges. Every energy remains a sum of pairwise edge terms; a later observation order does not introduce a new three-body interaction.

Calculation SNG-A13 — binary-census reference pair. Take

$$E(G) = \{03, 04, 06, 12, 13, 23, 24, 35\},$$
$$E(H) = \{01, 04, 06, 15, 16, 26, 36, 45\}.$$

Both have degree multiset $\{4, 3, 3, 2, 2, 1, 1\}$. The exact finite binary census agrees for every subset size and cut size. The local coefficients agree through volume two, but their volume-three difference is

$$\begin{aligned}\delta_3(q) &= q^3 - q^4 - 2q^6 + 2q^8 + q^9 + 3q^{10} - 3q^{11} - q^{12} - 2q^{13} \\ &\quad + 2q^{15} + q^{17} - q^{18} \\ &= q^3(1-q)^5(1+q)^2(1 + 2q + 5q^2 + 6q^3 + 8q^4 \\ &\qquad + 6q^5 + 5q^6 + 2q^7 + q^8).\end{aligned} \tag{7.9}$$

The factorization proves positivity throughout $0 < q < 1$; it is not a numerical sign test. At $q = 1/2$, the discrepancy is $11025/262144$. Therefore

$$R_{G,H}(s, 1/2) \sim C_3(3s-1)^{-11025/262144} \qquad (s \downarrow 1/3). \tag{7.10}$$

The finite calculation can be reproduced without a symbolic algebra system. Generate every weak composition of volumes zero through three into seven labeled coordinates, evaluate the integer energy from the supplied edges, and accumulate multiplicities by energy. There are only $1, 7, 28, 84$ profiles at those volumes. Subtract the two energy histograms. Multiplication of the explicit factors in (7.9) independently checks the polynomial identity. The full binary claim requires the separate exhaustive census of 2^7 subsets; it is not inferred from the first four height coefficients.

This pair does not match the full joint degree table on edges. Its degree-sum polynomial agrees, but the more refined table does not. Consequently the witness proves that the unequal-height sector carries information beyond the aggregated binary census. It does not prove survival after every degree-assortativity statistic has been matched. A stronger search should match that full table as well and seek discrepancies at volume four or later. The first-discrepancy theorem will still apply; the difficult new work is finding and interpreting the local witness.

7.4 Effective control of the finite-prime approximation

A singularity law is more useful when its regular factor can be bounded rather than merely named. The following bounds require only local coefficient estimates and ordinary convergent sums. They do not require assumptions about prime gaps or an unproved asymptotic formula for primes.

For an integer cutoff $P \geq 2$, define $Z_{\leq P}$ by retaining primes at most P. For real $s > 1$, put $\rho = P^{-s} < 1$. Since

$$0 \leq \log L(t,q) \leq L(t,q) - 1 \leq \frac{kt}{(1-\rho)^{k+1}} \qquad (0 \leq t \leq \rho),$$

we obtain the explicit interval

$$0 \leq \log Z_G(s,q) - \log Z_{\leq P}(s,q) \leq \frac{k}{(1-\rho)^{k+1}} \frac{P^{1-s}}{s-1}. \tag{7.11}$$

Here the tail over primes was bounded by the tail over all integers, followed by the integral test. Exponentiating gives a multiplicative enclosure. The bound is deliberately conservative near $s = 1$, where raw Euler products converge slowly.

For the regularized ratio factor a faster bound is available. Set $A_n = \binom{n+k-1}{k-1}$, choose $0 < \rho < 1$, and define the explicit nonnegative constants

$$D = \frac{(1-\rho)^{-k}-1}{\rho}, \qquad T = \frac{2}{\rho^{r+1}}\left((1-\rho)^{-k} - \sum_{n=0}^{r} A_n \rho^n\right),$$

$$C = T + |\delta| D + \frac{|\delta|\rho^{r-1}}{2(1-\rho^r)}. \tag{7.12}$$

Theorem SNG-A14 — computable regularized tail. Under the assumptions of Theorem SNG-A11, if real $s > 1/(r+1)$ and $P^{-s} \le \rho$, then

$$\left|\log K_{G,H}(s,q) - \log K_{G,H,\le P}(s,q)\right| \le C \frac{P^{1-(r+1)s}}{(r+1)s-1}. \tag{7.13}$$

Proof. Write $A = L_G(t,q)$, $B = L_H(t,q)$, and $D_t = A - B$. Matching the coefficients below r gives

$$|D_t - \delta t^r| \le T t^{r+1}, \qquad \max(A-1, B-1) \le Dt.$$

Because $A, B \ge 1$,

$$\log A - \log B = \int_0^1 \frac{D_t}{B + u D_t}\, du.$$

The denominator is at least one. Subtracting δt^r therefore gives an error at most $(T + |\delta| D) t^{r+1}$. Finally

$$|\log(1-t^r) + t^r| \le \frac{t^{2r}}{2(1-\rho^r)} \le \frac{\rho^{r-1} t^{r+1}}{2(1-\rho^r)}.$$

Adding these estimates bounds the absolute logarithm of one regularized factor by $C t^{r+1}$. Sum over primes beyond P, enlarge to all integers, and apply the integral test. □

Local factors themselves can also be enclosed by finitely many height coefficients. If $L_N(t,q) = \sum_{n=0}^{N} \ell_n(q) t^n$, choose any $t < R < 1$. Then

$$0 \le L(t,q) - L_N(t,q) \le (t/R)^{N+1} (1-R)^{-k}. \tag{7.14}$$

Indeed each omitted term gains at most $(t/R)^{N+1}$, and the complete series at R is bounded by $(1-R)^{-k}$. Since $L_N \ge 1$, its logarithmic error is at most the same bound. Finite-prime and finite-height errors can thus be propagated separately and added in logarithmic coordinates. A numerical implementation claiming a rigorous enclosure must also bound its arithmetic rounding; the inequalities alone do not certify an unchecked floating-point product.

These results establish controllable local witnesses, arithmetic visibility, and finite approximation. They leave a genuine open inverse question: which graph distinctions are invisible to the entire univariate volume-marked local series? The examples show strict separations from specified coarser observations. They do not answer that complete question or establish historical priority for the construction.

7.5 The complete binary comparison and the remaining elementary distinction

For the second reference pair, the common binary polynomial at each side size is small enough to display completely. Let $C_j(q) = \sum_{|S|=j} q^{|\partial S|}$. Complementation gives $C_j = C_{7-j}$, and the common values are

Side sizes	Common cut-energy polynomial
0,7	1
1,6	$2q+2q^2+2q^3+q^4$
2,5	$2q^2+8q^3+5q^4+4q^5+2q^6$
3,4	$q^2+4q^3+13q^4+10q^5+5q^6+2q^7$

Evaluating a row at $q = 1$ gives respectively $1, 7, 21, 35$, the appropriate binomial subset counts. This check does not prove equality of the polynomials, but it catches a missing subset in a proposed census. The actual equality is coefficientwise, obtained by evaluating each labeled subset against the original edges.

The edge joint-degree table identifies information the common binary census has merged. Entries not displayed are zero in both graphs.

Unordered endpoint degrees	Edges in G	Edges in H
(1,3)	1	0
(1,4)	1	2
(2,2)	0	1
(2,3)	3	2
(2,4)	1	0
(3,3)	0	1
(3,4)	2	2

Both columns total eight edges. Collapsing endpoint degrees to their sum makes the tables agree: sums four, five, six, seven occur with multiplicities one, four, one, two. This explains agreement at volume two. Unequal heights two and one weight the endpoint degrees differently, through $4d_u + d_v$, so the collapse no longer hides the difference. The height witness can therefore be understood without asserting an unexplained new source of graph information.

The distinction also lies in a specific span sector. Volume-three profiles of the form $3e_v$ have span three and agree because degrees agree. Binary profiles have span one and agree because the full binary census agrees. The entire discrepancy (7.9) is consequently in span two. Retaining a span mark would isolate that sector directly. After setting the span mark to one, the same discrepancy survives, but the proof that it came from that sector uses the separate matched observations.

7.6 A secondary boundary layer at an endpoint

The first-discrepancy exponent itself can become small near an endpoint. This produces a second controlled scaling law, now for graph ratios rather than the original synchronized partition function.

Theorem SNG-A20 — endpoint scaling of visibility. Suppose the polynomial local coefficients of two graphs agree below volume r for every $q \in [0, 1]$. Let $q_* \in [0, 1]$ be

a value where the entire local series agree, and write $\delta(q) = \ell_{G,r}(q) - \ell_{H,r}(q)$. Let $q_j \to q_*$, $\varepsilon_j \downarrow 0$, with

$$\delta(q_j)\log(1/\varepsilon_j) \longrightarrow \tau \in \mathbb{R}.$$

Then

$$R_{G,H}((1+\varepsilon_j)/r, q_j) \longrightarrow e^{\tau}. \tag{7.15}$$

Proof. The normalized local ratio in (7.3) has its first r nonconstant coefficients canceled identically in q. Its $O(x^{r+1})$ estimate can be chosen uniformly on the compact weight interval. Therefore the regularized prime product is jointly continuous near $(1/r, q_*)$, by uniform convergence of the large-prime logarithmic tail and positivity of the finitely many remaining factors. At q_*, every local ratio is one and $\delta(q_*) = 0$, so $K(1/r, q_*) = 1$. Finally

$$\log R = \delta(q_j)\log\zeta(1+\varepsilon_j) + \log K.$$

The second term tends to zero, while $\log\zeta(1+\varepsilon) - \log(1/\varepsilon) \to 0$ and δ stays bounded. This proves the limit. □

For the volume-two pair, $\delta_2(q) \sim 4(1-q)^3$ as $q \uparrow 1$. For the volume-three pair, $\delta_3(q) \sim 144(1-q)^5$, because the final polynomial factor in (7.9) has value 36 at one. Thus $(1-q)^5\log(1/\varepsilon) \to t$ yields the ratio limit e^{144t} for the second pair at $s = (1+\varepsilon)/3$. Near zero the corresponding leading behaviors are $\delta_2 \sim 2q^2$ and $\delta_3 \sim q^3$. These are proved parameter scales for particular exact witnesses. They do not establish a universal exponent for all graph pairs.

Chapter 8

Synchronization limits, uniform scaling, and exact contractions

Setting the energy weight to zero and letting the arithmetic size parameter approach its divergence point are different operations. At zero weight a connected graph forces every coordinate integer to agree, producing $\zeta(ks)$. At any fixed positive weight the system has rare unequal valuations at every prime, and their combined contribution diverges at $s = 1$. There are infinitely many opportunities for a rare local event. A small penalty need not remain negligible when those opportunities accumulate. This chapter turns that intuition into a uniform theorem and then develops optional extensions that retain the same exact arithmetic meaning.

Assume first that G is connected with $k \geq 2$, minimum degree $d \geq 1$, and edge connectivity $\lambda \geq 1$. Write m_d for the number of vertices of degree d, and $\alpha(q) = \sum_v q^{d_v}$. All limits below use real positive parameters. No complex continuation is needed for the double-scaling statements.

8.1 An explicit estimate uniform in the energy weight

We begin with constants that are intentionally simple rather than optimized. Define

$$D_k = 2(2^k - 1), \qquad C_k = 4(2^k - 1 - k/2), \qquad A_k = C_k + D_k^2/2 + k. \tag{8.1}$$

Let

$$J_k = \sum_{n \geq 2} \frac{\log n}{n^k} < \infty.$$

This last constant is computable with an elementary integral remainder. For a cutoff $N \geq 2$, the omitted tail is at most

$$\int_N^\infty \frac{\log t}{t^k}\, dt = N^{1-k} \left(\frac{\log N}{k-1} + \frac{1}{(k-1)^2} \right). \tag{8.2}$$

The summand is decreasing on $[2, \infty)$ for $k \geq 2$, so the integral bound applies directly.

Theorem SNG-A15 — uniform synchronization estimate. For $0 \le q \le 1$ and $\varepsilon > 0$,

$$\begin{aligned} &|\log Z_G(1+\varepsilon, q) - \log \zeta(k) - \alpha(q)\log(1/\varepsilon)| \\ &\qquad \le A_k q^\lambda(\zeta(2)-1) + (kJ_k + k)\varepsilon. \end{aligned} \tag{8.3}$$

Proof. Write $B_G(x,q) = 1 + u(x,q)$. Every nonzero minimum-zero profile has energy at least λ. Therefore, for real $0 \le x \le 1/2$,

$$0 \le u(x,q) \le q^\lambda D_k x,$$

and, since the volume-one coefficient is $\alpha(q)$,

$$0 \le u(x,q) - \alpha(q)x \le q^\lambda C_k x^2.$$

To verify the constants, bound each minimum-zero coefficient by the number of all weak compositions. At $x \le 1/2$, dividing the first bound by x and the second by x^2 leaves geometric-composition sums bounded respectively by D_k and C_k. These bounds discard graph information but retain the crucial factor q^λ.

For $u \ge 0$, integration of $1 - 1/(1+u)$ gives

$$0 \le u - \log(1+u) \le u^2/2.$$

Also $|\log(1-x) + x| \le x^2/[2(1-x)]$, and $\alpha(q) \le kq^\lambda$. Combining the estimates gives

$$|\log B_G(x,q) + \alpha(q)\log(1-x)| \le A_k q^\lambda x^2. \tag{8.4}$$

Here $q^{2\lambda} \le q^\lambda$ was used in the quadratic term. Substitute $x = p^{-s}$, with real $s \ge 1$, and sum over primes. By common-shift factorization,

$$\log H_G(s,q) - \log \zeta(ks) = \sum_p \{\log B_G(p^{-s}, q) + \alpha(q)\log(1-p^{-s})\}.$$

Thus its absolute value is at most $A_k q^\lambda(\zeta(2)-1)$. The derivative bound $|(\log\zeta)'(t)| \le \sum_{n\ge 2}(\log n)n^{-k} = J_k$ for $t \ge k$ gives

$$|\log\zeta(k(1+\varepsilon)) - \log\zeta(k)| \le kJ_k\varepsilon.$$

Finally the integral test yields $1/\varepsilon \le \zeta(1+\varepsilon) \le 1 + 1/\varepsilon$. Hence

$$0 \le \log\zeta(1+\varepsilon) - \log(1/\varepsilon) \le \varepsilon.$$

Multiply by $\alpha(q) \le k$, use the leading factorization, and add the three error bounds. □

This proof provides more than continuity of a regular factor. It shows exactly which uniform estimate permits the two parameters to move together. The graph-dependent

minimum-cut exponent controls the error in replacing the regular factor by its synchronized value. The minimum-degree exponent will control the rare unequal valuations that are amplified by the singular factor. Those exponents need not be equal, and they play different mathematical roles.

8.2 The double-scaling law

Theorem SNG-A16 — synchronization boundary layer. Let $q_j \to 0$, $\varepsilon_j \downarrow 0$. If

$$\alpha(q_j)\log(1/\varepsilon_j) \longrightarrow \tau \quad \text{in } [0,\infty],$$

then

$$Z_G(1+\varepsilon_j, q_j) \longrightarrow \zeta(k)e^{\tau}, \tag{8.5}$$

where $e^{+\infty} = +\infty$. In particular, if

$$q_j^d \log(1/\varepsilon_j) \longrightarrow t < \infty,$$

then the limit is $\zeta(k)e^{m_d t}$. If that product tends to infinity, the partition function tends to infinity.

Proof. The error in (8.3) tends to zero. Exponentiating gives the first statement, including the infinite case. For the second, use

$$\alpha(q) = m_d q^d + R(q), \qquad 0 \le R(q) \le (k-m_d)q^{d+1}.$$

When $q^d \log(1/\varepsilon)$ has a finite limit, the extra contribution $R(q)\log(1/\varepsilon)$ tends to zero. In the infinite case, $\alpha(q) \ge m_d q^d$ is enough. □

The scaling variable is logarithmic in the distance to the arithmetic singularity. For a graph with minimum degree one, maintaining a nontrivial finite limit requires q of order $1/\log(1/\varepsilon)$. With minimum degree two it requires order $1/\sqrt{\log(1/\varepsilon)}$. This is a conclusion of the exact estimate, not a power-law ansatz chosen from numerical plots. The coefficient m_d counts the cheapest single-vertex disagreements.

The theorem also explains the noncommuting iterated limits. At fixed $\varepsilon > 0$, sending $q \to 0$ gives $\zeta(k(1+\varepsilon))$, and then sending $\varepsilon \to 0$ gives $\zeta(k)$. At fixed $q > 0$, sending $\varepsilon \to 0$ gives infinity, so subsequently shrinking q does not recover that finite iterated limit. A family can therefore look synchronized throughout a large finite range and still have a different limiting divergence. The double-scaling variable states how close to one the size parameter must be before that discrepancy matters.

8.3 Optional extension: vertex masses

Positive integer vertex masses allow exact contraction without pretending that a merged vertex represents only one original coordinate. Assign a mass $m_v \ge 1$ to each vertex and let $M = \sum_v m_v$. Define

$$L_{G,m}(x,q)=\sum_{a\geq 0} q^{E_G(a)}x^{\sum_v m_v a_v}.$$

Its arithmetic counterpart weights an integer tuple by $\prod_v n_v^{-m_v s}$. All coefficients remain finite, and a common shift increases weighted volume by M.

Theorem SNG-A17 — weighted endpoints and first divergence. For connected G,

$$L_{G,m}=\frac{B_{G,m}}{1-x^M},\quad Z_{G,m}(s,0)=\zeta(Ms),\quad Z_{G,m}(s,1)=\prod_v \zeta(m_v s).$$

Let $r=\min_v m_v$ and fix $0<q\leq 1$. Put $\gamma(q)=\sum_{m_v=r} q^{d_v}$. Then the first nonconstant local term is $\gamma(q)x^r$, the Euler product converges for real $s>1/r$, and

$$Z_{G,m}(s,q)=\zeta(rs)^{\gamma(q)}H_{G,m}(s,q),$$

with a positive nonzero regular factor near $s=1/r$. Therefore its leading real singular exponent there is $\gamma(q)$.

Proof. The shift and endpoint arguments are unchanged after adding masses. The least positive weighted volume is r, attained exactly by the profiles e_v with $m_v=r$. Their energies are d_v. Every remaining weighted volume is an integer at least $r+1$. Multiplication by $(1-x^r)^{\gamma(q)}$ leaves $1+O(x^{r+1})$, and the same proof as Theorem SNG-A11 gives the regular factor. At least one vertex has minimum mass and $q>0$, so $\gamma(q)>0$; logarithmic divergence at $s=1/r$ also proves the sharp real convergence threshold. □

This extension changes the arithmetic size observable while retaining the same graph energy. It need not be interpreted as adding physical masses. Its exact purpose is to remember how many original coordinates have been identified, or how strongly their integer sizes are weighted. Ignoring those weights after contraction would move the synchronized zeta factor to the wrong argument.

8.4 Optional extension: partial edge synchronization

Allow one energy weight $q_e\in[0,1]$ for every edge. The local weight is $\prod_e q_e^{(a_u-a_v)^2}$. Setting a selected edge weight to zero imposes equality of its endpoint heights. Equality propagates through the connected components of the selected-edge subgraph.

Theorem SNG-A18 — exact contraction under zero edge weights. Let $F\subseteq E$, and set $q_e=0$ for $e\in F$. Let $C_1,\dots,C_t$ be the components of (V,F). Replace each component by one vertex of mass $|C_i|$. An unselected edge between C_i and C_j contributes its original factor $q_e^{(b_i-b_j)^2}$; an internal unselected edge contributes one. Then the resulting weighted, possibly multiple-edge local series equals the original specialized local series exactly. The equality also holds for the Euler products on every common domain of absolute convergence.

Proof. A profile survives the zero weights exactly when it is constant on each C_i. Such profiles correspond bijectively to $b\in\mathbb{Z}^t_{\geq 0}$. Its original volume is $\sum_i |C_i|b_i$. Internal

edges have zero difference, and every remaining edge retains its stated difference and weight. This is equality term by term. Euler assembly follows on an absolutely convergent domain. □

Parallel surviving edges can be combined: their weights multiply, since $q_1^{d^2} q_2^{d^2} = (q_1 q_2)^{d^2}$. Their presence is therefore supported mathematically even if the core graph class remains simple. The contraction description is an optional extended representation, not a request to accept duplicate edges silently in the ordinary constructor.

Contraction order does not matter. Applying zero constraints in stages produces the equivalence relation generated by all selected edges. Vertex masses add associatively, internal edges disappear, and external weights multiply associatively. This is a direct consequence of the termwise bijection, not an appeal to an unspecified categorical equivalence. If the selected edges connect the entire graph, the result is one vertex of mass k, recovering $\zeta(ks)$.

One can retain separate complex size parameters s_v as well. A synchronized component then has parameter $s_{C_i} = \sum_{v \in C_i} s_v$. A sufficient domain before contraction is $\Re s_v > 1$ for every vertex; after contraction the less restrictive sufficient condition is $\Re s_{C_i} > 1$ for every component. The exact local identity remains valid wherever the relevant sums converge. This multivariate refinement records how contraction redistributes size exponents without claiming a complete several-variable continuation theory.

8.5 Positive ordering and probabilistic derivatives

There is a further useful consequence of retaining positive weights. If $G \subseteq H$ on the same vertex set, then $E_H(a) \geq E_G(a)$. Hence, for $q \in [0,1]$, every local coefficient and every real convergent arithmetic value satisfies $L_H \leq L_G$ and $Z_H \leq Z_G$. Adding constraints reduces the positive population weight. The corresponding ratio need not be monotone in s, and no such stronger ordering is asserted.

For real $s > 1$ and $q = e^{-\beta}$, normalize the integer-tuple weights by Z_G. This defines an ordinary probability distribution. Its sufficient statistics are $U(n) = \log \prod_v n_v$ and $V(n) = \mathcal{E}_G(n)$. On compact parameter sets with $s > 1$ and $\beta \geq 0$, all their polynomial moments are finite. Differentiating gives

$$\partial_s \log Z = -\mathbb{E}[U], \qquad \partial_\beta \log Z = -\mathbb{E}[V],$$

$$\nabla^2_{(s,\beta)} \log Z = \begin{pmatrix} \operatorname{Var}(U) & \operatorname{Cov}(U,V) \\ \operatorname{Cov}(U,V) & \operatorname{Var}(V) \end{pmatrix} \succeq 0. \tag{8.6}$$

To justify finiteness, note that local energy is at most a graph-dependent constant times squared local volume, while the local geometric coefficient majorant has every polynomial volume moment at p^{-s}. Summing their prime contributions and logarithmic size factors is uniformly convergent when s stays above one. Alternatively, global energy is bounded by a constant times $(\log \prod_v n_v)^2$, and products of powers of logarithms remain summable against $\prod_v n_v^{-s}$. Dominated differentiation follows. The covariance identity is a general property of a normalized exponential family, used here with an explicitly defined arithmetic population.

These derivative identities supply rigorous sensitivity observations. They do not add a physical interpretation, empirical model, or claim of a thermodynamic phase transition. The proved limiting phenomenon is the uniform arithmetic boundary layer in Theorem SNG-A16. Its higher-order corrections, sharp graph-dependent constants, and possible multivariate visibility geometries remain separate analytic research questions.

8.6 A worked contraction sequence

Take the four-vertex path $0-1-2-3$, initially with mass one at every vertex and a common positive edge weight q. Its leading exponent at $s=1$ is $2q+2q^2$, because its endpoint degrees are one and its internal degrees are two. Now set the weight on edge 01 to zero while retaining weight q on the other edges. The surviving local profiles are exactly (a,a,b,c), and their local series is

$$\sum_{a,b,c\geq 0} x^{2a+b+c} q^{(a-b)^2+(b-c)^2}. \tag{8.7}$$

The contracted path has masses $2,1,1$. Its light vertices have degrees two and one, so Theorem SNG-A17 gives leading exponent q^2+q at $s=1$. The heavy vertex is still part of the graph energy, but it cannot occur in the lowest-volume profile. This is why simply counting contracted vertices would give the wrong answer.

Next synchronize edge 23 as well. The surviving profiles are (a,a,b,b). There are two contracted vertices, each of mass two, joined by the original middle edge. The local factor is exactly $L_{K_2}(x^2,q)$, and its arithmetic product is $Z_{K_2}(2s,q)$. The first divergence moves to $s=1/2$, with exponent $2q$. Finally synchronize the middle edge; all four heights become equal and the product becomes $\zeta(4s)$, whose first divergence is at $s=1/4$.

The changes in threshold have a direct combinatorial explanation. A minimal positive shift of a synchronized block raises every original coordinate in that block. Its arithmetic denominator is therefore the corresponding power of an integer, rather than the first power associated with a single original vertex. The contraction operation preserves that information through its masses. The example also shows why a statement about an exactly zero coupling should not be inferred from small positive couplings without a uniform estimate: rare violations of the equality constraint can restore a divergence at a different location.

For the derivative identities (8.6), values at $\beta=0$ mean derivatives from the allowed side $\beta\geq 0$. At an interior value $\beta>0$, ordinary derivatives are available in a neighborhood. The one-sided convention is necessary because rewarding unbounded quadratic energy with $\beta<0$ makes even a local height series diverge. This is a domain issue in the mathematical definition, not a numerical instability that can be fixed by increasing a cutoff.

Chapter 9

Exact examples: what the new population distinguishes

This chapter records the finite evidence on which the first release rests. The examples are deliberately small enough that their defining populations can be examined without numerical approximation. They serve a precise purpose: to show that integer heights retain relational information that certain familiar summaries discard. They do not establish that the resulting invariant determines every graph, that the Euler product adds information to its local factor, or that the construction has historical priority. The claims below distinguish direct arguments from exact finite computations. The computations can be repeated through the standard-library package accompanying this manuscript.

Throughout the chapter a graph is simple, undirected, and has the labeled vertex set $V = \{0, 1, \dots, 6\}$. Labels specify the input and the assignments being counted. The resulting polynomials are unchanged by relabeling the graph, because a permutation of vertices induces a bijection on all height assignments. An edge written 03 means the unordered pair $\{0, 3\}$. The height energy is

$$E_G(a) = \sum_{\{i,j\} \in E(G)} (a_i - a_j)^2, \qquad |a| = \sum_i a_i.$$

The unnormalized local population and the normalized marked population are

$$P_G(x, q) = \sum_{a \in \mathbb{N}^7} x^{|a|} q^{E_G(a)},$$

$$B_G(x, q, z) = \sum_{\substack{a \in \mathbb{N}^7 \\ \min_i a_i = 0}} x^{|a|} q^{E_G(a)} z^{\max_i a_i}.$$

Volume means total integer height. The variable z records height span after the common minimum has been removed. Consequently $B_G(x, q, 1) = (1 - x^7) P_G(x, q)$. For volumes below seven, the coefficients of the unmarked normalized and unnormalized series agree. Every assignment of such a volume already has a zero coordinate. This observation lets us discuss the first local coefficients and their span decomposition without changing normalization halfway through an argument.

9.1 Two different comparisons

The first pair has equal Laplacian spectra. The second pair has equal complete binary cut-volume populations. These are separate examples, and keeping them separate prevents an unsupported inference about a single pair defeating every existing invariant.

Name in this chapter	Package name	Edge set	Degrees in vertex order
G_{sp}	`spectral_g`	01,02,03,04,12,35,36	(4,2,2,3,1,1,1)
H_{sp}	`spectral_h`	01,02,03,04,15,16,56	(4,3,1,1,1,2,2)
G_{bin}	`binary_g`	03,04,06,12,13,23,24,35	(3,2,3,4,2,1,1)
H_{bin}	`binary_h`	01,04,06,15,16,26,36,45	(3,3,1,1,2,2,4)

All four graphs are connected. Each spectral graph has one triangle and four bridges. Each binary graph has the same two-connected block, a theta graph with three internally disjoint paths of lengths one, two, and three, together with two attached bridges. The attachments differ. This distinction between a common block structure and different vertex attachments will matter when we compare the marked population with an ordinary cut-lattice theta series.

Proposition SNG-E001: the spectral graphs are nonisomorphic. Each graph has a unique vertex of degree four, namely vertex zero. In G_{sp}, that vertex lies in the triangle $0, 1, 2$. In H_{sp}, the only triangle is $1, 5, 6$, which excludes vertex zero. An isomorphism preserves vertex degrees and triangle membership. It would therefore have to send the degree-four vertex to the degree-four vertex while preserving a property that differs, which is impossible.

Proposition SNG-E002: the binary graphs are nonisomorphic. The unique degree-four vertex of G_{bin} is vertex three, and its neighbors are $0, 1, 2, 5$. Exactly one of these neighbors, vertex five, is a leaf. The unique degree-four vertex of H_{bin} is vertex six, with neighbors $0, 1, 2, 3$. Exactly two of those neighbors are leaves. The number of leaf neighbors of a specified degree class is invariant under isomorphism. The graphs are therefore nonisomorphic.

These short arguments are logically independent of the population computation. A differing graph invariant would also prove nonisomorphism, but an elementary structural witness makes the examples easier to inspect. It also guards against a mistaken interpretation in which a relabeling of one graph has merely been counted in a different order.

9.2 The exact Laplacian comparison

The graph Laplacian has entries $L_{ii} = d_i$ and $L_{ij} = -\epsilon_{ij}$, where d_i is vertex degree and ϵ_{ij} is the adjacency indicator. We use the characteristic-polynomial convention $\chi_G(t) = \det(tI - L_G)$. The following values are exact finite computational results, obtained using integer arithmetic. They are not floating-point eigenvalue comparisons and are not described as proof-assistant-checked theorems.

Graphs	Characteristic polynomial
Both spectral graphs	$t^7-14t^6+73t^5-178t^4+209t^3-112t^2+21t$
G_{bin}	$t^7-16t^6+98t^5-290t^4+430t^3-300t^2+77t$
H_{bin}	$t^7-16t^6+98t^5-290t^4+432t^3-302t^2+77t$

The spectral pair is therefore Laplacian-cospectral. The binary pair is not. In particular, the second example must not be advertised as a simultaneous collision of the Laplacian spectrum and the binary cut polynomial. Its role is to isolate a different information gap. Both members of a pair have the same number of spanning trees: three for the spectral pair and eleven for the binary pair. These counts follow independently from the matrix-tree theorem or from the coefficient of t, whose absolute value is seven times the spanning-tree count in these connected seven-vertex graphs.

An integer implementation can compute the characteristic polynomial through traces of powers and Newton identities. If $p_j = \operatorname{tr}(L^j)$ and $c_0 = 1$, the descending coefficients satisfy

$$c_r = -\frac{1}{r}\sum_{j=1}^{r} c_{r-j}p_j.$$

Every division is exact for a matrix characteristic polynomial. This supplies one transparent computational route, although a determinant calculation is equally legitimate. The manuscript's conclusion depends on the exact resulting polynomial, not on a choice of numerical eigensolver or an eigenvalue tolerance.

9.3 All binary cut-volume profiles

For a subset $S \subseteq V$, let ∂S be its edge boundary and define

$$I_{G,r}(q) = \sum_{\substack{S\subseteq V\\ |S|=r}} q^{|\partial S|}.$$

The full binary polynomial is $C_G(x,q) = \sum_{r=0}^{7} x^r I_{G,r}(q)$. Its coefficients count every original subset. A graph automorphism does not reduce a coefficient by identifying subsets in the same orbit. Complementation gives $I_{G,r} = I_{G,7-r}$, because the two sides of a cut have the same boundary. At $q = 1$, the row sum is $\binom{7}{r}$. Both relations provide simple consistency checks on the tables.

r	$I_{G_{\mathrm{sp}},r}(q)$	$I_{H_{\mathrm{sp}},r}(q)$
0	1	1
1	$3q+2q^2+q^3+q^4$	$3q+2q^2+q^3+q^4$
2	$6q^2+7q^3+3q^4+5q^5$	$4q^2+11q^3+3q^4+q^5+2q^6$
3	$q+q^2+8q^3+15q^4+7q^5+3q^6$	$q+3q^2+4q^3+15q^4+11q^5+q^6$
4	$q+q^2+8q^3+15q^4+7q^5+3q^6$	$q+3q^2+4q^3+15q^4+11q^5+q^6$
5	$6q^2+7q^3+3q^4+5q^5$	$4q^2+11q^3+3q^4+q^5+2q^6$
6	$3q+2q^2+q^3+q^4$	$3q+2q^2+q^3+q^4$
7	1	1

The spectral graphs already differ in the binary sector. Their local height distinction does not require heights above one. This is useful evidence beyond the spectrum, but it does not yet demonstrate a benefit from nested thresholds. The binary pair was selected to make that second issue explicit.

r	Common value $I_{G_{\mathrm{bin}},r}=I_{H_{\mathrm{bin}},r}$	Row sum at $q=1$
0	1	1
1	$2q+2q^2+2q^3+q^4$	7
2	$2q^2+8q^3+5q^4+4q^5+2q^6$	21
3	$q^2+4q^3+13q^4+10q^5+5q^6+2q^7$	35
4	$q^2+4q^3+13q^4+10q^5+5q^6+2q^7$	35
5	$2q^2+8q^3+5q^4+4q^5+2q^6$	21
6	$2q+2q^2+2q^3+q^4$	7
7	1	1

Exact computation SNG-E003. Enumeration of all $2^7 = 128$ binary assignments gives the second table identically for both binary graphs. This is equality of the complete two-variable binary polynomial, not merely equality at a chosen value of q, equality of a few moments, or equality of one minimum-cut statistic. Under the usual changes of variables it is the same binary information encoded by an Ising polynomial with a field. The connection to that established literature does not make the next height sector redundant: it identifies exactly what the example must improve upon.

9.4 The first local coefficients and their span decomposition

The coefficients of P_G are easy to derive at small volume. A volume-two vector either puts height two at one vertex or height one at each of two vertices. A volume-three vector has one of the patterns 3, $2+1$, or $1+1+1$. Evaluating the edge energy on each pattern gives

$$A_1(q) := [x]P_G = \sum_i q^{d_i},$$

$$A_2(q) := [x^2]P_G = \sum_i q^{4d_i} + \sum_{i<j} q^{d_i+d_j-2\epsilon_{ij}},$$

$$A_3(q) := [x^3]P_G = \sum_i q^{9d_i} + \sum_{i\neq j} q^{4d_i+d_j-4\epsilon_{ij}} + \sum_{i<j<\ell} q^{d_i+d_j+d_\ell-2(\epsilon_{ij}+\epsilon_{i\ell}+\epsilon_{j\ell})}.$$

These identities are **Proposition SNG-E004**, valid for every finite simple graph. For the interpretation as separate normalized span sectors at these volumes, there must be a zero coordinate; seven vertices certainly suffice. In the marked population, the three summands of A_3 belong respectively to z^3, z^2, z^1. The middle sum is over ordered pairs: the vertex receiving height two is distinguished from the vertex receiving height one. Its population size at $q = 1$ is $7 \cdot 6 = 42$. The other two populations have sizes seven and thirty-five, adding to the eighty-four weak compositions of three into seven parts.

For the spectral pair the common linear coefficient is

$$A_1 = 3q + 2q^2 + q^3 + q^4.$$

The two quadratic coefficients are

$$A_{2,G_{\mathrm{sp}}} = 6q^2 + 7q^3 + 6q^4 + 5q^5 + 2q^8 + q^{12} + q^{16},$$

$$A_{2,H_{\mathrm{sp}}} = 4q^2 + 11q^3 + 6q^4 + q^5 + 2q^6 + 2q^8 + q^{12} + q^{16}.$$

Subtracting and factoring gives the exact identity

$$\boxed{[x^2](P_{G_{\mathrm{sp}}} - P_{H_{\mathrm{sp}}}) = 2q^2(1-q)^3(1+q).} \qquad \text{(SNG-E005)}$$

The height-two-at-one-vertex terms cancel because the degree multisets coincide. The difference is entirely in the binary span-one sector. It is positive for every real $0 < q < 1$, with exact value $3/32$ at $q = 1/2$. Thus the distinction is not a numerical coincidence at one parameter or a sign-sensitive coefficient observed only after rounding.

The binary pair has common first and second coefficients

$$A_1 = 2q + 2q^2 + 2q^3 + q^4,$$

$$A_2 = 2q^2 + 8q^3 + 7q^4 + 4q^5 + 2q^6 + 2q^8 + 2q^{12} + q^{16}.$$

The span-one part of its cubic coefficient agrees because the binary table agrees. Its span-three part also agrees, since it only depends on degree:

$$[x^3z^3]B_{G_{\mathrm{bin}}} = [x^3z^3]B_{H_{\mathrm{bin}}} = 2q^9 + 2q^{18} + 2q^{27} + q^{36}.$$

It remains to inspect exactly forty-two ordered height patterns in the span-two part. The complete nonzero energy profiles are as follows; omitted energies have multiplicity zero in both graphs.

Energy	G_{bin} multiplicity	H_{bin} multiplicity	Difference
3	1	0	1
4	1	2	-1
5	2	2	0
6	4	6	-2
7	6	6	0
8	2	0	2
9	5	4	1
10	5	2	3
11	1	4	-3
12	3	4	-1
13	4	6	-2
14	2	2	0
15	4	2	2
17	1	0	1
18	1	2	-1

Energy	G_{bin} multiplicity	H_{bin} multiplicity	Difference
Total	42	42	0

Exact computation SNG-E006. The difference polynomial obtained from this profile is

$$\begin{aligned}\Delta(q) &= q^3 - q^4 - 2q^6 + 2q^8 + q^9 + 3q^{10} - 3q^{11} \\ &\quad - q^{12} - 2q^{13} + 2q^{15} + q^{17} - q^{18} \\ &= \boxed{q^3(1-q)(1-q^3)^2(1-q^4)^2}.\end{aligned}$$

The factorization is an exact polynomial identity, checkable by multiplication. Enumeration over all ternary assignments in $\{0,1,2\}^7$ independently reproduces the same x^3 span-two difference. Therefore

$$[x^3z^2](B_{G_{\text{bin}}} - B_{H_{\text{bin}}}) = [x^3](P_{G_{\text{bin}}} - P_{H_{\text{bin}}}) = \Delta(q).$$

For every $0 < q < 1$, this is strictly positive. At one half it is exactly $11025/262144$. The factors at zero and one have straightforward meanings: zero eliminates all non-constant heights on a connected graph, while one forgets energy and leaves the same number of assignments on seven coordinates. The additional factors at cube and fourth roots are an algebraic feature of this finite difference; no universal cyclotomic cancellation rule is inferred from them.

9.5 What the analytic signatures see

For the primewise Euler lift, the singularity exponent at $s = 1$ is the common degree polynomial $c_G(q) = \sum_i q^{d_i}$. Accordingly neither pair is distinguished by that leading exponent. The spectral pair shares

$$c_{\text{sp}}(q) = 3q + 2q^2 + q^3 + q^4,$$

and the binary pair shares

$$c_{\text{bin}}(q) = 2q + 2q^2 + 2q^3 + q^4.$$

Both zero-parameter Euler products are $\zeta(7s)$, because all four graphs are connected. The first minimum-cut correction also agrees within each pair. For the spectral graphs, the oriented minimum-cut side sizes occur with multiplicities three, one, one, three at sizes one, three, four, six. For the binary graphs only sizes one and six occur, each twice. If $\mathcal{P}(s) = \sum_p p^{-s}$, the respective first corrections to $\log(Z_G/\zeta(7s))$ are

$$q\big(3\mathcal{P}(s) + \mathcal{P}(3s) + \mathcal{P}(4s) + 3\mathcal{P}(6s)\big)$$

and

$$2q\big(\mathcal{P}(s) + \mathcal{P}(6s)\big).$$

Those common signatures illustrate why a hierarchy is necessary. A graph can share degree data, a leading singular exponent, and its entire minimum-cut volume census with another graph while differing at a later local coefficient. After cancellation of the lower equal local coefficients, the first ratio singularity occurs at $s = 1/2$ for the spectral pair and $s = 1/3$ for the binary pair. Its exponent is respectively $2q^2(1 - q)^3(1 + q)$ or $\Delta(q)$. This is a local continuation statement on the positive-real branch for the ratio after cancellation; it is not a blanket meromorphic continuation of either original product.

The finite evidence supports a sharply bounded conclusion. The height-span marked invariant is strictly finer than its binary sector, and its unmarked specialization can distinguish a displayed Laplacian-cospectral pair. Both examples also distinguish vertex realizations with isometric ordinary cut lattices. The evidence does not prove completeness, does not show that the unmarked series always recovers the binary sector, and does not establish an efficient algorithm for general graph isomorphism. What has been established is a concrete, exact relational distinction and an analytic language that records where that distinction first appears.

9.6 A regular calibration: the cube and the Möbius ladder

A third comparison tests the low-volume blindness of regular graphs proved in the inverse-problems chapter. Let Q_3 be the cube graph on binary vertex labels $0, \dots, 7$, with an edge whenever two labels differ in one binary digit. Let M_8 be the eight-cycle together with the four opposite-vertex chords. Their edge sets are fixed here to make the computation reproducible.

Graph	Edge set
Q_3	01,02,04,13,15,23,26,37,45,46,57,67
M_8	01,04,07,12,15,23,26,34,37,45,56,67

Both graphs have eight vertices, twelve edges, degree three at every vertex, and no triangles. They are nonisomorphic for a simple reason. The cube is bipartite: the parity of the number of ones in the binary label changes across every edge. The ladder contains the five-cycle $0, 1, 2, 3, 4, 0$, using its opposite chord 04, and therefore is not bipartite. No spectral calculation is needed for that nonisomorphism witness.

The common local coefficients through volume three are

$$A_0 = 1, \qquad A_1 = 8q^3,$$

$$A_2 = 12q^4 + 16q^6 + 8q^{12},$$

$$A_3 = 24q^5 + 24q^7 + 8q^9 + 24q^{11} + 32q^{15} + 8q^{27}.$$

Their span decompositions also agree. This agreement is predicted by the theorem that order, regular degree and triangle count determine the marked coefficients through volume three. It is therefore a mathematically controlled delay, rather than an unexplained failure of the population calculation. At volume four the binary subsets already distinguish the graphs:

Graph	Binary four-vertex cut polynomial	Number of subsets
Q_3	$6q^4+32q^6+30q^8+2q^{12}$	70
M_8	$4q^4+40q^6+18q^8+8q^{10}$	70

The six cube subsets of boundary size four are its six square faces. Its two four-vertex subsets of boundary size twelve are its bipartition classes, each containing no internal edges. The ladder has no such boundary-twelve four-subset, consistently with the absence of a bipartition into independent four-sets. The other coefficients count the remaining labeled subsets, rather than quotienting them by graph symmetries.

Exact computation SNG-E007. The first full local difference is

$$[x^4](P_{Q_3} - P_{M_8}) = [x^4 z](B_{Q_3} - B_{M_8}) = \boxed{2q^4(1-q^2)^4}.$$

All other span sectors at volume four agree. The value at $q = 1/2$ is $81/2048$. At the next volume the difference moves to span two:

$$[x^5](P_{Q_3} - P_{M_8}) = [x^5 z^2](B_{Q_3} - B_{M_8}) = \boxed{8q^9(1-q^2)^2(1-q^4)^2}.$$

Its value at one half is $2025/262144$. These identities were replayed using exact coefficient enumeration with the displayed edges. Their status is exact finite computation, followed by exact polynomial factorization. The first ratio singularity of this pair is at $s = 1/4$, with exponent $2q^4(1-q^2)^4$, after the equal lower-volume terms are cancelled. The common leading exponent at $s = 1$ is $8q^3$.

This pair is not another collision of the complete binary polynomial. It agrees only in the early coefficients and separates in a binary sector at volume four. The nonbinary discrepancy at volume five adds an example of how span information is distributed, but is not needed to distinguish the graphs once the binary four-subset census is available. Its role is calibration: it demonstrates a predicted limitation of very early coefficients and provides a positive check that the next level responds to a concrete structural difference.

9.7 Reading the examples as a controlled experiment

The two pairs answer related but different objections. One might first suspect that the construction is simply a complicated way to reproduce the eigenvalues of a graph Laplacian. The spectral pair rules out that interpretation: the entire Laplacian characteristic polynomial agrees, while a positive difference appears at volume two for every interior real parameter. One might then suspect that the extra information is entirely the familiar information obtained by considering all binary cuts and remembering their side sizes. The binary pair rules out that second interpretation: every binary coefficient agrees, while a positive difference appears in the next height span. The experimental sequence is therefore a sequence of exact controls, not a ranking based on a few numerical scores.

The controls are intentionally strong within their stated scope. Both pairs have the same degree multiset, so a singleton degree statistic cannot explain either separation. Both also share their number of components, spanning-tree count, edge connectivity and minimum-cut volume census. Those equalities do not make the examples universally difficult. A simple neighborhood statistic separates each pair, and the second

pair has different Laplacian spectra. The point is to identify a definite piece of information that the height population retains, not to pretend that no other mathematical description can retain it.

For the binary pair, the first additional statistic is particularly transparent. An ordered pair (i, j) receives heights two and one. Its energy is $4d_i + d_j - 4\epsilon_{ij}$. A binary pair only sees $d_i + d_j - 2\epsilon_{ij}$, without distinguishing which endpoint is emphasized. The complete binary triple census adds many constraints, but it still does not force the displayed ordered-pair energy distribution. The table supplies a direct counterexample. The new information is a degree and adjacency mixing statistic that is visible through an asymmetric height pattern and is then aggregated without labels.

The analytic ratio does not create that statistic. It transports an already established local discrepancy into a location on the real singularity ladder. The spectral pair first differs at an exponent sum of two and therefore has its first ratio singularity at one half. The binary pair first differs at an exponent sum of three and therefore has its first ratio singularity at one third. This interpretation is helpful because it connects a combinatorial order of distinction to an analytic scale, while keeping the logical direction clear: the exact coefficient calculation comes first, and the analytic statement follows from it.

Chapter 10

Inverse problems and the depth of a height population

An invariant is a way of asking questions about an object while discarding its presentation. The inverse problem asks how much of the object can be recovered from the answers. For relational height populations, this question must be posed at several levels. A finite depth, the full marked series, an unmarked specialization, and a primewise Euler product are different data structures. Their relationships can be proved in some directions and remain open in others. Confusing them would replace a promising research question with an unsupported completeness claim.

The purpose of this chapter is to define those relationships and identify useful targets. The small examples show that the construction preserves some information absent from binary cut populations and the Laplacian spectrum. They do not tell us how many nonisomorphic graphs remain indistinguishable. In particular, a hierarchy can improve at one level and still stabilize long before it becomes complete. Determining whether that happens is a central mathematical task, rather than a detail to be deferred until after a universal reconstruction claim.

10.1 Which equivalence is being studied?

Fix the number of vertices n. Graphs are considered up to ordinary graph isomorphism, but every coefficient counts assignments on the original vertex set before relabeling. Let

$$B_G(x,q,z) = \sum_{\min a = 0} x^{|a|} q^{E_G(a)} z^{\max a}$$

be the normalized marked series. We say that $G \equiv_{\mathrm{mark}} H$ when these formal series agree. We say that $G \equiv_{\mathrm{unmark}} H$ when their specializations at $z = 1$ agree. The latter is equivalent to equality of the unnormalized local series P_G for graphs of the same order, because $B_G(x,q,1) = (1-x^n)P_G(x,q)$. These are definitions, not claims that either relation is graph isomorphism.

For an integer $D \geq 0$, define the depth-D population by

$$B_G^{\leq D}(x,q,z) = \sum_{h=0}^{D} z^h [z^h] B_G(x,q,z).$$

Equality of these finite polynomials is written $G \equiv_D H$. The depth is the largest height span retained; it is not the total volume and is not an iteration count in a numerical algorithm. For fixed D, every coordinate lies between zero and D, at least one coordinate is zero, and the number of assignments is exactly $(D+1)^n - D^n$. Thus each depth comparison is a finite exact question.

Proposition SNG-I001: hierarchy implications. Graph isomorphism implies marked equality. Marked equality implies equality at every finite depth and implies unmarked equality. Equality at depth $D+1$ implies equality at depth D. Conversely, equality at every finite depth implies equality of the full marked series, because every individual coefficient has a finite span. These statements follow directly from coefficient extraction and specialization. No converse from unmarked equality to marked equality is asserted.

The first depth is precisely the binary cut sector, after its two constant assignments are restored. If

$$C_G(x,q) = \sum_{S \subseteq V} x^{|S|} q^{|\partial S|},$$

then

$$C_G(x,q) = 1 + x^n + [z]B_G(x,q,z). \tag{SNG-I002}$$

The term one is the empty subset and the term x^n is the full subset. Both have zero boundary. The normalized span-one population excludes them because a representative must attain both zero and one. The binary pair in the preceding chapter agrees at depth one and differs at depth two, proving that this particular hierarchy step is strict. This is a theorem about the defined hierarchy, supported by an exact counterexample to equality of its first two levels.

For two graphs define their first marked distinguishing depth by

$$d_{\text{span}}(G,H) = \min\{D : G \not\equiv_D H\},$$

with value infinity when no finite depth distinguishes them. One may separately define a first distinguishing volume by the smallest r at which a coefficient of x^r differs. The two measures answer different questions. A large span can be concentrated on one vertex, while a low span can involve many vertices. The binary pair has distinguishing span two and distinguishing volume three. Its difference at volume three belongs specifically to span two; it is not evidence that every depth-two distinction must first occur at volume three.

10.2 Why retaining labels makes reconstruction easy

The intended inverse problem is difficult because vertex labels have been summed out. It is important to explain why a seemingly stronger multivariate formulation would trivialize part of that question. Define the fully labeled local series

$$\mathcal{P}_G(x_1, \dots, x_n; q) = \sum_{a \in \mathbb{N}^n} q^{E_G(a)} \prod_i x_i^{a_i}.$$

Its singleton coefficient is $[x_i]\mathcal{P}_G = q^{d_i}$, so it determines the degree of every named vertex. Its coefficient at two distinct variables is

$$[x_i x_j]\mathcal{P}_G = q^{d_i + d_j - 2\epsilon_{ij}}.$$

The exponent therefore reveals whether the named pair is an edge. This proves **Proposition SNG-I003**: the labeled formal series determines the entire labeled graph using coefficients of total degree at most two. A formal parameter avoids questions of numerical recognition. At a fixed exact real $q \in (0,1)$, the distinct integer powers are also distinct, but a floating-point implementation should not be substituted for that exact statement without an error analysis.

This easy reconstruction is not the problem Singularity is claiming to solve. The specialization $x_1 = \cdots = x_n = x$ combines many assignments into one volume class. Individual pair identities disappear, and their relational distribution remains. The challenge is to recover a graph from that distribution without quietly reintroducing one coordinate variable per vertex. This distinction also explains why a large formal generating function need not supply an efficient isomorphism algorithm. One can encode the adjacency matrix in a generating function almost immediately; the mathematical question is what survives a specified loss of labels.

10.3 Universal relations within each depth

Several identities are available before any difficult inverse problem is solved. They constrain every valid output and reveal how the hierarchy is organized. They also provide exact checks that are different from merely running the same enumeration twice.

Proposition SNG-I004: span reflection. Put $B_{G,h}(x,q) = [z^h]B_G(x,q,z)$. Then

$$B_{G,h}(x,q) = x^{nh} B_{G,h}(x^{-1},q).$$

For a normalized assignment of span h, the reflected assignment $a_i' = h - a_i$ again has minimum zero and maximum h. Every edge difference changes sign, so energy is unchanged, while volume becomes $nh - |a|$. Reflection is an involution, proving the identity coefficient by coefficient. For $h = 1$, this is the usual complement symmetry of binary cuts. For the seven-vertex binary pair, its proved x^3z^2 difference therefore also occurs at $x^{11}z^2$, since $14 - 3 = 11$. This second location follows from a bijection, not from a new empirical claim.

For $h \geq 1$, an assignment of exact span h has at least one coordinate zero and at least one equal to h. Consequently its volume lies between h and $(n-1)h$. Its energy is an integer between zero and $|E|h^2$. If the graph is connected and $h > 0$, the energy is positive. These bounds make $B_{G,h}$ a finite polynomial with explicitly bounded support. The reflection center $nh/2$ is consistent with the two volume bounds.

There is an inclusion-exclusion construction that does not require enforcing minimum and maximum conditions separately in a transfer algorithm. Define

$$T_{G,h}(x,q) = \sum_{a \in \{0,\ldots,h\}^n} x^{|a|} q^{E_G(a)}, \qquad T_{G,-1} = 0.$$

For $h \geq 1$, shift invariance of energy gives

$$B_{G,h} = T_{G,h} - (1 + x^n)T_{G,h-1} + x^n T_{G,h-2}. \tag{SNG-I005}$$

Assignments missing the top height contribute $T_{G,h-1}$. Assignments missing zero can be shifted down by one, contributing $x^n T_{G,h-1}$. Assignments missing both endpoints have been subtracted twice and contribute $x^n T_{G,h-2}$. The formula holds for every graph because only common-shift invariance was used. At $q = 1$, it reduces to a universal volume polynomial independent of graph edges. In particular,

$$B_{G,h}(1,1) = (h+1)^n - 2h^n + (h-1)^n.$$

Summing through depth D gives $(D+1)^n - D^n$, as required. At unmarked $q = 1$, one obtains the further check

$$B_G(x,1,1) = \frac{1-x^n}{(1-x)^n}.$$

These identities explain why all examples coincide when energy is forgotten. A successful distinguishing polynomial must vanish at that specialization. They also clarify why a zero at $q = 1$ is a necessary population constraint, rather than evidence by itself of a special physical or cyclotomic mechanism.

10.4 Connected components and exact recovery at zero parameter

Disconnected graphs require a different normalization argument from connected graphs. Energy zero means that height is constant on each connected component, not that all vertices share a common height. If the component sizes are $n_1, \dots, n_c$, then

$$P_G(x,0) = \prod_{j=1}^{c} \frac{1}{1-x^{n_j}}, \qquad Z_G(s,0) = \prod_{j=1}^{c} \zeta(n_j s). \tag{SNG-I006}$$

This statement follows by assigning an arbitrary nonnegative common height to each component. It includes isolated vertices, whose factors are $(1-x)^{-1}$ and $\zeta(s)$. It shows why the connected-graph claim that the zero-parameter value is finite at $s = 1$ cannot be copied unchanged to a graph with isolated vertices.

Proposition SNG-I007: the zero-parameter local series determines the multiset of component sizes. Let m_d be the number of components of size d, and put $b_r = r[x^r] \log P_G(x,0)$. Expanding the logarithm gives

$$b_r = \sum_{d \mid r} d\, m_d.$$

Möbius inversion yields

$$m_n = \frac{1}{n}\sum_{d|n}\mu(n/d)b_d.$$

All operations here are formal power-series operations; no analytic continuation is needed. This recovers component sizes, not the internal graph on each component. Within a fixed connected component size, all connected graphs have the same zero-energy population.

For unmarked populations, ordinary disjoint union gives $P_{G\sqcup H} = P_G P_H$. The globally normalized marked population does not simply factor as $B_G B_H$: the two components may be shifted by different heights before their common global minimum is removed, and the global span depends on those relative shifts. A depth algorithm must retain this information or work first with the bounded boxes $T_{G,h}$, which do multiply over disjoint components, and then use the inclusion-exclusion formula. Treating marked disjoint union as an automatic product would erase legitimate assignments.

10.5 Easy classes and the limits of degree information

The first local coefficient is the degree polynomial $c_G(q) = \sum_v q^{d_v}$. It determines the degree multiset, because its coefficient of q^d is the number of degree-d vertices. This immediately resolves several inverse questions under explicit class assumptions. Such small results are useful calibration points; they must not be expanded into a claim about arbitrary graphs with the same degree sequence.

For a star $K_{1,n-1}$,

$$c_G(q) = (n-1)q + q^{n-1}.$$

For a path P_n with $n \geq 3$,

$$c_G(q) = 2q + (n-2)q^2.$$

The two polynomials differ for $n \geq 4$, while the three-vertex star and path are the same graph. A degree multiset consisting of one degree-$n-1$ vertex and $n-1$ leaves forces the star even without a prior star-family promise. A connected graph whose vertices all have degree two is a cycle, so the degree polynomial determines it within that connected class. If connectedness is omitted, a disjoint union of cycles has the same degree polynomial for every partition of the total order into allowed cycle lengths. The zero-parameter component-size series then resolves those different cycle decompositions.

A connected tree of maximum degree at most two is a path: there can be no branch and no cycle. Complete graphs and edgeless graphs are also forced by their degree multisets. Within the family of complete bipartite graphs, the polynomial $aq^b + bq^a$ determines the unordered pair of part sizes $\{a, b\}$. This last statement has a family restriction. Other graph families may share that degree sequence, so the polynomial by itself does not certify complete bipartiteness.

The early coefficients become less informative on regular graphs. If G is d-regular on n vertices with $m = nd/2$, then

$$A_2 = nq^{4d} + mq^{2d-2} + \left(\binom{n}{2} - m\right) q^{2d}.$$

Thus no two d-regular graphs of the same order can be distinguished at volume two. At volume three, the ordered $2+1$ sector is

$$2mq^{5d-4} + (n(n-1) - 2m)q^{5d}.$$

The remaining binary triple sector depends on the numbers N_j of three-vertex subsets spanning exactly j edges. If $T = N_3$ is the triangle count, then

$$N_2 + 3T = n\binom{d}{2}, \qquad N_1 + 2N_2 + 3T = m(n-2),$$

$$N_0 + N_1 + N_2 + T = \binom{n}{3}.$$

These identities count wedges, edge-containing triples, and all triples. They show that n, d, T determine every N_j. Therefore **Proposition SNG-I008** states that regular graphs with the same order, degree and triangle count have identical marked coefficients through total volume three. This result supplies a useful negative prediction for searches. The heterogeneous degrees of the displayed binary pair are part of why an ordered height pattern can distinguish it so early.

There are also useful support bounds on the coefficients at arbitrary volume. Suppose the graph is connected on $n \geq 2$ vertices, with maximum degree $d_{\max} > 0$ and edge connectivity λ. For a nonnegative assignment of volume r,

$$E_G(a) = \sum_i d_i a_i^2 - 2\sum_{ij \in E} a_i a_j \leq d_{\max} \sum_i a_i^2 \leq d_{\max} r^2.$$

The upper bound is attained by placing all height r at a maximum-degree vertex. For $r \geq 1$, equality in $\sum_i a_i^2 \leq r^2$ requires that only one coordinate be positive. Therefore the coefficient of the highest possible energy $q^{d_{\max} r^2}$ in $[x^r]P_G$ is exactly the number of maximum-degree vertices. This is **Proposition SNG-I009**, an exact boundary constraint on every local population. It should not be mistaken for a new distinguishing statistic: it is already determined by the first coefficient's degree data.

For a normalized nonzero assignment, let $h = \max a$. The threshold-cut identity and integrality give $E_G(a) \geq \lambda h$. At least one coordinate is zero, so $r \leq (n-1)h$. Consequently every occupied coefficient of the normalized population at positive volume satisfies

$$E_G(a) \geq \lambda \left\lceil \frac{r}{n-1} \right\rceil.$$

This lower bound is easy to evaluate and does not require solving an optimization problem over all assignments. It is not asserted to be sharp at every volume. A spectral bound can be stronger at large volume: if λ_2 is the first nonzero Laplacian eigenvalue, subtract the mean of a to obtain

$$E_G(a) \geq \lambda_2 \left(\sum_i a_i^2 - \frac{r^2}{n} \right) \geq \frac{\lambda_2 r^2}{n(n-1)}.$$

The last step follows because a zero coordinate leaves at most $n-1$ nonzero coordinates and Cauchy's inequality gives $\sum_i a_i^2 \geq r^2/(n-1)$. These inequalities connect familiar spectral control to the support of the new population without equating the two objects. They help delimit an exact search region, while the examples show that knowledge of the spectrum does not determine the multiplicities within that region.

10.6 What should be searched next?

The first inverse question is whether full marked equality implies graph isomorphism on any substantial class. Trees are an attractive target because the ordinary cut lattice of every tree is a standard integer lattice and therefore discards almost all attachment geometry. The marked volume population could preserve some of that geometry. Completeness on trees is a question, not a conclusion. A single pair of nonisomorphic trees with equal full marked series would refute it; a finite census without a collision would only establish a bounded result.

A second question is whether the full hierarchy eventually stops improving on a fixed graph order. Since there are finitely many isomorphism classes on n vertices, if the full marked invariant is complete on that order then some finite maximum distinguishing depth exists. This existence is purely finite and does not supply a useful bound. If the invariant has collisions, the same argument instead gives a finite depth at which its equivalence partition stabilizes to its final, possibly nontrivial partition. A search can record that stabilization for a finite census, while a theorem must control all higher depths.

A third question concerns the loss of the span marker. It is possible that the unmarked series is complete on a class where no simple proof recovers the binary sector. It is also possible that different marked populations collapse to the same unmarked series. Neither possibility is settled by the current examples. The appropriate experiment compares marked equality and unmarked equality separately, rather than assuming that agreement or disagreement of one automatically decides the other.

A useful search ledger should retain graph order, component sizes, degree polynomial, binary profile, first different volume, first different span, and exact difference polynomial. It should separate pairs selected to collide under a background invariant from arbitrary pairs that happen to be easy to distinguish. Regular graphs with equal triangle count offer controlled difficult cases after the easy volume-three obstructions have been removed. Trees offer a different challenge, in which lattice isometry is automatic but attachment structure varies. Graphs related by changes of attachment within a common block structure test whether the volume marker preserves the intended vertex realization.

These are falsifiable targets. The hypothesis that depth two distinguishes all binary-profile collisions can be rejected by one depth-two collision. The hypothesis that a finite volume cutoff suffices for all graphs of a stated order needs a census whose completeness is itself justified. The hypothesis that a coefficient depends only on a known motif census can be tested by deriving its assignment patterns before scanning data. Every negative result would locate an exact limit of the representation. Such a limit would still be useful mathematics, because it would say what relational information the population has actually retained.

Chapter 11

Archival bridges: what is inherited and what changes

Singularity did not arise from an empty mathematical landscape. PerfectPower, PerfectJet, the SOE program, and older conductor and lattice investigations already contain several of its ingredients. The appropriate historical claim is therefore narrow: the present project selects a different object and a different inverse question from a substantial existing toolkit. It does not claim to invent prime valuations, graph Laplacians, discrete Gaussian heights, cut lattices, binary Ising polynomials, or Euler products. Those established subjects are part of the mathematical background [BHN1997; AM2009; BPR2024]. The value of the new direction must be demonstrated through its precise marking, its exact examples, and the questions those examples make available.

This chapter explains the bridges in mathematical terms. A shared vocabulary such as population, shadow, quotient or phase is not enough to establish a bridge. There must be a defined map, a preserved quantity, an exact identity, or a specifically stated proposal. The distinction matters because the archives contain both successful computations and ambitious descriptions whose full objects were not implemented. Reusing a phrase should not silently upgrade the status of either the old claim or the new one.

11.1 From multiplicative integers to relational exponent geometry

The oldest relevant arithmetic step is the valuation embedding

$$n \longmapsto v(n) = (v_2(n), v_3(n), v_5(n), ...).$$

Every positive integer gives a finitely supported nonnegative integer vector, and multiplication becomes addition: $v(mn) = v(m) + v(n)$. The June collision work used this geometry to define

$$\kappa(m, n) = \sum_p v_p(m) v_p(n).$$

It related shared prime support to a dot product and represented divisors of an integer as a product of finite chains. Its analytic deformation rewarded or penalized shared

exponents with $e^{t\kappa(m,n)}$. The exact Euler factor was $\sum_{a,b\geq 0} e^{tab}p^{-s(a+b)}$. A positive real reward is unstable because the terms with $a = b = N$ contain quadratic exponential growth against only a linear exponential penalty. This is a specific mathematical limitation, not a general failure of the valuation viewpoint [PJ2026].

The present construction uses exponent differences instead of that attractive dot product. For a graph G, put

$$\mathcal{E}_G(n_1, \dots, n_k) = \sum_p \sum_{ij\in E(G)} (v_p(n_i) - v_p(n_j))^2.$$

The sum is finite for every tuple because only finitely many primes divide the product. For $0 \leq q \leq 1$, the weight $q^{\mathcal{E}_G}$ is bounded by one. The resulting series is therefore dominated by the independent tuple series in its elementary convergence domain. This change does more than repair a divergent sign. It creates a zero-energy relation: two adjacent integer positions must have equal valuations at every prime, hence equal integers. On a connected graph, zero energy means the entire tuple is constant.

Proposition SNG-I020: exact synchronization at zero parameter. For a connected graph on k vertices,

$$\mathcal{E}_G(n_1, \dots, n_k) = 0 \quad \Longleftrightarrow \quad n_1 = \cdots = n_k.$$

The forward direction follows because each squared difference must vanish and equality propagates along graph paths. The reverse direction is immediate. For disconnected graphs, equality is required separately on each component. Thus the earlier multiplicative linearization now supports a relational arithmetic population. The graph specifies which exponent coordinates must agree; the primewise product turns that specification into a Dirichlet series.

There is no information gain in the Euler lift itself. Once $P_G(x, q)$ is known, $\prod_p P_G(p^{-s}, q)$ is fixed in its convergence domain. Conversely, the coefficient of a prime power p^r in the Dirichlet series recovers $[x^r]P_G$, because a product equal to p^r forces every tuple entry to be a power of that single prime. The arithmetic series is an exact re-expression with useful analytic operations, not an additional sensor that distinguishes graphs whose full local factors agree.

11.2 From exact populations to navigable height assignments

PerfectPower repeatedly studies a large finite population through counts, conditional counts, rankings and samples instead of explicitly listing every member. That approach transfers directly to bounded height populations [PP2026]. At depth D, the original objects are vectors $a \in \{0, \dots, D\}^k$ with minimum zero. Their number is $(D+1)^k - D^k$. If energy, volume and span constraints are imposed, the retained population is smaller but remains a population of original assignments. A conditional count can determine how many completions remain after the first few coordinates have been fixed.

The released `HeightPopulation` supplies exact completion counts; lexicographic ranking and unranking follow from the usual branching identity. At a prefix u, the total completion count is the sum of completion counts after each allowed next height. A

rank selects the first branch whose cumulative count exceeds it. An original assignment receives a rank by adding all earlier branch populations. This supplies an exact navigation mechanism for height configurations without identifying assignments that merely share a profile. It also makes uniform sampling possible by drawing a uniform integer rank in the feasible population.

The distinction between counting assignments and counting profile types is essential. The energy-volume-span histogram may have relatively few occupied cells, but a cell's coefficient is its multiplicity. Replacing that coefficient by one changes the invariant. The seven-vertex span-two cubic sector has forty-two ordered assignments even when only a modest set of energies occurs. Its successful distinction depends on the distribution of those multiplicities. The population machinery is therefore relevant as an exact computational method, while its prior application domains need not be imported into the present mathematical target.

An exact residual decomposition also becomes possible. A partial assignment contributes the energy of edges whose endpoints are both processed. Edges crossing to unprocessed vertices remain symbolic functions of their boundary heights. Their contributions should not be guessed from the already accumulated scalar energy. Keeping the boundary state allows future assignments to complete the squared differences exactly. This is the natural finite-state interface for dynamic programming over a path or tree decomposition. For a fixed cutoff D, a boundary of size b has at most $(D+1)^b$ raw height states before any further reduction. This is a conditional complexity observation about a chosen decomposition, not a universal small-state theorem.

11.3 From all-future behavioral equivalence to weighted transfer states

The SOE program formalizes the principle that two partial states should be identified exactly when every allowed future continuation gives the same observable behavior [SOE2026]. A direct analog exists for a fixed finite height-transfer computation. The observation is now a polynomial population, and a continuation adds vertices, edges, weights, or terminal conditions according to a declared processing scheme. Two boundary states can be merged only when all continuations permitted by that scheme produce the same remaining generating polynomial.

That sentence is a proposed computational use of the SOE idea, not a claim that the previous Lean theorem has already verified this graph algorithm. Several details must be fixed before a formal transport is legitimate. The state must include any boundary heights required by future edges, the accumulated or deferred volume information, and sufficient data to enforce the global span convention. Polynomial weights must be handled over a specified coefficient structure. A quotient of a two-state partial machine does not automatically verify a family of graph transfers with arbitrarily many boundary states.

The proposal nevertheless has a concrete exact form. Let a processing alphabet encode the legal remaining transfer steps for a fixed decomposition. To a partial state s, associate the function $f_s(w)$ giving the remaining polynomial after continuation word w. Define $s \sim t$ when $f_s(w) = f_t(w)$ for every permitted w. This equivalence is correct by definition and is stable under a common legal next step. The research problem is to compute it efficiently and provide a finite certificate that the retained tests suffice. On a fixed finite continuation space, exhaustive comparison is available. Across unbounded decompositions, the finite characterization requires an additional argument.

The distinction connects usefully to PerfectJet's exact representation work. A vector of sufficient boundary quantities should remain closed under every declared transfer. If it does not, the missing interaction is part of the returned mathematical diagnosis. In the present examples the binary population is insufficient for a specific higher-height question, and the span-two population supplies an explicit witness. This is an actual object-level result, stronger than merely observing that a chosen summary might omit information.

11.4 The relation to classical cut lattices

For a connected graph choose an orientation of its edges and write d for the vertex-to-edge incidence map. Then $E_G(a) = \|da\|^2$, and $\ker d$ consists of common constant shifts. The image of integer vertex heights under d is the integral cut lattice. Every integer height class modulo common shifts has a unique representative of minimum zero. Consequently

$$B_G(1, q, 1) = \sum_{a \in \mathbb{Z}^k / \mathbb{Z}\mathbf{1}} q^{\|da\|^2}$$

is its ordinary theta series for $0 < q < 1$. This is classical lattice territory. The relation between a cut lattice's minimum norm and graph edge connectivity is established background [BHN1997]. Singularity's elementary minimum-cut proof should be read as a proof in its chosen coordinates, not as a historical priority claim about that relationship.

The added volume and span markers retain the position of a height class relative to the vertex-height coordinate system. They are not invariants of an abstract Euclidean lattice by itself. An isometry of cut lattices need not preserve the minimum-normalized sum of vertex heights or their maximum. This is the precise sense in which the new object can distinguish vertex realizations that an ordinary theta series identifies.

The examples give a concrete instance. Gluing two graph blocks at a single vertex makes their cut lattices an orthogonal direct sum. To see this, independently choose integer heights on each block modulo constants, and shift their representatives so the common vertex agrees. The edge-energy sum splits into the two blocks, and the correspondence is bijective. A bridge contributes a one-dimensional standard integer lattice. Each spectral graph therefore has a cut lattice isometric to the triangle cut lattice plus four such bridge factors. Each binary graph has the cut lattice of its common theta block plus two bridge factors. The placement of those bridge attachments does not change this abstract lattice decomposition, but it does change the marked populations established in the preceding chapter.

This distinction also cautions against interpreting the construction as a new kind of Euclidean spectral statistic. The Laplacian spectrum, the abstract cut lattice, and the marked height population forget different information. Their comparison is valuable precisely because they are not interchangeable. The examples show a difference after some conventional data have been fixed; they do not prove a total ordering among every graph invariant.

11.5 Finite conductors and the part that is already old

The conductor archive studied rational phases and exact residue closure. The same algebraic mechanism works for the present unmarked local factor. Let $q = \zeta_L$ be an L-th root of unity. Changing a single coordinate by L changes energy by

$$E_G(a + Le_i) - E_G(a) = 2L \sum_{j \sim i} (a_i - a_j) + L^2 d_i,$$

which is divisible by L. The phase is therefore periodic in each height coordinate modulo L. Splitting $a_i = r_i + Lb_i$ gives the exact identity

$$P_G(x, \zeta_L) = \frac{N_{G,L}(x)}{(1 - x^L)^k}, \qquad N_{G,L}(x) = \sum_{r \in \{0,\dots,L-1\}^k} \zeta_L^{E_G(r)} x^{|r|}. \tag{SNG-I021}$$

The equality holds as a formal series and analytically for $|x| < 1$. It is a finite cyclotomic calculation. The archive already used this kind of residue decomposition for collision and Cartan quadratic forms [PJ2026]. Its presence here is an inherited bridge, not the new claim. The difference is that a graph Laplacian has a common-shift null direction and the project retains coordinate-sensitive marks around that direction.

The marked span variable prevents one from copying the same numerator formula unchanged. Adding L to one coordinate can alter the common minimum and maximum, so its effect on the span marker is not determined by that coordinate's residue alone. One may specialize to $z = 1$, keep a finite height cutoff, or develop an additional boundary description. The unmarked rational formula does not establish a correspondingly simple rational formula for the full marked series.

There is a further exact bridge to spanning trees. For an odd prime p, let

$$\Gamma_G(p) = \sum_{a \in \mathbb{F}_p^k} \exp(2\pi i E_G(a)/p).$$

Fix one vertex height to zero and separate the free common shift. The remaining quadratic form has a reduced Laplacian matrix with determinant $\tau(G)$, the spanning-tree count. If $p \nmid \tau(G)$, the reduced form is nondegenerate. Standard finite-field diagonalization then gives

$$\Gamma_G(p) = p \left(\frac{\tau(G)}{p}\right) g_p^{k-1}, \qquad g_p = \sum_{u \in \mathbb{F}_p} e^{2\pi i u^2/p}. \tag{SNG-I022}$$

In particular its absolute value is $p^{(k+1)/2}$. When p divides the spanning-tree count, the reduced quadratic form is singular and this nondegenerate formula must be replaced by its rank-sensitive version. This provides a defined arithmetic role for discriminant primes: they signal additional modular degeneracy. It does not make every finite-conductor phase statistic complete for graph reconstruction.

11.6 What the new direction actually adds

The archival programs often sought an invariant that survived a passage from a large object to a smaller representation. That ambition remains useful, but the present project asks a more relational inverse question: what is preserved when one counts every nested height pattern while forgetting which vertex carried it? A binary subset census counts one boundary at a time. A two-level height configuration simultaneously places two nested boundaries and records edges that cross both. The exact co-binary example proves that this added relation is not determined by the entire one-boundary census.

The difference should not be called a higher-body interaction in the energy. The energy remains a sum of pairwise edge terms. Higher relational information appears because the observation organizes those same pairwise terms across several nested threshold sets. This distinction matters both mathematically and historically. The project has changed what is retained about a population, rather than positing a new force or claiming a new physical realization.

The resulting research program is therefore internal to mathematics at this stage. It studies a hierarchy of marked populations, their inverse problems, exact computation, and arithmetic singularities after a precisely defined Euler lift. Older physical, commercial, or external ambitions neither prove its significance nor limit the objects it can now examine. The useful inheritance is the ability to calculate exact populations, distinguish insufficient representations, and move consistently between valuation, lattice and conductor descriptions. The new work must earn its further claims through theorems or explicit counterexamples about the height hierarchy itself.

One further lesson from the archive is that a change of coefficient domain and a change of question are separate moves. Exact cyclotomic arithmetic can certify a finite phase sum that floating-point recognition only suggests. Exact integer arithmetic can certify the histogram of a large finite population. Neither act automatically changes what the histogram measures. A perfectly certified binary census remains a binary census. The co-binary example makes that point without relying on measurement noise, numerical instability, unverified physical premises or imperfect training data: even exact binary information can be insufficient for an exact higher-height question.

Conversely, enlarging the question does not require abandoning the old exact arithmetic. The same graph can be studied through an integer coefficient table, a real parameter $0 < q < 1$, a cyclotomic specialization, a cut-lattice theta series, or a Dirichlet series. Each view should carry an explicit account of the markers it retains. A statement about the full marked hierarchy cannot be replaced by a statement about one root-of-unity specialization. A statement about an Euler singularity cannot substitute for equality of every local coefficient. The creative move is the selection of the relational population, while the archival discipline supplies the maps that let its different views be compared honestly.

Chapter 12

Exact computation, certificates, and reproducible experiments

12.1 What a computation is intended to establish

The computational object is a population of labeled integer assignments. For a simple graph $G = (V, E)$, with $V = \{0, \dots, n-1\}$, an assignment $a \in \mathbb{N}^n$ has volume $v(a) = \sum_i a_i$, energy $E_G(a) = \sum_{\{i,j\} \in E} (a_i - a_j)^2$, and span $\max_i a_i - \min_i a_i$. The local series is $L_G(x, q) = \sum_a x^{v(a)} q^{E_G(a)}$. Computing through volume R means computing every coefficient of every polynomial $[x^v]L_G$ for $0 \le v \le R$. It does not mean evaluating the series at a few floating-point points or recording only energies below a second cutoff. An additional energy cutoff is legitimate only when it is part of the declared output and its omitted region is explicit.

The distinction between labeled profiles and isomorphism classes is essential. Graph relabeling does not change the resulting polynomial, but assignments related by a graph automorphism remain distinct members of the population. Dividing counts by an automorphism-group order generally gives incorrect coefficients because stabilizers vary. Orbit enumeration can accelerate computation, provided each representative carries its exact orbit size. The first implementation should retain labeled assignments; a later orbit implementation should reproduce those counts before becoming a source of larger examples.

Finite coefficient equality is a finite claim. Equality through volume R is evidence about a truncation and proves equality of that truncation. It does not prove equality of the full local series. Conversely, a single differing coefficient is a complete witness that two full series differ. The software should make this asymmetry visible in its result types: a discrepancy record is conclusive, whereas an unresolved collision retains its tested bound. A graph pair may be cospectral, binary-census equivalent, or equal through a stated volume without being equivalent for the relational height population.

12.2 A reference enumerator whose meaning is transparent

At a fixed volume v, enumerate all weak compositions of v into n parts. There are exactly $\binom{v+n-1}{n-1}$ such assignments when $n > 0$. For each tuple, calculate energy directly from the normalized edge list, then increment the corresponding integer histogram. Python integers suffice: coefficients are arbitrary-precision integers and no

rounding is involved. A sparse dictionary indexed by energy is preferable to a dense array when the attainable energies occupy a small subset of their possible range. The mathematical definition, rather than an optimization, remains the specification of this enumerator.

An exact and useful bound is $E_G(a) \le \Delta(G)v^2$ for nonnegative assignments of volume v. Indeed, $(a_i - a_j)^2 \le a_i^2 + a_j^2$, so the energy is at most $\sum_i d_i a_i^2 \le \Delta \sum_i a_i^2 \le \Delta v^2$. Concentrating all volume at a vertex of maximum degree attains the bound. The bound therefore checks both data validity and the maximum polynomial degree. It is not a reason to allocate every intermediate degree. At zero volume, the unique assignment is zero and contributes one at energy zero.

The reference cost through R is proportional to $|E|\binom{R+n}{n}$, up to tuple-generation and integer-arithmetic costs. This follows from summing the weak-composition counts, not from an assumption that all energies are distinct. Memory is controlled by the number of nonzero histogram entries retained. Timings must report n, $|E|$, R, the number of assignments visited, histogram support sizes, interpreter version, and hardware. An isolated runtime without these quantities cannot distinguish a better algorithm from an easier instance.

12.3 Three independent representations of the same coefficients

Translation normalization supplies the first independent check. Write $a = b + c\mathbf{1}$, where $c = \min_i a_i$ and $\min_i b_i = 0$. Energy is unchanged and volume increases by nc. Consequently $B_G = (1 - x^n)L_G$, where B_G counts minimum-zero profiles. A direct minimum-zero enumerator should agree with the coefficient subtraction $B_v = L_v - L_{v-n}$, taking negative indices as zero. This tests a bijection and a polynomial operation against an independent predicate. It also catches the mistake of normalizing each connected component separately when the declared convention uses one global minimum.

The second representation uses nested level sets. For a normalized profile of span h, put $S_t = \{i : a_i \ge t\}$. Then $\emptyset \ne S_h \subseteq \cdots \subseteq S_1 \subsetneq V$, and repeated sets are permitted. Volume is $\sum_t |S_t|$. Define $Q(S,T) = \sum_{\{i,j\}\in E}(1_S(i) - 1_S(j))(1_T(i) - 1_T(j))$. For $T \subseteq S$, this is the number of edges between T and $V \setminus S$. Expanding a square gives

$$E_G(a) = \sum_{t=1}^{h} |\delta S_t| + 2 \sum_{1\le u<t\le h} Q(S_u, S_t).$$

A level-chain enumerator therefore has a genuinely different failure surface from a weak-composition enumerator. It should reconstruct the tuple as well as its three statistics and compare complete histograms. Omitting repetitions silently loses profiles with height gaps. Omitting the mixed terms silently replaces squared differences by absolute differences. These are mathematically different models, even when a few tiny examples happen to agree.

The fully marked series $B_G(x,q,z)$ retains span. Its span-one sector is exactly the nonconstant binary population: $[z]B_G = \sum_{\emptyset\ne S\subsetneq V} x^{|S|}q^{|\delta S|}$. This identifies a direct check against a bit-mask cut enumerator. The empty and full subsets require separate treatment in an unnormalized binary census. The marked identity proves that the full

marked series contains the binary sector; setting $z = 1$ mixes sectors and does not establish a method for recovering them.

12.4 Exact polynomials before numerical evaluation

Each volume coefficient is stored as an integer polynomial in q. Addition, multiplication, subtraction, and comparison use degree-indexed sparse maps. Rational coefficients are needed for logarithms, division of formal series, and Euler-factor extraction, so those operations use reduced fractions. A numerical value such as $q = 1/2$ is substituted only after the polynomial identity has been established. Equality at one rational point is weaker than polynomial equality; a discrepancy can also disappear at an exceptional specialization.

The two supplied graph pairs illustrate this discipline. The spectral pair has its first local discrepancy at volume two, with polynomial $2q^2(1-q)^3(1+q)$. The binary-census pair agrees in its complete binary population but first differs at volume three, with polynomial $q^3(1-q)(1-q^3)^2(1-q^4)^2$. The corresponding values at one half are $3/32$ and $11025/262144$. A release should retain expanded coefficient maps, factorizations, and exact substitutions. The factorization explains signs and exceptional endpoints; the expanded maps make recomputation straightforward. The factored expressions alone do not certify that every lower-volume coefficient was checked.

For connected graphs, the endpoint checks are particularly strong. At $q = 1$, the local series is $(1 - x)^{-n}$; at $q = 0$, only constant profiles survive, giving $(1 - x^n)^{-1}$. Thus the normalized series becomes $(1 - x^n)(1 - x)^{-n}$ and 1, respectively. A disconnected graph has additional zero-energy profiles, so the connected endpoint formula must not be used as a universal assertion. Empty-edge graphs and isolated vertices are useful tests precisely because they expose hidden connectedness assumptions.

12.5 Ranking, unranking, and exact sampling

The finite population machinery can provide reproducible witnesses as well as totals. For fixed volume, order weak compositions lexicographically by labeled coordinates. Once a prefix is fixed, its unconstrained completion count is a binomial coefficient. Ranking sums counts of earlier possible next entries; unranking subtracts those counts until the selected branch is found. These operations establish an explicit bijection between assignments and integers in a known interval. A uniformly selected integer therefore gives an exactly uniform assignment, without floating-point probabilities.

Conditioning on energy or span requires completion counts for the conditioned population, not reuse of the unconstrained formula. A memoized state can retain the assigned boundary heights, remaining volume, accumulated energy, and any extrema required by the marking. Whenever an edge has both endpoints assigned, charge its energy once. The remaining state must retain every quantity on which future contributions depend. A compressed state justified only by empirical agreement is a heuristic until its completion-count equivalence is proved. Tree decompositions offer a principled route to smaller boundaries, but the correctness argument is still the same conditional partition of the original labeled population.

The released `HeightPopulation` implementation supports ranking for finite bounded classes whose completion counts are exact and independently checked. The round-trip identities $\mathrm{unrank}(\mathrm{rank}(a)) = a$ and $\mathrm{rank}(\mathrm{unrank}(j)) = j$ must hold over the com-

plete small population. Sampling is useful for exploring typical profiles, but sampling cannot replace complete enumeration when the reported result is a full coefficient or an absence claim. Each sampled witness should include its rank, population size, seed provenance when applicable, and the exact conditioning predicate.

12.6 Certificates and the trusted boundary

The implemented certificate format is `singularity.certificate/1`. It records the finite graph source and its digest, the maximum retained volume, local coefficients, minimum-zero coefficients, fully marked coefficients, total enumerated profile count, and formal Euler exponents. Local integer counts are JSON integers. Exact rational polynomial data use triples `[degree, numerator, denominator]` with integer entries and a positive denominator. The released Python reader preserves these integers exactly. A JavaScript transport should convert large counts to decimal strings or BigInt before arithmetic, because ordinary JavaScript numbers do not preserve integers beyond $2^{53} - 1$; that transport change is a future schema extension rather than a property of the current format.

A pair certificate adds the second graph and digest, its retained local coefficients, the first discrepancy within the common truncation, and an optional supplied factorization. Replay independently enumerates original labeled profiles, checks their energies against the nested-threshold identity, reconstructs all recorded sectors, and replays the formal Euler recomposition. The receipt records a total profile count rather than a separate count for every volume. It does not encode a q-specialization or assert an analytic continuation. Those conclusions belong to the separately stated mathematical theorem. A source digest identifies the declared graph bytes; correctness comes from the independent calculation and mathematical identities.

There are several different assurance levels. Direct Python enumeration is an exact finite computation relative to the interpreter and implementation. Independent enumerators reduce common implementation risk. A proof-assistant theorem about the enumeration algorithm is stronger, but does not automatically verify a particular exported table. A kernel-checked replay of the concrete table crosses that remaining boundary. Documentation should state which level each claim has reached. A schema and a future formalization plan are not present proof-assistant verification.

12.7 Analytic experiments and graph boundaries

For a first discrepancy at volume r, a numerical prime product should evaluate the normalized ratio after extracting $\zeta(rs)^{\delta}$, not multiply divergent raw products and interpret numerical overflow as evidence. The remainder product converges on the positive real region $s > 1/(r+1)$ under the stated local cancellation argument. A plot near $1/r$ should therefore compare the normalized remainder with a certified tail bound. Finite prime cutoffs need their own error estimates. Exact local coefficients establish the exponent; a fitted slope does not establish it independently.

Graph isomorphism is a separate task. Degree sequences, Laplacian spectra, binary census, and truncated height populations are filters, not general isomorphism decision procedures. Small examples can be checked by exhaustive permutation with an explicit witness or exhaustive exclusion. Larger censuses should use a documented canonical-labeling implementation and preserve its version and generation parameters. A claim that an invariant separates every graph in a census requires an inde-

pendently specified complete graph population. A claim that it reconstructs all finite graphs requires a theorem. Neither follows from many successful pair comparisons.

The reproducible unit is a command, its declared finite input population, a complete output artifact, and an explanation of what that artifact proves. Small deterministic fixtures belong in routine checks. Larger censuses belong in explicit experiment scripts with resumable checkpoints and resource limits. The reference enumerator should remain available even after faster methods are introduced: its value is interpretability and independent agreement. The project becomes stronger when a reader can reproduce a modest decisive computation in minutes and then understand exactly which larger statements remain mathematical questions.

Chapter 13

A mathematical research program with distinct proof routes

13.1 The question that organizes the program

The program asks how much relational information survives when labeled integer height profiles are counted by volume and quadratic edge energy, and how a first finite discrepancy becomes visible in an arithmetic Euler product. It is a question about explicit combinatorial and analytic objects. It does not require a universal theory of every obstruction, a claim that all graph information has been recovered, or an interpretation of the Euler product as a source of additional graph data. The local series supplies the data; the arithmetic product changes where and how a discrepancy is expressed.

Three levels of progress should remain distinct. A structural theorem explains an entire graph class or a transformation law. An exact counterexample separates two stated invariants. A complete finite census establishes a bounded classification relative to a documented input population. Each can support a paper, but they answer different questions. The 36 pushes in `docs/development/PUSHES.md` are a dependency plan for obtaining these results, not a list of results already achieved. Negative outcomes are legitimate endpoints when they are recorded with the same precision as successful separations.

13.2 Tree messages and separator geometry

Trees provide the first substantial algorithmic theorem target because conditioning on a vertex height separates the descendant subtrees. Root a tree and let $F_v(t; x, q)$ count assignments on the subtree at v, with $a_v = t$. With heights bounded by H, the exact recursion is

$$F_v(t) = x^t \prod_{u \text{ child of } v} \left(\sum_{j=0}^{H} q^{(t-j)^2} F_u(j) \right).$$

The local bounded-height polynomial is $\sum_t F_{\text{root}}(t)$. Truncating every product modulo x^{R+1} preserves all coefficients through volume R, because every volume exponent is

nonnegative. Taking $H = R$ therefore computes the unrestricted local series through that volume. A proof should specify where the root vertex weight is charged and where each edge is charged; double-counting either invalidates an otherwise plausible recursion.

The same bounded-height computation recovers span marking without tracking both extrema in every message. Let C_H count profiles with every height in $\{0, \dots, H\}$, and set $C_{-1} = 0$. The polynomial $D_H = C_H - x^n C_{H-1}$ counts profiles with minimum zero and maximum at most H. Hence $D_H - D_{H-1}$ counts exact span H. This follows by subtracting profiles whose minimum is positive and shifting them down by one. It is a useful proposed implementation path because its correctness reduces to a simple bijection rather than an enlarged and difficult-to-audit message state.

For general graphs, a tree decomposition replaces a single parent height by a tuple of heights on a separator. The number of assignments to a bag of size $w+1$ is $(H+1)^{w+1}$; this is only the state-count contribution, not the full arithmetic complexity. Volume and energy polynomials still require exact convolutions. The research objective is a correctness theorem together with measured support growth on paths, stars, balanced trees, cycles, and bounded-width constructions. A speed claim should compare complete identical coefficient ranges against the reference enumerator, not compare a marked computation with an unmarked one.

13.3 What nested boundaries remember

The span-one sector records a cut-size distribution indexed by the size of a side. Span two introduces ordered nested pairs $T \subseteq S$, with volume $|S| + |T|$ and energy $|\delta S| + |\delta T| + 2e(T, V \setminus S)$. Thus two cuts with the same individual sizes and boundary sizes can contribute differently because the graph places their boundaries differently. This is an explicit refinement mechanism. It should not be described as a higher-body interaction: the original energy remains a sum over graph edges.

A useful new intermediate object is the finite nested-pair census indexed by $(|S|, |T|, |\delta S|, |\delta T|, e(T, V \setminus S))$. Its purpose is diagnostic. If binary-equivalent graphs first separate at span two, the census can identify which mixed boundary statistic causes that separation. Higher nested censuses are defined similarly, but their dimension grows quickly. The right theorem is not that more marks are always better; it is an explicit projection identity and, when available, a pair of graphs demonstrating that one projection loses information retained by another.

Minimum cuts provide a complementary asymptotic direction. On a connected graph, a nonconstant binary profile has positive energy, and the least such energy is the edge-connectivity. Its side-size multiplicities are visible in the marked binary sector. Nonbinary profiles can have energy tied to several nested boundaries, so identifying the first low-energy terms of the complete normalized series requires an argument about all heights. A promising target is a characterization of the minimum positive energy and its minimizing profiles, followed by a classification of the next energy layer. Each statement should distinguish energy ordering from volume ordering, since the Euler visibility theorem uses the latter.

13.4 Matroids, cut lattices, and the role of a positive chamber

The quadratic form $a \mapsto a^\top L_G a$ factors through integer height vectors modulo constants. Sending a height vector to its oriented edge differences identifies this quotient, in the connected case, with the integral cut lattice equipped with its Euclidean energy. This is established territory. The additional marks in this project are not automatically invariants of an abstract lattice: choosing the minimum-zero representative and taking its coordinate sum uses the vertex-height realization of that lattice. Span also uses that realization.

The corresponding research question is precise: which parts of $B_G(x, q, z)$ are determined by the metric cut lattice, which are determined by the graphic matroid, and which require additional vertex information? Whitney switches are a natural test population because they preserve cycle-matroid structure under the appropriate hypotheses. A complete search can either produce a separating pair, thereby disproving a proposed factorization through the matroid, or identify a transformation law worth proving. Failure to find a small separating pair is not proof of matroid invariance.

This path is especially useful because it can explain a collision rather than merely record one. If an equality comes from an integral lattice isometry that also preserves the chosen height chamber and marks, then a structural bijection may exist. If an isometry preserves energy but moves the chamber or the volume functional, the resulting failure identifies exactly which extra structure the series observes. The intended publication contribution would be a theorem describing such preservation or failure, with an explicit worked graph pair, rather than a broad claim that graph polynomials and lattices are connected.

13.5 Root-of-unity specializations as finite exact laboratories

Let ω have order m. Write every nonnegative height uniquely as $a_i = r_i + mb_i$, with $0 \le r_i < m$. Since energy is an integer quadratic form, $E_G(a) \equiv E_G(r) \pmod m$. Summing the independent nonnegative b_i gives the exact identity

$$L_G(x, \omega) = \frac{\sum_{r \in \{0,\dots,m-1\}^n} x^{\sum_i r_i} \omega^{E_G(r)}}{(1 - x^m)^n}.$$

An infinite height population therefore becomes a finite residue-cube numerator at each root-of-unity specialization. This identity is a direct consequence of residue decomposition; it is not a claim of new arithmetic theory. It offers a particularly clean computational bridge because all coefficients can be represented exactly in a cyclotomic ring, and equality can be decided by reduction modulo its defining polynomial.

At $q = -1$, the result collapses further. Squares agree with their bases modulo two, so $E_G(a) \equiv \sum_i d_i a_i \pmod 2$, and

$$L_G(x, -1) = \prod_i (1 - (-1)^{d_i} x)^{-1}.$$

This specialization sees only degree parity counts. It supplies both a theorem and a warning: an elegant arithmetic specialization can discard most graph information. At

odd primes, evaluating the finite numerator at $x = 1$ produces a quadratic Gauss sum for the Laplacian over a finite field. Rank and the discriminant of its nonsingular part become relevant. The singular constant direction must be accounted for, and prime powers need their own lifting argument; an odd-prime formula must not be reused at two.

There is a separate root-of-unity filter in the volume variable. Averaging $L_G(\omega^j x, q)$ with character weights isolates volumes in a chosen congruence class. It does not isolate binary profiles or recover span after setting $z = 1$. Both kinds of specialization are worthwhile because they provide finite exact comparison problems with different information loss. The positive-real visibility theorem for $0 < q < 1$ does not automatically extend to complex q, where local factors may vanish and a logarithm needs an explicitly chosen domain.

13.6 Order chambers and weighted lattice-point geometry

Every height profile has a unique order obtained by sorting vertices first by height and then by label. For a permutation π in that order, introduce $d_0 = a_{\pi_1}$ and $d_j = a_{\pi_{j+1}} - a_{\pi_j}$ for $1 \le j < n$. These gaps are nonnegative, with $d_j > 0$ whenever $\pi_j > \pi_{j+1}$, because an adjacent descent cannot occur inside a tied block. The resulting chambers are disjoint, including their prescribed boundaries. Volume becomes $nd_0 + \sum_{j=1}^{n-1}(n-j)d_j$, and energy becomes an explicit quadratic form in the remaining gaps.

This description links the population to familiar order and lattice-point methods while retaining a concrete accounting benefit. It can expose which cross terms depend on the relative placement of cuts, and it gives another independent enumerator. The marks are not automatically Ehrhart polynomials. Unweighted integer points in a dilated fixed rational polytope admit established Ehrhart theory; weighting them by q to a quadratic energy introduces an additional structure. Any reciprocity statement must specify whether it concerns unweighted counts, a formal weighted series, or a transformed analytic function, and must prove the claimed boundary terms.

The intended order-theoretic theorem target is a chamber formula with a correct treatment of ties, followed by reductions for graph families with a controlled set of chamber types. An algorithm summing all $n!$ orders may be a useful check without being a useful large-scale algorithm. A later symmetry reduction must preserve the original labeled multiplicities. Successful chamber identities would provide exact explanations for some coefficient factorizations, rather than relying on numerical pattern recognition.

13.7 Visibility strata and formal proof priorities

For fixed graph pairs, coefficient differences are polynomials in q. Their common vanishing defines exceptional parameter values where the first discrepancy moves to a later volume. The generic first discrepancy and its exceptional set should be computed separately. At a rational parameter, direct substitution determines whether a proposed exponent survives. At an algebraic parameter, reduction by a minimal polynomial gives an exact test. Sampling several real values cannot prove that a discrepancy has no interior zero.

The analytic path starts with a finite certified coefficient discrepancy, then proves a local factor expansion and a uniformly convergent normalized prime product in

the stated real domain. More cancellations can move the remainder convergence boundary; they do not justify a global meromorphic continuation of the original product. Nonintegral exponents require positive-real branches or explicitly described local complex branches. The project should seek quantitative remainder bounds uniform on compact parameter intervals away from exceptional points, so numerical demonstrations become reproducible approximations to an already stated theorem.

Formalization should follow the dependency order of the mathematics. First prove finite graph energy definitions, translation normalization, binary extraction, and the nested-level identity. Next verify finite coefficient tables and polynomial discrepancy certificates. Only then formalize the analytic comparison theorem against whatever zeta and infinite-product library is actually available. If a library theorem is assumed or imported externally, state that boundary. The central objective is a short chain from an explicit labeled population to a specific real singularity, with every link visible to a reader and no broader reconstruction or formal-verification claim added by implication.

Chapter 14

Literature, attribution, and publication claims

14.1 The contribution must be stated at the right scale

The ingredients of this project have substantial precedents. Graph Laplacians, integer cut lattices, Ising partition functions, Gaussian fields, order polytopes, Euler transforms, and zeta-factor extraction are established mathematics. Combining their names is not a novelty claim. A defensible contribution must identify a specific population, a theorem about information retained by its particular marks, a verified separation from a stated invariant, a new algorithm with a correctness argument, or a classification result that was not already supplied by the background literature.

The present formulation counts nonnegative labeled vertex heights by total height and squared edge differences, with an optional minimum-zero normalization and span mark. Its immediate value is that this object supports a short exact route from finite profiles to an arithmetic comparison. That route is worth explaining even when individual steps are elementary. Whether the resulting invariant, a specific identity, or a particular example has historical priority remains a literature question. The repository should claim what it establishes, not use the absence of a familiar name as evidence of originality.

There is also a distinction between mathematical novelty and a useful research artifact. A reproducible pair of examples with explicit edges, exact coefficient maps, an independent implementation, and a readable proof can be valuable without being advertised as a new general theory. A literature-ready manuscript should explain the closest known objects first, then state the narrower result. It should give readers enough detail to decide whether the distinction is substantial. Publication merit is strengthened by a theorem that explains a family of examples, rather than by a larger vocabulary around one computation.

14.2 Binary heights and the established Ising polynomial

Define the complete binary census by $C_G(u,v) = \sum_{S \subseteq V} u^{|S|} v^{|\delta S|}$. This is a conventional form of a bivariate Ising polynomial. With spins $\sigma_i \in \{-1, 1\}$, take S to be the negative-spin set. Then $\sum_i \sigma_i = n - 2|S|$ and $\sum_{\{i,j\} \in E} \sigma_i \sigma_j = |E| - 2|\delta S|$. The standard Ising partition sum with uniform coupling and field is therefore $e^{\beta|E| + hn} C_G(e^{-2h}, e^{-2\beta})$, under this sign convention. This change of variables should

accompany comparisons with the literature; polynomial names alone can conceal differing normalizations.

Andrén and Markström studied the bivariate Ising polynomial explicitly. Sokal's account of the multivariate Tutte polynomial explains the larger Potts and matroid setting. Neither reference justifies claiming that every multilevel integer-height population is determined by the binary census. The supplied binary-census pair provides an exact finite separation in the opposite direction for this particular series. The appropriate statement is that these two specified graphs have equal binary census and unequal height coefficients, with the precise certificate supplied. It is not that the project first discovered that distinct graphs can share an Ising polynomial.

The span mark clarifies the information relation. The full marked normalized series contains the nonconstant binary census at span one. The unmarked specialization sums all spans together. No recovery theorem for the binary sector from that specialization has been established here. This boundary belongs in a paper because it distinguishes a proved inclusion of information from a plausible but unproved relationship between two projections.

14.3 Cut lattices, spectra, and Gaussian comparisons

Bacher, de la Harpe, and Nagnibeda develop the integral flow and cut lattices associated with graphs. Their work is a direct antecedent for interpreting squared edge differences as a lattice energy. In a connected graph, integer heights modulo constants map to integral edge gradients; changing an edge orientation changes signs of coordinates, not the energy. A manuscript should cite this framework before discussing its additional chamber and volume marks. The existence of a cut-lattice description is background, while any theorem describing which of these additional marks survive a lattice isometry is a separate contribution requiring proof.

The usual continuous Gaussian integral on the zero-mean subspace depends only on the nonzero Laplacian eigenvalues. For example, integrating $\exp(-tx^{\top}L_G x)$ with Euclidean measure on that subspace gives $\pi^{(n-1)/2}t^{-(n-1)/2}/\sqrt{\det{}' L_G}$. This makes cospectrality a natural comparison, but does not imply that every discrete Gaussian or lattice theta series is spectral. Integrality, a positive-height chamber, and a volume mark retain additional structure. The distinction is classical in spirit and should not be presented as a discovery that continuous spectra can lose combinatorial information.

Rue and Held provide an authoritative account of Gaussian Markov random fields and precision-matrix methods. Their subject helps explain why Laplacians appear, not why this specific height series must be novel. The spectral example in this repository establishes a concrete strict comparison with the Laplacian characteristic polynomial on the given graph pair. A statement about all graph spectra, all lattice theta functions, or complete reconstruction would go beyond that evidence.

14.4 Orders, matroids, and Euler products

Stanley's order-polytope and chain-polytope theory is relevant when height inequalities are organized by partial orders. His work on acyclic orientations is relevant when edge directions are induced by vertex orders. The project should use these theories where their hypotheses actually hold. A finite height box can be decomposed into ordered chambers; this does not make a quadratic-energy weighted sum an ordinary

Ehrhart polynomial. A claimed reciprocity theorem must specify its weights, strictness convention, and boundary strata. An analogy with a familiar theorem is a guide to a proof, not a substitute for one.

The matroid question likewise needs a precise target. Sokal treats graph and matroid partition functions, while cut-lattice theory supplies another route to graph equivalences. A relational height series may retain information about the vertex realization that a matroid forgets. To establish this, one should either produce a certified pair related by an appropriate matroid-preserving operation and separated by the series, or prove a transformation theorem. Neither the phrase 'matroid refinement' nor a generic graph collision is sufficient.

Euler transforms and Möbius inversion explain how local formal coefficients can be reorganized into cyclotomic factors. For a scalar formal series $A(x) \in 1 + xK[[x]]$, with K of characteristic zero, write $\log A(x) = \sum_{n\geq 1} c_n x^n$. There are unique formal exponents $b_n \in K$ with $A(x) = \prod_{n\geq 1}(1-x^n)^{-b_n}$, and $nc_n = \sum_{d|n} db_d$. Moree's work on the formal series Witt transform is a relevant account of the associated cyclotomic identities and Möbius transforms. When q is a parameter, scalar exponents need not be nonnegative integers or even integer-coefficient polynomials in q.

The analytic Euler product requires a separate argument. Extracting finitely many zeta factors can improve the convergence of a remainder, but a formal infinite factorization is not automatically a global analytic continuation. In this project, the first-discrepancy statement uses a finite local expansion and a normalized remainder converging in a specified region. With a nonintegral discrepancy exponent, a real positive branch is available; a blanket meromorphic-continuation claim is not. The arithmetic lift reexpresses local information and supplies a singularity comparison. It does not increase the invariant's graph-reconstruction power.

14.5 A claim register for papers and releases

An established-background claim should name its source and retain its hypotheses. A proved proposition should have a complete argument in the manuscript, with graph order, connectedness, and parameter restrictions visible. An exact finite claim should give the finite input population and a reproducible coefficient or graph certificate. A conjecture should be phrased so that a counterexample or a proof can settle it. A proposed algorithm should not be described as implemented until its code and validation are present. These categories are editorial tools for keeping a mathematical narrative honest, rather than an additional software framework.

For the current examples, acceptable language is: 'For the two graphs with the displayed edge lists, the Laplacian characteristic polynomials agree, while the local height series first differ at volume two by the displayed polynomial.' The binary example has the analogous finite census statement and a first difference at volume three. If independent enumeration has been run, that fact can be added with its method. 'The invariant reconstructs graphs,' 'all graphs were checked,' 'the proof is kernel verified,' and 'this is the first such invariant' require evidence of a different kind and should be absent unless that evidence is supplied.

The strongest candidate conjectures concern explicit comparison boundaries: whether a marked series factors through a given lattice or matroid structure, whether a specified family has a bounded first-separation volume, or whether a nested-cut hierarchy is strict at arbitrarily many levels. An exhaustive finite experiment can guide these conjectures and may disprove them. It cannot establish an

unrestricted theorem. If a proposed direction reproduces an already known object after a change of variables, the correct outcome is an attribution and a revised question.

14.6 Candidate paper abstracts with stated prerequisites

Candidate A: Relational height populations and arithmetic visibility. The following abstract is suitable only when the finite examples and analytic comparison have their full proofs and released certificates: 'We study labeled nonnegative integer heights on a finite graph, weighted by total height and quadratic edge energy. Translation normalization gives minimum-zero representatives, and a span marking resolves nested boundary contributions. Explicit graph pairs separate this population from the Laplacian spectrum and from a complete binary cut census. For a fixed real energy parameter, we show that the first differing local coefficient determines a zeta-factor exponent in the ratio of the associated prime products. The remainder is controlled in a specified real convergence region. The examples and coefficient identities are supplied with exact reproducible data.' This abstract claims no general reconstruction theorem or historical priority.

Candidate B: Exact height enumeration on trees and bounded separators. This is a future methods paper, conditional on the dynamic programs, correctness proofs, and measured experiments being completed: 'We develop exact volume-truncated algorithms for graph height populations using separator-conditioned generating functions. A bounded-height formulation recovers minimum-zero and span-marked coefficients by translation subtraction. Independent composition and level-chain enumerators establish finite reference data, and exact ranking provides reproducible conditioned witnesses. We analyze state growth and polynomial-convolution costs separately and compare complete coefficient ranges on tree and bounded-width families.' Speedups and maximum instance sizes must be inserted from actual measurements, not anticipated results.

Candidate C: What graph structure survives a height population? This is a structural question rather than an available result: 'We compare marked integer-height populations with metric cut lattices and graph matroids, identifying transformation laws and explicit obstructions to preservation of the volume chamber.' A paper with this abstract needs an actual theorem or a decisive classification. The current program supplies the question and test apparatus. It does not yet supply the answer. A negative result showing that a proposed refinement collapses to known data would also be publishable if its statement and proof are substantive.

14.7 Verified references and their intended use

Roland Bacher, Pierre de la Harpe, and Tatiana Nagnibeda, 'The lattice of integral flows and the lattice of integral cuts on a finite graph,' Bulletin de la Société Mathématique de France **125** (1997), 167–198. DOI: 10.24033/bsmf.2303. The Numdam record and article provide the primary source. Use this for the established lattice setting, not as an attribution for the particular marked series defined here.

Daniel Andrén and Klas Markström, 'The bivariate Ising polynomial of a graph,' Discrete Applied Mathematics **157** (2009), 2515–2524. DOI: 10.1016/j.dam.2009.02.021. Use this for the binary comparison and for checking normalization conventions. Equality of a binary census is an established equivalence relation; its failure to determine this particular multilevel series is the narrower issue tested here.

Alan D. Sokal, ‘The multivariate Tutte polynomial (alias Potts model) for graphs and matroids,’ in Surveys in Combinatorics 2005, 173–226. Primary author manuscript: arXiv:math/0503607. Use this for partition-function and matroid background. The document explicitly addresses the flexibility of multivariate weights; any claim of improvement must compare actual specializations and information, rather than compare names.

Håvard Rue and Leonhard Held, Gaussian Markov Random Fields: Theory and Applications, Chapman & Hall/CRC, 2005. DOI: 10.1201/9780203492024. The publisher record verifies authorship and edition. Use this for Gaussian precision-matrix context; the elementary restricted Gaussian integral should still be derived with the measure convention used in the manuscript.

Richard P. Stanley, ‘Two poset polytopes,’ Discrete & Computational Geometry **1** (1986), 9–23. DOI: 10.1007/BF02187680. The author-hosted article is the primary accessible text. Use this for order and chain polytopes and their lattice-point interpretation, with a separate proof for any quadratic-energy extension.

Richard P. Stanley, ‘Acyclic orientations of graphs,’ Discrete Mathematics **5** (1973), 171–178. DOI: 10.1016/0012-365X(73)90108-8. The 2006 reprint has a different DOI; cite the original year when discussing the original theorem. Use this for the classical orientation and reciprocity background, not for a weighted identity that has not been established.

Pieter Moree, ‘The formal series Witt transform,’ Discrete Mathematics **295** (2005), 143–160. DOI: 10.1016/j.disc.2005.03.004. The author’s institutional publication record supplies an accessible description and published version. Use this for formal cyclotomic and Möbius-transform context. The real analytic remainder argument in this project must be stated and proved independently.

Brendan D. McKay and Adolfo Piperno, ‘Practical graph isomorphism, II,’ Journal of Symbolic Computation **60** (2014), 94–112. DOI: 10.1016/j.jsc.2013.09.003. Primary author manuscript: arXiv:1301.1493. Use this when a census relies on canonical labeling or on nauty/Traces. A citation does not substitute for recording the actual executable version, graph-generation command, and completeness boundary of a particular experiment.

These references were checked against primary author, publisher, or journal records for this release. They are a starting bibliography, not a completed priority search. Before journal submission, the author should search directly for integer-valued graph Gaussian models, graph height partition functions, multivariate theta series, and refinements of Ising-equivalent graphs, then compare definitions term by term. A responsible priority statement belongs at the end of that process.

Chapter 15

Implementation and retained computational results

The mathematical manuscripts describe a research program, while this chapter records the executable boundary of version 0.1.0. The core package requires Python 3.10 or newer and uses only the standard library. Documentation generation additionally uses Pandoc and a LaTeX installation. The rendered PDF and HTML are distributed so that a reader does not need those authoring tools merely to read the work.

15.1 Installing and running the project

From the repository root, `python -m pip install -e .` installs the command `singularity`. An installation-free alternative is to set `PYTHONPATH=src` and invoke `python -m singularity`. The commands emit structured JSON so a retained calculation can be consumed by another experiment. Human explanations and proofs live in the monograph; JSON field names are not intended to replace the mathematical exposition.

`python -m singularity examples` prints the four original seven-vertex sources. `python -m singularity compare binary_g binary_h --max-volume 3` recovers their binary-census equality and first local difference. The parameter `--q 1/2` uses an exact rational, and the reported value is an exact fraction. A vanishing specialization is reported without pretending that a symbolic discrepancy remained visible at that parameter.

`python -m singularity profile --example binary_g --max-volume 4` returns the degree data, exact Laplacian polynomial, binary census, local coefficients, normalized coefficients and marked height sectors. The input can instead be a JSON object with an integer vertex count and an explicit edge list. Original vertex labels are retained throughout all finite populations. Isomorphism is a separate mathematical quotient, never an implicit division of multiplicities.

15.2 Implemented mathematical modules

Module	Exact object	Current capability
`graph`	Finite simple graph and integer quadratic energy	Validation, components, relabeling, cuts, Laplacian, signed integer height energy

Module	Exact object	Current capability
`heights`	Total-height and span-marked populations	Weak compositions, local coefficients, minimum-zero normalization, marked sectors
`invariants`	Binary and linear-algebraic comparisons	Complete binary cut census, minimum-cut side sizes, integer characteristic polynomial, first finite difference
`population`	Constrained original-labeled height tuples	Exact counts, lexicographic rank/unrank, feasibility witnesses and finite bound certificates
`polynomials`	Sparse rational q-polynomials and formal x-series	Exact arithmetic, logarithms, specialization, JSON conversion and factor replay
`euler`	Formal zeta exponents	Recursive and Mobius inversion, generalized-binomial recomposition, parameter-specific first discrepancy
`transfer`	Bounded-height forest populations	Separator messages on trees/forests, volume pruning, joint volume-energy histograms and minimum-zero subtraction
`conductor`	Root-of-unity local series	Integer cyclotomic rings, finite residue numerators, arbitrary-index exact coefficients with explicit residue-box budget
`weighted`	Positive integral vertex masses and edge weights	Parallel-weight aggregation and hard equality contraction preserving original energy and mass-weighted height
`arithmetic`	Prime-valuation populations	Positive integer factorization, finite Dirichlet coefficients, conservative exact local remainder and tail constants
`primitive`	Gcd-one normalized height profiles	Mobius extraction, independent gcd enumeration and exact scaling recomposition
`certificates`	Source-bound finite mathematical claims	Independent profile replay, nested-cut checks, normalization and factorization checks

The primitive module implements a useful extension of the foundational scaling identity. For every nonzero normalized profile a, its positive coordinate gcd d is unique; dividing by d produces a primitive profile. Total height and span scale by d, while energy scales by d squared. Consequently full marked populations are recovered from primitive populations by a finite divisor sum in every retained volume. The inverse operation uses the ordinary Mobius function on the scaling factor. The released test checks both directions on every simple graph through four vertices at a declared finite depth.

15.3 Population navigation and exact bounds

The command `population --example binary_g --volume 3 --span 2 --sample 5` compiles the 42 ordered profiles of type 2e_i+e_j. Additional exact or upper-bound constraints may restrict energy, total height, coordinate height and span. The com-

piler records why its search domain is finite. A bound supplied by the total height is different from a bound inferred from connectedness and energy; that distinction remains visible in the report.

The ranking order is lexicographic in the original coordinate tuple. `--rank` checks a proposed tuple and reports its membership and index; `--unrank` reconstructs an original tuple from an index. Prefix completion counts make these operations possible without storing every completed tuple. Their computational cost still depends on the graph's frontier and on the active constraints. The package does not advertise a uniform polynomial-time procedure for arbitrary graphs or arbitrary energy budgets.

15.4 Forest transfer experiment

The forest engine conditions each subtree on the height at its root. Combining a child charges the connecting edge exactly once and convolves the child's volume-energy histogram. A final convolution joins disconnected components. The implementation rejects cycles instead of applying a tree formula to an unsupported graph. A Cartesian enumerator provides an independent oracle on small graphs.

For the path on 96 vertices, heights in 0,1,2, and total height at most 12, the retained population has exactly 2,506,262,962,660,245 original labeled profiles. This is a restricted low-volume population, rather than the entire height cube of size 3^96. Every volume marginal is independently checked against

$$[x^v](1+x+x^2)^{96} = \sum_{j\geq 0}(-1)^j \binom{96}{j}\binom{v-3j+95}{95},$$

with inadmissible binomial coefficients treated as zero. The report in `results/transfer_benchmark.json` records both counts and measured execution times. Timing is an environmental observation. The arithmetic identities and retained population definition are the reproducible mathematical conclusions.

15.5 Complete small-graph census

The script `scripts/census.py` enumerates every labeled simple graph through six vertices, filters connected graphs, and takes an exact isomorphism quotient using degree-respecting permutations. This is an intentionally modest exhaustive baseline. It avoids a hidden external graph database or an unrecorded canonical-labeling dependency. Its work limit is explicit: graph orders one through six.

Vertices	Connected labeled graphs	Connected isomorphism classes	Degree-layer classes	Through volume 2	Through volume 3
1	1	1	1	1	1
2	1	1	1	1	1
3	4	2	2	2	2
4	38	6	6	6	6
5	728	21	19	21	21
6	26704	112	68	100	112

Here the degree layer is the complete coefficient polynomial at volume one, not a single numeric evaluation. Local and fully marked prefixes produce the same class

counts on this retained census. All 112 connected six-vertex classes separate by volume three, and the complete binary census also separates them. The seven-vertex binary collision demonstrates why this finite observation cannot be promoted to a reconstruction theorem. The census gives a reproducible starting point for measuring how visibility changes as the graph population expands.

15.6 Reproduction and document generation

`make test` runs the standard-library suite. `make examples` regenerates the three reference comparisons, graph inputs, finite certificates and a remote conductor coefficient. `python scripts/reproduce.py --check` recomputes these results and compares them with their retained JSON. `python scripts/benchmark_transfer.py` reruns the path experiments. The exhaustive census is separate because its search scope is broader than the routine example replay.

`make book` assembles the chapter sources into a navigable monograph, generates LaTeX and HTML, and compiles the mathematical PDF. The shorter article in `docs/papers/visibility.md` provides a paper-shaped entry to the central construction and visibility theorem. `make release` performs the checks and creates a complete archive with source, documents, tests, inputs and results. Generated archives exclude earlier archives, temporary files and the local git object directory, preventing recursive packaging.

The released finite calculations are computational evidence for their declared inputs. The proofs in the manuscript establish the mathematical identities and analytic transfer statements with their stated hypotheses. Neither the software tests nor a source digest imply a proof-assistant verification of the implementation. This separation lets later pushes strengthen formal coverage without rewriting the meaning of the existing results.

Chapter 16

Thirty-six concrete research pushes

This is a proposed sequence after the initial release. No entry is marked complete by its presence here. A push closes when its mathematical output and acceptance criterion are both met; a negative theorem or certified counterexample can be a successful outcome. Pushes are ordered for review and integration, while the explicit dependencies identify which preparatory work can proceed independently. Every completed push should update the manuscript, release its finite inputs and outputs, and record unresolved questions in ordinary prose. A larger computation is justified by the question it settles, not by its runtime.

The sequence follows six linked paths: exact populations, separator algorithms, nested boundaries and matroids, arithmetic specializations, order geometry, and visibility. Formal proof work follows concrete definitions and certificates. General graph reconstruction is a question to be investigated, never an acceptance criterion satisfied by a finite census. The two supplied graph pairs remain fixed regression examples throughout, so an expansion cannot silently change their labeling, normalization, or first-discrepancy statements.

16.1 P01 — Freeze the coefficient semantics

Inputs: the graph and height definitions of the initial release. **Outcome:** a versioned mathematical data specification covering labels, minimum-zero normalization, span, and truncation. **Acceptance:** every public function has an explicit finite meaning, and disconnected examples distinguish global from componentwise translation. **Dependencies:** the initial release. This push settles conventions before optimization introduces alternative representations.

16.2 P02 — Add independent level-chain enumeration

Inputs: P01 and the nested-boundary identity. **Outcome:** a second coefficient enumerator using nonempty nested level sets with repetitions. **Acceptance:** complete marked histograms agree with weak compositions on a declared small graph population, including profiles with height gaps. **Dependencies:** P01. The implementation must charge mixed boundary terms rather than substitute absolute-difference energy.

16.3 P03 — Publish portable coefficient certificates

Inputs: both enumerators and canonical graph serialization. **Outcome:** `relational-height/1` records with exact integer or rational coefficients and source digests. **Acceptance:** an independent reader rejects malformed degrees, duplicate terms, invalid edges, and altered coefficients, while reproducing all released small tables. **Dependencies:** P01–P02. Hash agreement alone does not close this push.

16.4 P04 — Give the spectral pair a short coefficient proof

Inputs: the fixed seven-vertex spectral pair. **Outcome:** a human-readable volume-two classification yielding $2q^2(1-q)^3(1+q)$. **Acceptance:** classify every profile type at volumes zero, one, and two; compare exact Laplacian characteristic polynomials; match both enumerators. **Dependencies:** P03. The proof should explain the difference through explicit degree and adjacency contributions.

16.5 P05 — Explain the binary-census separation

Inputs: the second fixed graph pair and its 128 binary assignments. **Outcome:** a full binary equality certificate and a volume-three proof of the displayed discrepancy. **Acceptance:** identify the span-two profile types responsible, account for every lower coefficient, and recover the exact value at $q = 1/2$. **Dependencies:** P02–P03. Binary sampling is insufficient.

16.6 P06 — Derive universal low-volume formulas

Inputs: arbitrary simple graphs and profile partitions through volume three. **Outcome:** formulas using degree statistics and explicitly defined adjacent-vertex statistics. **Acceptance:** prove each formula by a disjoint profile classification and compare it against complete small-graph enumeration. **Dependencies:** P04–P05. Any proposed graph statistic must have a definition independent of the formula it explains.

16.7 P07 — Formalize translation normalization

Inputs: the finite graph energy definition and minimum-zero bijection. **Outcome:** a proof-assistant theorem for $B = (1 - x^n)L$ at coefficient level. **Acceptance:** prove injectivity, surjectivity, volume shift, and energy preservation without admitted lemmas; publish the actual build command and dependencies. **Dependencies:** P01. No analytic infinite-product library is needed for this push.

16.8 P08 — Formalize the nested-level energy identity

Inputs: finite height tuples and threshold sets. **Outcome:** kernel-checked volume, span, and quadratic energy identities. **Acceptance:** allow repeated levels, prove the mixed term equals the stated edge count for nested sets, and derive binary extraction. **Dependencies:** P07. The theorem should expose the pairwise edge model rather than introduce a different interaction law.

16.9 P09 — Implement bounded-height tree messages

Inputs: rooted trees, height bound H, and volume bound R. **Outcome:** exact separator-conditioned message polynomials. **Acceptance:** prove the child-product recursion, count every vertex and edge once, and match reference coefficients on paths, stars, and branched trees. **Dependencies:** P02. Report polynomial support sizes separately from height-state counts.

16.10 P10 — Prove safe volume truncation

Inputs: P09's recursion and nonnegative volume exponents. **Outcome:** a theorem that intermediate reduction modulo x^{R+1} preserves the requested output. **Acceptance:** show that $H = R$ suffices for unrestricted coefficients through R, then demonstrate larger tree computations with identical semantics. **Dependencies:** P09. Benchmarks must compare complete coefficient ranges.

16.11 P11 — Recover span by bounded-box subtraction

Inputs: the bounded-height polynomials C_H. **Outcome:** a marked tree algorithm using $D_H = C_H - x^n C_{H-1}$ and exact-span differences. **Acceptance:** prove the shift subtraction and compare every (v, e, h) count with direct marked enumeration. **Dependencies:** P09–P10. Negative intermediate coefficients must be investigated rather than clipped to zero.

16.12 P12 — Extend messages to bounded separators

Inputs: a graph with an explicitly supplied tree decomposition. **Outcome:** a bag-conditioned exact algorithm and decomposition checker. **Acceptance:** verify the decomposition axioms, prove local charge bookkeeping, and compare cycles and bounded-width examples with reference data. **Dependencies:** P10–P11. State-count and sparse polynomial-convolution costs must remain separate in the complexity discussion.

16.13 P13 — Classify the first positive-energy layer

Inputs: connected graphs and marked coefficients. **Outcome:** a theorem describing minimum-energy nonconstant profiles and their relation to minimum cuts. **Acceptance:** handle possible nonbinary competitors explicitly, prove the stated multiplicities, and compare with exhaustive profiles. **Dependencies:** P06 and P08. Energy order must not be confused with volume order.

16.14 P14 — Compute the nested-pair census

Inputs: all $T \subseteq S \subseteq V$ in a declared graph population. **Outcome:** the five-statistic nested-boundary tensor described in Chapter 13. **Acceptance:** recover the span-two generating function by an explicit projection and explain the binary-census example's separation. **Dependencies:** P05 and P08. A new tensor name is not itself a structural result.

16.15 P15 — Separate lattice information from chamber information

Inputs: the cut-lattice realization and minimum-zero volume functional. **Outcome:** a precise preservation question with proved sufficient conditions for marked-series equality. **Acceptance:** specify which lattice isometries preserve height volume and span, and exhibit a nontrivial worked instance or a failure of a proposed condition. **Dependencies:** P08 and P14. Abstract energy preservation alone is insufficient.

16.16 P16 — Test Whitney switches systematically

Inputs: a documented family of admissible two-terminal gluings and switches. **Outcome:** exact comparison tables and either a separating pair or a transformation conjecture. **Acceptance:** verify the matroid-preserving hypotheses independently and retain tested volume bounds for unresolved pairs. **Dependencies:** P15. No finite failure to separate may be reported as universal matroid invariance.

16.17 P17 — Add exact volume congruence filters

Inputs: local polynomials and a positive modulus. **Outcome:** character-filtered populations for volume residue classes. **Acceptance:** prove the root-of-unity averaging identity and compare it with direct modular filtering of assignments. **Dependencies:** P03. This filter must not be described as extracting span or the binary sector after those marks have been summed away.

16.18 P18 — Prove root-of-unity rationality in energy

Inputs: an energy root of unity of order m. **Outcome:** the residue-cube numerator over $(1 - x^m)^n$, represented in an exact cyclotomic ring. **Acceptance:** prove the coordinate residue decomposition and reproduce local coefficients through an independent bound. **Dependencies:** P03 and P17. Complex specialization does not inherit positive-real branch assertions.

16.19 P19 — Analyze finite quadratic Gauss sums

Inputs: P18 numerators at $x = 1$ and odd-prime moduli. **Outcome:** formulas in terms of Laplacian rank and nonsingular discriminant, with singular directions accounted for. **Acceptance:** compare direct residue enumeration with diagonalized forms and give explicit exceptional-prime examples. **Dependencies:** P18. Prime powers and characteristic two remain separate until proved.

16.20 P20 — Build the disjoint order-chamber formula

Inputs: vertex permutations with label-based tie breaking. **Outcome:** a gap-coordinate expression for volume and quadratic energy. **Acceptance:** prove that every profile belongs to exactly one chamber, including tied heights, and match complete histograms. **Dependencies:** P02 and P06. Summing closed chambers without boundary corrections fails this criterion.

16.21 P21 — Relate constrained heights to order polytopes

Inputs: an explicit finite poset of height inequalities. **Outcome:** a comparison between constrained populations and established order-polytope lattice counts. **Acceptance:** derive the unweighted bounded-height specialization with correct weak and strict inequalities, then identify what quadratic weighting adds. **Dependencies:** P20. Ordinary Ehrhart conclusions require an ordinary fixed-polytope counting problem.

16.22 P22 — Establish one genuine reciprocity statement

Inputs: a precisely chosen bounded order or chamber population. **Outcome:** a proved reciprocity identity with explicit boundary terms, or a counterexample to the proposed weighted extension. **Acceptance:** verify complete small instances and give a full argument for the stated scope. **Dependencies:** P21. An analogy to Ehrhart or chromatic reciprocity does not close the push.

16.23 P23 — Implement formal logarithmic extraction

Inputs: exact local coefficient polynomials through volume R. **Outcome:** formal logarithm, inverse series, and finite Euler exponents using rational arithmetic. **Acceptance:** reconstruct the input truncation by multiplication and verify the Möbius coefficient identities. **Dependencies:** P03. Distinguish scalar exponents depending on q from integer combinatorial multiplicities.

16.24 P24 — Determine exceptional energy parameters

Inputs: discrepancy polynomials for a graph pair. **Outcome:** exact rational or algebraic parameter strata where the first differing volume changes. **Acceptance:** isolate every real root in $(0, 1)$ for the tested leading polynomials and recompute at each exceptional value. **Dependencies:** P23. A plot or a finite parameter grid cannot establish absence of roots.

16.25 P25 — Organize successive visibility levels

Inputs: coefficient differences through a declared volume bound. **Outcome:** a filtration by first discrepancy, including parameter-dependent refinements. **Acceptance:** prove every reported equality below the first differing coefficient and label unresolved collisions with their tested bound. **Dependencies:** P23–P24. Claims of an unrestricted separation depth require an additional theorem.

16.26 P26 — Make analytic remainder bounds explicit

Inputs: local coefficient growth bounds and a first discrepancy at r. **Outcome:** a quantified normalized-factor estimate sufficient for convergence when $s > 1/(r + 1)$. **Acceptance:** display constants, parameter restrictions, and treatment of finitely many small primes. **Dependencies:** P25. A formal expansion without a controlled remainder is not the analytic theorem.

16.27 P27 — Produce controlled real singularity experiments

Inputs: P26 bounds and exact discrepancy exponents. **Outcome:** prime-product evaluations of the normalized positive-real remainder near $s = 1/r$. **Acceptance:** attach finite-cutoff error intervals and recover the predicted asymptotic within them. **Dependencies:** P26. Numerical slope fitting illustrates an established exponent; it does not determine the exponent's proof.

16.28 P28 — State the available complex domain exactly

Inputs: fixed admissible q, local factor zeros, and normalized products. **Outcome:** a local complex branch statement on an explicitly described nonvanishing domain, or a documented obstruction. **Acceptance:** specify logarithm choices and convergence uniformly on compact subsets. **Dependencies:** P26. Do not infer global meromorphic continuation from real-axis control.

16.29 P29 — Specify a complete finite graph census

Inputs: a chosen graph class, order bound, and graph-generation method. **Outcome:** a canonical input manifest with isomorphism handling documented. **Acceptance:** independently reconcile population counts, generation options, and graph hashes. **Dependencies:** P03. Degree and spectrum filters are not a substitute for a complete census or a graph-isomorphism decision.

16.30 P30 — Measure collision classes across invariants

Inputs: P29's fixed graph population and exact coefficient algorithms. **Outcome:** collision partitions for spectra, binary census, and specified height truncations. **Acceptance:** release full class memberships, first-discrepancy witnesses, resource limits, and unresolved classes. **Dependencies:** P12, P14, and P29. Report bounded separation percentages with their exact denominator.

16.31 P31 — Resolve targeted collisions structurally

Inputs: the smallest unresolved classes from P30. **Outcome:** an explicit bijection, a higher-volume separation, or a precise remaining conjecture for each selected pair. **Acceptance:** explain the mechanism rather than only increase a cutoff; retain negative results. **Dependencies:** P16 and P30. A full-series equality claim needs an all-volume argument.

16.32 P32 — Seek a separation bound for one graph family

Inputs: low-volume formulas, tree messages, and collision evidence. **Outcome:** a proved first-separation bound for a narrowly stated family, or an explicit family defeating the proposed bound. **Acceptance:** parameterize the family and prove the statement uniformly. **Dependencies:** P06, P12, and P31. Complete graph reconstruction remains outside the claim unless proved separately.

16.33 P33 — Verify concrete coefficient certificates in a kernel

Inputs: P03 certificates and the formal finite identities. **Outcome:** a checker connecting released tables to the mathematical coefficient definition. **Acceptance:** kernel-check both decisive pair certificates and reject a changed coefficient or graph edge. **Dependencies:** P07–P08 and P23. A verified parser alone does not verify enumerated multiplicities.

16.34 P34 — Formalize the visibility comparison boundary

Inputs: the finite first-discrepancy theorem and available analytic libraries. **Outcome:** a formal statement connecting the normalized local expansion to the real zeta-factor asymptotic. **Acceptance:** expose every imported theorem and any unformalized premise; claim complete verification only when the full chain compiles. **Dependencies:** P26 and P33. Missing library infrastructure is recorded explicitly.

16.35 P35 — Release the first focused manuscript

Inputs: the exact examples, finite proofs, and real comparison theorem. **Outcome:** a literature-positioned paper matching Candidate A in Chapter 14, with source and reproducible artifacts. **Acceptance:** independently replay decisive coefficients, reconcile every claim with its proof or certificate, and remove unsupported priority language. **Dependencies:** P04–P05, P26–P27, and P33; formal analytics are included only to their achieved scope.

16.36 P36 — Choose the second paper from actual results

Inputs: the tree, matroid, arithmetic, and order outcomes. **Outcome:** one coherent methods or structural paper, plus an updated open-problem ledger. **Acceptance:** select a theorem or decisive negative classification as its center, complete the relevant literature comparison, and package reproducible data. **Dependencies:** P19, P22, P31–P32, and P35. Do not combine unfinished paths merely to make the manuscript larger.

The final decision is mathematical. If the work yields only a useful enumerator and two small examples, the release should say so clearly. If it yields a family theorem, a genuine comparison boundary, or a sharp analytic statement with quantitative control, the paper should lead with that result. The plan is designed to make these outcomes distinguishable early enough that the next few dozen pushes accumulate knowledge rather than accumulate terminology.